SM+ϕ_1+ϕ_2+ϕ_3+ϕ_4+ϕ_5

Model Report

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1 Field Content and Anomalies

1.1 Standard Model Field Content

$$\begin{aligned} q =& \quad \text{weyl fermion} \quad (3, 2, 1/6) \\ l =& \quad \text{weyl fermion} \quad (1, 2, -1/2) \\ d^C =& \quad \text{weyl fermion} \quad (\overline{3}, 1, 1/3) \\ u^C =& \quad \text{weyl fermion} \quad (\overline{3}, 1, -2/3) \\ e^C =& \quad \text{weyl fermion} \quad (1, 1, 1) \\ H =& \quad \text{scalar} \quad (1, 2, 1/2) \end{aligned}$$

1.2 BSM Field Content

$$\begin{aligned} \phi_1 =& \quad \text{scalar} \quad (3, 1, -1/3) \\ \phi_2 =& \quad \text{scalar} \quad (3, 1, -4/3) \\ \phi_3 =& \quad \text{scalar} \quad (3, 2, 7/6) \\ \phi_4 =& \quad \text{scalar} \quad (3, 2, 1/6) \\ \phi_5 =& \quad \text{scalar} \quad (3, 3, -1/3) \end{aligned}$$

1.3 Anomaly Summary

Anomalies: free.

Channel	Value
$SU(3)c - SU(3)c - SU(3)c$	0
$SU(2)L - SU(2)L - U(1)Y$	0
$SU(3)c - SU(3)c - U(1)Y$	0
$U(1)Y - gravity$	0
$U(1)Y - U(1)Y - U(1)Y$	0

2 Compact Form of the Full Lagrangian

2.1 Yukawa and Fermion-Mass Terms

$$Y_1\, qq\phi_1 + \text{h.c.}$$

$$Y_2\, qq\phi_5 + \text{h.c.}$$

$$Y_3\, {\phi_1}^\dagger lq + \text{h.c.}$$

$$Y_4\, {\phi_5}^\dagger lq + \text{h.c.}$$

$$Y_5\, H^\dagger d^C q + \text{h.c.}$$

$$Y_6\, u^C qH + \text{h.c.}$$

$$Y_7\, {\phi_3}^\dagger e^C q + \text{h.c.}$$

$$Y_8\, d^C l\phi_4 + \text{h.c.}$$

$$Y_9\, u^C l\phi_3 + \text{h.c.}$$

$$Y_{10}\, H^\dagger e^C l + \text{h.c.}$$

$$Y_{11}\, {\phi_1}^\dagger d^C u^C + \text{h.c.}$$

$$Y_{12}\, e^C d^C \phi_2 + \text{h.c.}$$

$$Y_{13}\, {\phi_2}^\dagger u^C u^C + \text{h.c.}$$

$$Y_{14}\, e^C u^C \phi_1 + \text{h.c.}$$

2.2 Scalar Potential

$$m_1^2\, H H^\dagger$$

$$m_2^2\, \phi_1 {\phi_1}^\dagger$$

$$m_3^2\, \phi_2 {\phi_2}^\dagger$$

$$m_4^2\, \phi_3 {\phi_3}^\dagger$$

$$m_5^2\, \phi_4 {\phi_4}^\dagger$$

$$m_6^2\, \phi_5 {\phi_5}^\dagger$$

$$\mu_7\, H \phi_1 {\phi_4}^\dagger + \text{h.c.}$$

$$\mu_8\, H {\phi_4}^\dagger \phi_5 + \text{h.c.}$$

$$\mu_9\, \phi_1 \phi_4 \phi_4 + \text{h.c.}$$

$$\mu_{10}\, \phi_2 \phi_3 \phi_4 + \text{h.c.}$$

$$\lambda_{11}\, H H H^\dagger H^\dagger$$

$$\lambda_{12}\, H H \phi_2 {\phi_5}^\dagger + \text{h.c.}$$

$$\lambda_{13}\, H H {\phi_3}^\dagger \phi_4 + \text{h.c.}$$

$$\lambda_{14}\, H H^\dagger \phi_1 {\phi_1}^\dagger$$

$$\lambda_{15}\, H H^\dagger \phi_1 {\phi_5}^\dagger + \text{h.c.}$$

$$\lambda_{16}\, H H^\dagger \phi_2 {\phi_2}^\dagger$$

$$\lambda_{17,c1}\, H H^\dagger \phi_3 {\phi_3}^\dagger$$

$$\lambda_{17,c2}\, H H^\dagger \phi_3 {\phi_3}^\dagger$$

$$\lambda_{19,c1}\, H H^\dagger \phi_4 {\phi_4}^\dagger$$

$$\lambda_{19,c2}\, H H^\dagger \phi_4 {\phi_4}^\dagger$$

$$\lambda_{21,c1}\, H H^\dagger \phi_5 {\phi_5}^\dagger$$

$$\lambda_{21,c2}\, H H^\dagger \phi_5 {\phi_5}^\dagger$$

$$\lambda_{23}\, H \phi_1\phi_2\phi_3 + \text{h.c.}$$

$$\lambda_{24}\, H \phi_1\phi_4\phi_5 + \text{h.c.}$$

$$\lambda_{25}\, H \phi_1{}^\dagger\phi_3{}^\dagger\phi_5{}^\dagger + \text{h.c.}$$

$$\lambda_{26}\, H \phi_2\phi_3\phi_5 + \text{h.c.}$$

$$\lambda_{27}\, H \phi_3{}^\dagger\phi_5{}^\dagger\phi_5{}^\dagger + \text{h.c.}$$

$$\lambda_{28}\, H \phi_4\phi_5\phi_5 + \text{h.c.}$$

$$\lambda_{29}\, \phi_1\phi_1\phi_1{}^\dagger\phi_1{}^\dagger$$

$$\lambda_{30}\, \phi_1\phi_1\phi_5{}^\dagger\phi_5{}^\dagger + \text{h.c.}$$

$$\lambda_{31,c1}\, \phi_1\phi_1{}^\dagger\phi_2\phi_2{}^\dagger$$

$$\lambda_{31,c2}\, \phi_1\phi_1{}^\dagger\phi_2\phi_2{}^\dagger$$

$$\lambda_{33,c1}\, \phi_1\phi_1{}^\dagger\phi_3\phi_3{}^\dagger$$

$$\lambda_{33,c2}\, \phi_1\phi_1{}^\dagger\phi_3\phi_3{}^\dagger$$

$$\lambda_{35,c1}\, \phi_1\phi_1{}^\dagger\phi_4\phi_4{}^\dagger$$

$$\lambda_{35,c2}\, \phi_1\phi_1{}^\dagger\phi_4\phi_4{}^\dagger$$

$$\lambda_{37,c1}\, \phi_1\phi_1{}^\dagger\phi_5\phi_5{}^\dagger$$

$$\lambda_{37,c2}\, \phi_1\phi_1{}^\dagger\phi_5\phi_5{}^\dagger$$

$$\lambda_{39,c1}\, \phi_1\phi_2{}^\dagger\phi_3{}^\dagger\phi_4 + \text{h.c.}$$

$$\lambda_{39,c2}\, \phi_1\phi_2{}^\dagger\phi_3{}^\dagger\phi_4 + \text{h.c.}$$

$$\lambda_{41,c1}\, \phi_1\phi_3\phi_3{}^\dagger\phi_5{}^\dagger + \text{h.c.}$$

$$\lambda_{41,c2}\, \phi_1\phi_3\phi_3{}^\dagger\phi_5{}^\dagger + \text{h.c.}$$

$$\lambda_{43,c1}\, \phi_1\phi_4\phi_4{}^\dagger\phi_5{}^\dagger + \text{h.c.}$$

$$\lambda_{43,c2}\, \phi_1\phi_4\phi_4{}^\dagger\phi_5{}^\dagger + \text{h.c.}$$

$$\lambda_{45}\, \phi_1\phi_5\phi_5{}^\dagger\phi_5{}^\dagger + \text{h.c.}$$

$$\lambda_{46}\, \phi_2\phi_2{\phi_2}^\dagger{\phi_2}^\dagger$$

$$\lambda_{47,c1}\, \phi_2{\phi_2}^\dagger\phi_3{\phi_3}^\dagger$$

$$\lambda_{47,c2}\, \phi_2{\phi_2}^\dagger\phi_3{\phi_3}^\dagger$$

$$\lambda_{49,c1}\, \phi_2{\phi_2}^\dagger\phi_4{\phi_4}^\dagger$$

$$\lambda_{49,c2}\, \phi_2{\phi_2}^\dagger\phi_4{\phi_4}^\dagger$$

$$\lambda_{51,c1}\, \phi_2{\phi_2}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{51,c2}\, \phi_2{\phi_2}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{53,c1}\, \phi_2\phi_3{\phi_4}^\dagger{\phi_5}^\dagger + \text{h.c.}$$

$$\lambda_{53,c2}\, \phi_2\phi_3{\phi_4}^\dagger{\phi_5}^\dagger + \text{h.c.}$$

$$\lambda_{55,c1}\, \phi_3\phi_3{\phi_3}^\dagger{\phi_3}^\dagger$$

$$\lambda_{55,c2}\, \phi_3\phi_3{\phi_3}^\dagger{\phi_3}^\dagger$$

$$\lambda_{57,c1}\, \phi_3{\phi_3}^\dagger\phi_4{\phi_4}^\dagger$$

$$\lambda_{57,c2}\, \phi_3{\phi_3}^\dagger\phi_4{\phi_4}^\dagger$$

$$\lambda_{57,c3}\, \phi_3{\phi_3}^\dagger\phi_4{\phi_4}^\dagger$$

$$\lambda_{57,c4}\, \phi_3{\phi_3}^\dagger\phi_4{\phi_4}^\dagger$$

$$\lambda_{61,c1}\, \phi_3{\phi_3}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{61,c2}\, \phi_3{\phi_3}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{61,c3}\, \phi_3{\phi_3}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{61,c4}\, \phi_3{\phi_3}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{65,c1}\, \phi_4\phi_4{\phi_4}^\dagger{\phi_4}^\dagger$$

$$\lambda_{65,c2}\, \phi_4\phi_4{\phi_4}^\dagger{\phi_4}^\dagger$$

$$\lambda_{67,c1}\, \phi_4{\phi_4}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{67,c2}\, \phi_4{\phi_4}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{67,c3}\, \phi_4{\phi_4}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{67,c4}\, \phi_4{\phi_4}^\dagger\phi_5{\phi_5}^\dagger$$

$$\lambda_{71,c1}\, \phi_5\phi_5{\phi_5}^\dagger{\phi_5}^\dagger$$

$$\lambda_{71,c2}\, \phi_5\phi_5{\phi_5}^\dagger{\phi_5}^\dagger$$

$$\lambda_{71,c3}\, \phi_5\phi_5{\phi_5}^\dagger{\phi_5}^\dagger$$

2.3 Gauge and Kinetic Terms

$$-\frac{1}{4}B_{\mu\nu}B^{\mu\nu} + -\frac{1}{4}W^a_{\mu\nu}W^{a,\mu\nu} + -\frac{1}{4}G^A_{\mu\nu}G^{A,\mu\nu}$$

$$(D_\mu q)^\dagger D^\mu q + (D_\mu l)^\dagger D^\mu l + (D_\mu d^C)^\dagger D^\mu d^C$$

$$(D_\mu u^C)^\dagger D^\mu u^C + (D_\mu e^C)^\dagger D^\mu e^C + (D_\mu H)^\dagger D^\mu H$$

$$(D_\mu \phi_1)^\dagger D^\mu \phi_1 + (D_\mu \phi_2)^\dagger D^\mu \phi_2 + (D_\mu \phi_3)^\dagger D^\mu \phi_3$$

$$(D_\mu \phi_4)^\dagger D^\mu \phi_4 + (D_\mu \phi_5)^\dagger D^\mu \phi_5$$

3 Fully Expanded Form of the Lagrangian

3.1 Yukawa and Fermion-Mass Terms

$$\begin{aligned}Y_1(&-q_b^{+2/3}q_g^{-1/3}{\phi_1}_r^{-1/3} + q_b^{+2/3}q_r^{-1/3}{\phi_1}_g^{-1/3} + q_g^{+2/3}q_b^{-1/3}{\phi_1}_r^{-1/3} - q_g^{+2/3}q_r^{-1/3}{\phi_1}_b^{-1/3}\\ &- q_r^{+2/3}q_b^{-1/3}{\phi_1}_g^{-1/3} + q_r^{+2/3}q_g^{-1/3}{\phi_1}_b^{-1/3} + q_b^{-1/3}q_g^{+2/3}{\phi_1}_r^{-1/3} - q_b^{-1/3}q_r^{+2/3}{\phi_1}_g^{-1/3}\\ &- q_g^{-1/3}q_b^{+2/3}{\phi_1}_r^{-1/3} + q_g^{-1/3}q_r^{+2/3}{\phi_1}_b^{-1/3} + q_r^{-1/3}q_b^{+2/3}{\phi_1}_g^{-1/3} - q_r^{-1/3}q_g^{+2/3}{\phi_1}_b^{-1/3}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}Y_2(&-q_b^{+2/3}q_g^{+2/3}{\phi_5}_r^{-4/3} + q_b^{+2/3}q_r^{+2/3}{\phi_5}_g^{-4/3} + \frac{\sqrt{2}}{2}q_b^{+2/3}q_g^{-1/3}{\phi_5}_r^{-1/3} - \frac{\sqrt{2}}{2}q_b^{+2/3}q_r^{-1/3}{\phi_5}_g^{-1/3}\\ &+ q_g^{+2/3}q_b^{+2/3}{\phi_5}_r^{-4/3} - q_g^{+2/3}q_r^{+2/3}{\phi_5}_b^{-4/3} - \frac{\sqrt{2}}{2}q_g^{+2/3}q_b^{-1/3}{\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{2}q_g^{+2/3}q_r^{-1/3}{\phi_5}_b^{-1/3}\\ &- q_r^{+2/3}q_b^{+2/3}{\phi_5}_g^{-4/3} + q_r^{+2/3}q_g^{+2/3}{\phi_5}_b^{-4/3} + \frac{\sqrt{2}}{2}q_r^{+2/3}q_b^{-1/3}{\phi_5}_g^{-1/3} - \frac{\sqrt{2}}{2}q_r^{+2/3}q_g^{-1/3}{\phi_5}_b^{-1/3}\\ &+ \frac{\sqrt{2}}{2}q_b^{-1/3}q_g^{+2/3}{\phi_5}_r^{-1/3} - \frac{\sqrt{2}}{2}q_b^{-1/3}q_r^{+2/3}{\phi_5}_g^{-1/3} - q_b^{-1/3}q_g^{-1/3}{\phi_5}_r^{+2/3} + q_b^{-1/3}q_r^{-1/3}{\phi_5}_g^{+2/3}\\ &- \frac{\sqrt{2}}{2}q_g^{-1/3}q_b^{+2/3}{\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{2}q_g^{-1/3}q_r^{+2/3}{\phi_5}_b^{-1/3} + q_g^{-1/3}q_b^{-1/3}{\phi_5}_r^{+2/3} - q_g^{-1/3}q_r^{-1/3}{\phi_5}_b^{+2/3}\\ &+ \frac{\sqrt{2}}{2}q_r^{-1/3}q_b^{+2/3}{\phi_5}_g^{-1/3} - \frac{\sqrt{2}}{2}q_r^{-1/3}q_g^{+2/3}{\phi_5}_b^{-1/3} - q_r^{-1/3}q_b^{-1/3}{\phi_5}_g^{+2/3} + q_r^{-1/3}q_g^{-1/3}{\phi_5}_b^{+2/3}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}Y_3(&-{\phi_1}^{-1/3,\bar{b}\dagger}l^-q_b^{+2/3} + {\phi_1}^{-1/3,\bar{b}\dagger}l^0q_b^{-1/3} - {\phi_1}^{-1/3,\bar{g}\dagger}l^-q_g^{+2/3}\\ &+ {\phi_1}^{-1/3,\bar{g}\dagger}l^0q_g^{-1/3} - {\phi_1}^{-1/3,\bar{r}\dagger}l^-q_r^{+2/3} + {\phi_1}^{-1/3,\bar{r}\dagger}l^0q_r^{-1/3}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}Y_4(&{\phi_5}^{+2/3,\bar{b}\dagger}l^0q_b^{+2/3} + {\phi_5}^{+2/3,\bar{g}\dagger}l^0q_g^{+2/3} + {\phi_5}^{+2/3,\bar{r}\dagger}l^0q_r^{+2/3} + \frac{\sqrt{2}}{2}{\phi_5}^{-1/3,\bar{b}\dagger}l^-q_b^{+2/3}\\ &+ \frac{\sqrt{2}}{2}{\phi_5}^{-1/3,\bar{b}\dagger}l^0q_b^{-1/3} + \frac{\sqrt{2}}{2}{\phi_5}^{-1/3,\bar{g}\dagger}l^-q_g^{+2/3} + \frac{\sqrt{2}}{2}{\phi_5}^{-1/3,\bar{g}\dagger}l^0q_g^{-1/3} + \frac{\sqrt{2}}{2}{\phi_5}^{-1/3,\bar{r}\dagger}l^-q_r^{+2/3}\\ &+ \frac{\sqrt{2}}{2}{\phi_5}^{-1/3,\bar{r}\dagger}l^0q_r^{-1/3} + {\phi_5}^{-4/3,\bar{b}\dagger}l^-q_b^{-1/3} + {\phi_5}^{-4/3,\bar{g}\dagger}l^-q_g^{-1/3} + {\phi_5}^{-4/3,\bar{r}\dagger}l^-q_r^{-1/3}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}Y_5(&H^{+\dagger}d^{C+1/3,\bar{b}}q_b^{+2/3} + H^{+\dagger}d^{C+1/3,\bar{g}}q_g^{+2/3} + H^{+\dagger}d^{C+1/3,\bar{r}}q_r^{+2/3}\\ &+ H^{0\dagger}d^{C+1/3,\bar{b}}q_b^{-1/3} + H^{0\dagger}d^{C+1/3,\bar{g}}q_g^{-1/3} + H^{0\dagger}d^{C+1/3,\bar{r}}q_r^{-1/3}) + \text{h.c.}\end{aligned}$$

$$Y_6(u^{C-2/3,\bar{b}}q_b^{+2/3}H^0 - u^{C-2/3,\bar{b}}q_b^{-1/3}H^+ + u^{C-2/3,\bar{g}}q_g^{+2/3}H^0 - u^{C-2/3,\bar{g}}q_g^{-1/3}H^+ + u^{C-2/3,\bar{r}}q_r^{+2/3}H^0 - u^{C-2/3,\bar{r}}q_r^{-1/3}H^+) + \text{h.c.}$$

$$Y_7({\phi_3}^{+2/3,\bar{b}^\dagger}e^{C+}q_b^{-1/3} + {\phi_3}^{+2/3,\bar{g}^\dagger}e^{C+}q_g^{-1/3} + {\phi_3}^{+2/3,\bar{r}^\dagger}e^{C+}q_r^{-1/3} + {\phi_3}^{+5/3,\bar{b}^\dagger}e^{C+}q_b^{+2/3} + {\phi_3}^{+5/3,\bar{g}^\dagger}e^{C+}q_g^{+2/3} + {\phi_3}^{+5/3,\bar{r}^\dagger}e^{C+}q_r^{+2/3}) + \text{h.c.}$$

$$Y_8(-d^{C+1/3,\bar{b}}l^-{\phi_4}_b^{+2/3} + d^{C+1/3,\bar{b}}l^0{\phi_4}_b^{-1/3} - d^{C+1/3,\bar{g}}l^-{\phi_4}_g^{+2/3} + d^{C+1/3,\bar{g}}l^0{\phi_4}_g^{-1/3} - d^{C+1/3,\bar{r}}l^-{\phi_4}_r^{+2/3} + d^{C+1/3,\bar{r}}l^0{\phi_4}_r^{-1/3}) + \text{h.c.}$$

$$Y_9(-u^{C-2/3,\bar{b}}l^-{\phi_3}_b^{+5/3} + u^{C-2/3,\bar{b}}l^0{\phi_3}_b^{+2/3} - u^{C-2/3,\bar{g}}l^-{\phi_3}_g^{+5/3} + u^{C-2/3,\bar{g}}l^0{\phi_3}_g^{+2/3} - u^{C-2/3,\bar{r}}l^-{\phi_3}_r^{+5/3} + u^{C-2/3,\bar{r}}l^0{\phi_3}_r^{+2/3}) + \text{h.c.}$$

$$Y_{10}\left(H^{+\dagger}e^{C+}l^0 + H^{0\dagger}e^{C+}l^-\right) + \text{h.c.}$$

$$Y_{11}(-{\phi_1}^{-1/3,\bar{b}^\dagger}d^{C+1/3,\bar{g}}u^{C-2/3,\bar{r}} + {\phi_1}^{-1/3,\bar{b}^\dagger}d^{C+1/3,\bar{r}}u^{C-2/3,\bar{g}} + {\phi_1}^{-1/3,\bar{g}^\dagger}d^{C+1/3,\bar{b}}u^{C-2/3,\bar{r}} - {\phi_1}^{-1/3,\bar{g}^\dagger}d^{C+1/3,\bar{r}}u^{C-2/3,\bar{b}} - {\phi_1}^{-1/3,\bar{r}^\dagger}d^{C+1/3,\bar{b}}u^{C-2/3,\bar{g}} + {\phi_1}^{-1/3,\bar{r}^\dagger}d^{C+1/3,\bar{g}}u^{C-2/3,\bar{b}}) + \text{h.c.}$$

$$Y_{12}(e^{C+}d^{C+1/3,\bar{b}}{\phi_2}_b^{-4/3} + e^{C+}d^{C+1/3,\bar{g}}{\phi_2}_g^{-4/3} + e^{C+}d^{C+1/3,\bar{r}}{\phi_2}_r^{-4/3}) + \text{h.c.}$$

$$Y_{13}(-{\phi_2}^{-4/3,\bar{b}^\dagger}u^{C-2/3,\bar{g}}u^{C-2/3,\bar{r}} + {\phi_2}^{-4/3,\bar{b}^\dagger}u^{C-2/3,\bar{r}}u^{C-2/3,\bar{g}} + {\phi_2}^{-4/3,\bar{g}^\dagger}u^{C-2/3,\bar{b}}u^{C-2/3,\bar{r}} - {\phi_2}^{-4/3,\bar{g}^\dagger}u^{C-2/3,\bar{r}}u^{C-2/3,\bar{b}} - {\phi_2}^{-4/3,\bar{r}^\dagger}u^{C-2/3,\bar{b}}u^{C-2/3,\bar{g}} + {\phi_2}^{-4/3,\bar{r}^\dagger}u^{C-2/3,\bar{g}}u^{C-2/3,\bar{b}}) + \text{h.c.}$$

$$Y_{14}(e^{C+}u^{C-2/3,\bar{b}}{\phi_1}_b^{-1/3} + e^{C+}u^{C-2/3,\bar{g}}{\phi_1}_g^{-1/3} + e^{C+}u^{C-2/3,\bar{r}}{\phi_1}_r^{-1/3}) + \text{h.c.}$$

3.2 Scalar Potential

$$m_1^2\left(H^0H^{0\dagger} + H^+H^{+\dagger}\right)$$

$$m_2^2({\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger})$$

$$m_3^2({\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger})$$

$$m_4^2({\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger} + {\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger})$$

$$m_5^2({\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}^\dagger} + {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}^\dagger} + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}^\dagger} + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}^\dagger} + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}^\dagger} + {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}^\dagger})$$

$$m_6^2({\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}^\dagger} + {\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}^\dagger} + {\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}^\dagger} + {\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}^\dagger} + {\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}^\dagger} + {\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}^\dagger} + {\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}^\dagger} + {\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}^\dagger} + {\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}^\dagger})$$

$$\begin{aligned}&\mu_7(H^0{\phi_1}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + H^0{\phi_1}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger} + H^0{\phi_1}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}\\ &\quad + H^+{\phi_1}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger} + H^+{\phi_1}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger} + H^+{\phi_1}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}&\mu_8(-\sqrt{2}\,H^0{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3} - \sqrt{2}\,H^0{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3} - \sqrt{2}\,H^0{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}\\ &- H^0{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3} - H^0{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3} - H^0{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}\\ &+ H^+{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3} + H^+{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3} + H^+{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}\\ &+ \sqrt{2}\,H^+{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3} + \sqrt{2}\,H^+{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3} + \sqrt{2}\,H^+{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}&\mu_9(-2\,{\phi_1}_r^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}_b^{+2/3} + 2\,{\phi_1}_r^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}_g^{+2/3} + 2\,{\phi_1}_g^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}_b^{+2/3}\\ &\quad - 2\,{\phi_1}_g^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}_r^{+2/3} - 2\,{\phi_1}_b^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}_g^{+2/3} + 2\,{\phi_1}_b^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}_r^{+2/3}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}&\mu_{10}(-{\phi_2}_r^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}_b^{+2/3} + {\phi_2}_r^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}_g^{+2/3} + {\phi_2}_r^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}_b^{-1/3} - {\phi_2}_r^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}_g^{-1/3}\\ &+ {\phi_2}_g^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}_b^{+2/3} - {\phi_2}_g^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}_r^{+2/3} - {\phi_2}_g^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}_b^{-1/3} + {\phi_2}_g^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}_r^{-1/3}\\ &- {\phi_2}_b^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}_g^{+2/3} + {\phi_2}_b^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}_r^{+2/3} + {\phi_2}_b^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}_g^{-1/3} - {\phi_2}_b^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}_r^{-1/3}) + \text{h.c.}\end{aligned}$$

$$\lambda_{11}(H^0H^0{H^0}^\dagger{H^0}^\dagger + 2\,H^0H^+{H^+}^\dagger{H^0}^\dagger + H^+H^+{H^+}^\dagger{H^+}^\dagger)$$

$$\begin{aligned}&\lambda_{12}(H^0H^0{\phi_2}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + H^0H^0{\phi_2}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} + H^0H^0{\phi_2}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger}\\ &+ \sqrt{2}\,H^0H^+{\phi_2}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\,H^0H^+{\phi_2}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\,H^0H^+{\phi_2}_b^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger}\\ &+ H^+H^+{\phi_2}_r^{-4/3}{\phi_5}^{+2/3,\bar{r}\dagger} + H^+H^+{\phi_2}_g^{-4/3}{\phi_5}^{+2/3,\bar{g}\dagger} + H^+H^+{\phi_2}_b^{-4/3}{\phi_5}^{+2/3,\bar{b}\dagger}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}&\lambda_{13}(-H^0H^0{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{+2/3} - H^0H^0{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_g^{+2/3} - H^0H^0{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}\\ &- H^0H^+{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{+2/3} - H^0H^+{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{+2/3} - H^0H^+{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}\\ &+ H^0H^+{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{-1/3} + H^0H^+{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_g^{-1/3} + H^0H^+{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}\\ &+ H^+H^+{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{-1/3} + H^+H^+{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{-1/3} + H^+H^+{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}&\lambda_{14}(H^0{H^0}^\dagger{\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger} + H^0{H^0}^\dagger{\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger} + H^0{H^0}^\dagger{\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}\\ &\quad + H^+{H^+}^\dagger{\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger} + H^+{H^+}^\dagger{\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger} + H^+{H^+}^\dagger{\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger})\end{aligned}$$

$$\begin{aligned}&\lambda_{15}(\sqrt{2}\,H^0{H^+}^\dagger{\phi_1}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,H^0{H^+}^\dagger{\phi_1}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} + \sqrt{2}\,H^0{H^+}^\dagger{\phi_1}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger}\\ &- H^0{H^0}^\dagger{\phi_1}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} - H^0{H^0}^\dagger{\phi_1}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} - H^0{H^0}^\dagger{\phi_1}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger}\\ &+ H^+{H^+}^\dagger{\phi_1}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} + H^+{H^+}^\dagger{\phi_1}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} + H^+{H^+}^\dagger{\phi_1}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger}\\ &- \sqrt{2}\,H^+{H^0}^\dagger{\phi_1}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} - \sqrt{2}\,H^+{H^0}^\dagger{\phi_1}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger} - \sqrt{2}\,H^+{H^0}^\dagger{\phi_1}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}) + \text{h.c.}\end{aligned}$$

$$\begin{aligned}&\lambda_{16}(H^0{H^0}^\dagger{\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger} + H^0{H^0}^\dagger{\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger} + H^0{H^0}^\dagger{\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}\\ &\quad + H^+{H^+}^\dagger{\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger} + H^+{H^+}^\dagger{\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger} + H^+{H^+}^\dagger{\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger})\end{aligned}$$

$$
\begin{aligned}
\lambda^{(1)}_{17,c1}(&H^0 H^{+\dagger} {\phi_3}_r^{+5/3} {\phi_3}^{+2/3,\bar{r}\dagger} + H^0 H^{+\dagger} {\phi_3}_g^{+5/3} {\phi_3}^{+2/3,\bar{g}\dagger} + H^0 H^{+\dagger} {\phi_3}_b^{+5/3} {\phi_3}^{+2/3,\bar{b}\dagger} \\
&+ H^0 H^{0\dagger} {\phi_3}_r^{+2/3} {\phi_3}^{+2/3,\bar{r}\dagger} + H^0 H^{0\dagger} {\phi_3}_g^{+2/3} {\phi_3}^{+2/3,\bar{g}\dagger} + H^0 H^{0\dagger} {\phi_3}_b^{+2/3} {\phi_3}^{+2/3,\bar{b}\dagger} \\
&+ H^+ H^{+\dagger} {\phi_3}_r^{+5/3} {\phi_3}^{+5/3,\bar{r}\dagger} + H^+ H^{+\dagger} {\phi_3}_g^{+5/3} {\phi_3}^{+5/3,\bar{g}\dagger} + H^+ H^{+\dagger} {\phi_3}_b^{+5/3} {\phi_3}^{+5/3,\bar{b}\dagger} \\
&+ H^+ H^{0\dagger} {\phi_3}_r^{+2/3} {\phi_3}^{+5/3,\bar{r}\dagger} + H^+ H^{0\dagger} {\phi_3}_g^{+2/3} {\phi_3}^{+5/3,\bar{g}\dagger} + H^+ H^{0\dagger} {\phi_3}_b^{+2/3} {\phi_3}^{+5/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(2)}_{17,c2}(&-H^0 H^{+\dagger} {\phi_3}_r^{+5/3} {\phi_3}^{+2/3,\bar{r}\dagger} - H^0 H^{+\dagger} {\phi_3}_g^{+5/3} {\phi_3}^{+2/3,\bar{g}\dagger} - H^0 H^{+\dagger} {\phi_3}_b^{+5/3} {\phi_3}^{+2/3,\bar{b}\dagger} \\
&+ H^0 H^{0\dagger} {\phi_3}_r^{+5/3} {\phi_3}^{+5/3,\bar{r}\dagger} + H^0 H^{0\dagger} {\phi_3}_g^{+5/3} {\phi_3}^{+5/3,\bar{g}\dagger} + H^0 H^{0\dagger} {\phi_3}_b^{+5/3} {\phi_3}^{+5/3,\bar{b}\dagger} \\
&+ H^+ H^{+\dagger} {\phi_3}_r^{+2/3} {\phi_3}^{+2/3,\bar{r}\dagger} + H^+ H^{+\dagger} {\phi_3}_g^{+2/3} {\phi_3}^{+2/3,\bar{g}\dagger} + H^+ H^{+\dagger} {\phi_3}_b^{+2/3} {\phi_3}^{+2/3,\bar{b}\dagger} \\
&- H^+ H^{0\dagger} {\phi_3}_r^{+2/3} {\phi_3}^{+5/3,\bar{r}\dagger} - H^+ H^{0\dagger} {\phi_3}_g^{+2/3} {\phi_3}^{+5/3,\bar{g}\dagger} - H^+ H^{0\dagger} {\phi_3}_b^{+2/3} {\phi_3}^{+5/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(1)}_{19,c1}(&H^0 H^{+\dagger} {\phi_4}_r^{+2/3} {\phi_4}^{-1/3,\bar{r}\dagger} + H^0 H^{+\dagger} {\phi_4}_g^{+2/3} {\phi_4}^{-1/3,\bar{g}\dagger} + H^0 H^{+\dagger} {\phi_4}_b^{+2/3} {\phi_4}^{-1/3,\bar{b}\dagger} \\
&+ H^0 H^{0\dagger} {\phi_4}_r^{-1/3} {\phi_4}^{-1/3,\bar{r}\dagger} + H^0 H^{0\dagger} {\phi_4}_g^{-1/3} {\phi_4}^{-1/3,\bar{g}\dagger} + H^0 H^{0\dagger} {\phi_4}_b^{-1/3} {\phi_4}^{-1/3,\bar{b}\dagger} \\
&+ H^+ H^{+\dagger} {\phi_4}_r^{+2/3} {\phi_4}^{+2/3,\bar{r}\dagger} + H^+ H^{+\dagger} {\phi_4}_g^{+2/3} {\phi_4}^{+2/3,\bar{g}\dagger} + H^+ H^{+\dagger} {\phi_4}_b^{+2/3} {\phi_4}^{+2/3,\bar{b}\dagger} \\
&+ H^+ H^{0\dagger} {\phi_4}_r^{-1/3} {\phi_4}^{+2/3,\bar{r}\dagger} + H^+ H^{0\dagger} {\phi_4}_g^{-1/3} {\phi_4}^{+2/3,\bar{g}\dagger} + H^+ H^{0\dagger} {\phi_4}_b^{-1/3} {\phi_4}^{+2/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(2)}_{19,c2}(&-H^0 H^{+\dagger} {\phi_4}_r^{+2/3} {\phi_4}^{-1/3,\bar{r}\dagger} - H^0 H^{+\dagger} {\phi_4}_g^{+2/3} {\phi_4}^{-1/3,\bar{g}\dagger} - H^0 H^{+\dagger} {\phi_4}_b^{+2/3} {\phi_4}^{-1/3,\bar{b}\dagger} \\
&+ H^0 H^{0\dagger} {\phi_4}_r^{+2/3} {\phi_4}^{+2/3,\bar{r}\dagger} + H^0 H^{0\dagger} {\phi_4}_g^{+2/3} {\phi_4}^{+2/3,\bar{g}\dagger} + H^0 H^{0\dagger} {\phi_4}_b^{+2/3} {\phi_4}^{+2/3,\bar{b}\dagger} \\
&+ H^+ H^{+\dagger} {\phi_4}_r^{-1/3} {\phi_4}^{-1/3,\bar{r}\dagger} + H^+ H^{+\dagger} {\phi_4}_g^{-1/3} {\phi_4}^{-1/3,\bar{g}\dagger} + H^+ H^{+\dagger} {\phi_4}_b^{-1/3} {\phi_4}^{-1/3,\bar{b}\dagger} \\
&- H^+ H^{0\dagger} {\phi_4}_r^{-1/3} {\phi_4}^{+2/3,\bar{r}\dagger} - H^+ H^{0\dagger} {\phi_4}_g^{-1/3} {\phi_4}^{+2/3,\bar{g}\dagger} - H^+ H^{0\dagger} {\phi_4}_b^{-1/3} {\phi_4}^{+2/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(1)}_{21,c1}(&\sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_r^{-1/3} {\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_g^{-1/3} {\phi_5}^{-4/3,\bar{g}\dagger} + \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_b^{-1/3} {\phi_5}^{-4/3,\bar{b}\dagger} \\
&+ \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_r^{+2/3} {\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_g^{+2/3} {\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_b^{+2/3} {\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ H^0 H^{0\dagger} {\phi_5}_r^{-4/3} {\phi_5}^{-4/3,\bar{r}\dagger} + H^0 H^{0\dagger} {\phi_5}_g^{-4/3} {\phi_5}^{-4/3,\bar{g}\dagger} + H^0 H^{0\dagger} {\phi_5}_b^{-4/3} {\phi_5}^{-4/3,\bar{b}\dagger} \\
&- H^0 H^{0\dagger} {\phi_5}_r^{+2/3} {\phi_5}^{+2/3,\bar{r}\dagger} - H^0 H^{0\dagger} {\phi_5}_g^{+2/3} {\phi_5}^{+2/3,\bar{g}\dagger} - H^0 H^{0\dagger} {\phi_5}_b^{+2/3} {\phi_5}^{+2/3,\bar{b}\dagger} \\
&- H^+ H^{+\dagger} {\phi_5}_r^{-4/3} {\phi_5}^{-4/3,\bar{r}\dagger} - H^+ H^{+\dagger} {\phi_5}_g^{-4/3} {\phi_5}^{-4/3,\bar{g}\dagger} - H^+ H^{+\dagger} {\phi_5}_b^{-4/3} {\phi_5}^{-4/3,\bar{b}\dagger} \\
&+ H^+ H^{+\dagger} {\phi_5}_r^{+2/3} {\phi_5}^{+2/3,\bar{r}\dagger} + H^+ H^{+\dagger} {\phi_5}_g^{+2/3} {\phi_5}^{+2/3,\bar{g}\dagger} + H^+ H^{+\dagger} {\phi_5}_b^{+2/3} {\phi_5}^{+2/3,\bar{b}\dagger} \\
&+ \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_r^{-4/3} {\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_g^{-4/3} {\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_b^{-4/3} {\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_r^{-1/3} {\phi_5}^{+2/3,\bar{r}\dagger} + \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_g^{-1/3} {\phi_5}^{+2/3,\bar{g}\dagger} + \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_b^{-1/3} {\phi_5}^{+2/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(2)}_{21,c2}(&-\sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_r^{-1/3} {\phi_5}^{-4/3,\bar{r}\dagger} - \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_g^{-1/3} {\phi_5}^{-4/3,\bar{g}\dagger} \\
&- \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_b^{-1/3} {\phi_5}^{-4/3,\bar{b}\dagger} - \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_r^{+2/3} {\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_g^{+2/3} {\phi_5}^{-1/3,\bar{g}\dagger} \\
&- \sqrt{2}\, H^0 H^{+\dagger} {\phi_5}_b^{+2/3} {\phi_5}^{-1/3,\bar{b}\dagger} + H^0 H^{0\dagger} {\phi_5}_r^{-1/3} {\phi_5}^{-1/3,\bar{r}\dagger} + H^0 H^{0\dagger} {\phi_5}_g^{-1/3} {\phi_5}^{-1/3,\bar{g}\dagger} \\
&+ H^0 H^{0\dagger} {\phi_5}_b^{-1/3} {\phi_5}^{-1/3,\bar{b}\dagger} + 2\, H^0 H^{0\dagger} {\phi_5}_r^{+2/3} {\phi_5}^{+2/3,\bar{r}\dagger} + 2\, H^0 H^{0\dagger} {\phi_5}_g^{+2/3} {\phi_5}^{+2/3,\bar{g}\dagger} \\
&+ 2\, H^0 H^{0\dagger} {\phi_5}_b^{+2/3} {\phi_5}^{+2/3,\bar{b}\dagger} + 2\, H^+ H^{+\dagger} {\phi_5}_r^{-4/3} {\phi_5}^{-4/3,\bar{r}\dagger} + 2\, H^+ H^{+\dagger} {\phi_5}_g^{-4/3} {\phi_5}^{-4/3,\bar{g}\dagger} \\
&+ 2\, H^+ H^{+\dagger} {\phi_5}_b^{-4/3} {\phi_5}^{-4/3,\bar{b}\dagger} + H^+ H^{+\dagger} {\phi_5}_r^{-1/3} {\phi_5}^{-1/3,\bar{r}\dagger} + H^+ H^{+\dagger} {\phi_5}_g^{-1/3} {\phi_5}^{-1/3,\bar{g}\dagger} \\
&+ H^+ H^{+\dagger} {\phi_5}_b^{-1/3} {\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_r^{-4/3} {\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_g^{-4/3} {\phi_5}^{-1/3,\bar{g}\dagger} \\
&- \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_b^{-4/3} {\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_r^{-1/3} {\phi_5}^{+2/3,\bar{r}\dagger} \\
&- \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_g^{-1/3} {\phi_5}^{+2/3,\bar{g}\dagger} - \sqrt{2}\, H^+ H^{0\dagger} {\phi_5}_b^{-1/3} {\phi_5}^{+2/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
&\lambda_{23}\big(-H^0 {\phi_1}_r^{-1/3} {\phi_2}_g^{-4/3} {\phi_3}_b^{+5/3} + H^0 {\phi_1}_r^{-1/3} {\phi_2}_b^{-4/3} {\phi_3}_g^{+5/3} + H^0 {\phi_1}_g^{-1/3} {\phi_2}_r^{-4/3} {\phi_3}_b^{+5/3} \\
&\quad - H^0 {\phi_1}_g^{-1/3} {\phi_2}_b^{-4/3} {\phi_3}_r^{+5/3} - H^0 {\phi_1}_b^{-1/3} {\phi_2}_r^{-4/3} {\phi_3}_g^{+5/3} + H^0 {\phi_1}_b^{-1/3} {\phi_2}_g^{-4/3} {\phi_3}_r^{+5/3} \\
&\quad + H^+ {\phi_1}_r^{-1/3} {\phi_2}_g^{-4/3} {\phi_3}_b^{+2/3} - H^+ {\phi_1}_r^{-1/3} {\phi_2}_b^{-4/3} {\phi_3}_g^{+2/3} - H^+ {\phi_1}_g^{-1/3} {\phi_2}_r^{-4/3} {\phi_3}_b^{+2/3} \\
&\quad + H^+ {\phi_1}_g^{-1/3} {\phi_2}_b^{-4/3} {\phi_3}_r^{+2/3} + H^+ {\phi_1}_b^{-1/3} {\phi_2}_r^{-4/3} {\phi_3}_g^{+2/3} - H^+ {\phi_1}_b^{-1/3} {\phi_2}_g^{-4/3} {\phi_3}_r^{+2/3}\big) + \text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
&\lambda_{24}\big(H^0 {\phi_1}_r^{-1/3} {\phi_4}_g^{-1/3} {\phi_5}_b^{+2/3} - H^0 {\phi_1}_r^{-1/3} {\phi_4}_b^{-1/3} {\phi_5}_g^{+2/3} - \frac{\sqrt{2}}{2}\, H^0 {\phi_1}_r^{-1/3} {\phi_4}_g^{+2/3} {\phi_5}_b^{-1/3} \\
&+ \frac{\sqrt{2}}{2}\, H^0 {\phi_1}_r^{-1/3} {\phi_4}_b^{+2/3} {\phi_5}_g^{-1/3} - H^0 {\phi_1}_g^{-1/3} {\phi_4}_r^{-1/3} {\phi_5}_b^{+2/3} + H^0 {\phi_1}_g^{-1/3} {\phi_4}_b^{-1/3} {\phi_5}_r^{+2/3} \\
&+ \frac{\sqrt{2}}{2}\, H^0 {\phi_1}_g^{-1/3} {\phi_4}_r^{+2/3} {\phi_5}_b^{-1/3} - \frac{\sqrt{2}}{2}\, H^0 {\phi_1}_g^{-1/3} {\phi_4}_b^{+2/3} {\phi_5}_r^{-1/3} + H^0 {\phi_1}_b^{-1/3} {\phi_4}_r^{-1/3} {\phi_5}_g^{+2/3} \\
&- H^0 {\phi_1}_b^{-1/3} {\phi_4}_g^{-1/3} {\phi_5}_r^{+2/3} - \frac{\sqrt{2}}{2}\, H^0 {\phi_1}_b^{-1/3} {\phi_4}_r^{+2/3} {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{2}\, H^0 {\phi_1}_b^{-1/3} {\phi_4}_g^{+2/3} {\phi_5}_r^{-1/3} \\
&- \frac{\sqrt{2}}{2}\, H^+ {\phi_1}_r^{-1/3} {\phi_4}_g^{-1/3} {\phi_5}_b^{-1/3} + \frac{\sqrt{2}}{2}\, H^+ {\phi_1}_r^{-1/3} {\phi_4}_b^{-1/3} {\phi_5}_g^{-1/3} + H^+ {\phi_1}_r^{-1/3} {\phi_4}_g^{+2/3} {\phi_5}_b^{-4/3} \\
&- H^+ {\phi_1}_r^{-1/3} {\phi_4}_b^{+2/3} {\phi_5}_g^{-4/3} + \frac{\sqrt{2}}{2}\, H^+ {\phi_1}_g^{-1/3} {\phi_4}_r^{-1/3} {\phi_5}_b^{-1/3} - \frac{\sqrt{2}}{2}\, H^+ {\phi_1}_g^{-1/3} {\phi_4}_b^{-1/3} {\phi_5}_r^{-1/3} \\
&- H^+ {\phi_1}_g^{-1/3} {\phi_4}_r^{+2/3} {\phi_5}_b^{-4/3} + H^+ {\phi_1}_g^{-1/3} {\phi_4}_b^{+2/3} {\phi_5}_r^{-4/3} - \frac{\sqrt{2}}{2}\, H^+ {\phi_1}_b^{-1/3} {\phi_4}_r^{-1/3} {\phi_5}_g^{-1/3} \\
&+ \frac{\sqrt{2}}{2}\, H^+ {\phi_1}_b^{-1/3} {\phi_4}_g^{-1/3} {\phi_5}_r^{-1/3} + H^+ {\phi_1}_b^{-1/3} {\phi_4}_r^{+2/3} {\phi_5}_g^{-4/3} - H^+ {\phi_1}_b^{-1/3} {\phi_4}_g^{+2/3} {\phi_5}_r^{-4/3}\big) + \text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
&\lambda_{25}\big(\sqrt{2}\, H^0 \phi_1^{-1/3,\bar{r}\dagger} \phi_3^{+5/3,\bar{g}\dagger} \phi_5^{-4/3,\bar{b}\dagger} - \sqrt{2}\, H^0 \phi_1^{-1/3,\bar{r}\dagger} \phi_3^{+5/3,\bar{b}\dagger} \phi_5^{-4/3,\bar{g}\dagger} \\
&\quad - H^0 \phi_1^{-1/3,\bar{r}\dagger} \phi_3^{+2/3,\bar{g}\dagger} \phi_5^{-1/3,\bar{b}\dagger} + H^0 \phi_1^{-1/3,\bar{r}\dagger} \phi_3^{+2/3,\bar{b}\dagger} \phi_5^{-1/3,\bar{g}\dagger} \\
&\quad - \sqrt{2}\, H^0 \phi_1^{-1/3,\bar{g}\dagger} \phi_3^{+5/3,\bar{r}\dagger} \phi_5^{-4/3,\bar{b}\dagger} + \sqrt{2}\, H^0 \phi_1^{-1/3,\bar{g}\dagger} \phi_3^{+5/3,\bar{b}\dagger} \phi_5^{-4/3,\bar{r}\dagger} \\
&\quad + H^0 \phi_1^{-1/3,\bar{g}\dagger} \phi_3^{+2/3,\bar{r}\dagger} \phi_5^{-1/3,\bar{b}\dagger} - H^0 \phi_1^{-1/3,\bar{g}\dagger} \phi_3^{+2/3,\bar{b}\dagger} \phi_5^{-1/3,\bar{r}\dagger} \\
&\quad + \sqrt{2}\, H^0 \phi_1^{-1/3,\bar{b}\dagger} \phi_3^{+5/3,\bar{r}\dagger} \phi_5^{-4/3,\bar{g}\dagger} - \sqrt{2}\, H^0 \phi_1^{-1/3,\bar{b}\dagger} \phi_3^{+5/3,\bar{g}\dagger} \phi_5^{-4/3,\bar{r}\dagger} \\
&\quad - H^0 \phi_1^{-1/3,\bar{b}\dagger} \phi_3^{+2/3,\bar{r}\dagger} \phi_5^{-1/3,\bar{g}\dagger} + H^0 \phi_1^{-1/3,\bar{b}\dagger} \phi_3^{+2/3,\bar{g}\dagger} \phi_5^{-1/3,\bar{r}\dagger} \\
&\quad + H^+ \phi_1^{-1/3,\bar{r}\dagger} \phi_3^{+5/3,\bar{g}\dagger} \phi_5^{-1/3,\bar{b}\dagger} - H^+ \phi_1^{-1/3,\bar{r}\dagger} \phi_3^{+5/3,\bar{b}\dagger} \phi_5^{-1/3,\bar{g}\dagger} \\
&\quad - \sqrt{2}\, H^+ \phi_1^{-1/3,\bar{r}\dagger} \phi_3^{+2/3,\bar{g}\dagger} \phi_5^{+2/3,\bar{b}\dagger} + \sqrt{2}\, H^+ \phi_1^{-1/3,\bar{r}\dagger} \phi_3^{+2/3,\bar{b}\dagger} \phi_5^{+2/3,\bar{g}\dagger} \\
&\quad - H^+ \phi_1^{-1/3,\bar{g}\dagger} \phi_3^{+5/3,\bar{r}\dagger} \phi_5^{-1/3,\bar{b}\dagger} + H^+ \phi_1^{-1/3,\bar{g}\dagger} \phi_3^{+5/3,\bar{b}\dagger} \phi_5^{-1/3,\bar{r}\dagger} \\
&\quad + \sqrt{2}\, H^+ \phi_1^{-1/3,\bar{g}\dagger} \phi_3^{+2/3,\bar{r}\dagger} \phi_5^{+2/3,\bar{b}\dagger} - \sqrt{2}\, H^+ \phi_1^{-1/3,\bar{g}\dagger} \phi_3^{+2/3,\bar{b}\dagger} \phi_5^{+2/3,\bar{r}\dagger} \\
&\quad + H^+ \phi_1^{-1/3,\bar{b}\dagger} \phi_3^{+5/3,\bar{r}\dagger} \phi_5^{-1/3,\bar{g}\dagger} - H^+ \phi_1^{-1/3,\bar{b}\dagger} \phi_3^{+5/3,\bar{g}\dagger} \phi_5^{-1/3,\bar{r}\dagger} \\
&\quad - \sqrt{2}\, H^+ \phi_1^{-1/3,\bar{b}\dagger} \phi_3^{+2/3,\bar{r}\dagger} \phi_5^{+2/3,\bar{g}\dagger} + \sqrt{2}\, H^+ \phi_1^{-1/3,\bar{b}\dagger} \phi_3^{+2/3,\bar{g}\dagger} \phi_5^{+2/3,\bar{r}\dagger}\big) + \text{h.c.}
\end{aligned}
$$

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\begin{aligned}
&\lambda_{26}(H^0\phi_{2_r}^{-4/3}\phi_{3_g}^{+2/3}\phi_{5_b}^{+2/3} - H^0\phi_{2_r}^{-4/3}\phi_{3_b}^{+2/3}\phi_{5_g}^{+2/3} - \frac{\sqrt{2}}{2}\,H^0\phi_{2_r}^{-4/3}\phi_{3_g}^{+5/3}\phi_{5_b}^{-1/3}\\
&+\frac{\sqrt{2}}{2}\,H^0\phi_{2_r}^{-4/3}\phi_{3_b}^{+5/3}\phi_{5_g}^{-1/3} - H^0\phi_{2_g}^{-4/3}\phi_{3_r}^{+2/3}\phi_{5_b}^{+2/3} + H^0\phi_{2_g}^{-4/3}\phi_{3_b}^{+2/3}\phi_{5_r}^{+2/3}\\
&+\frac{\sqrt{2}}{2}\,H^0\phi_{2_g}^{-4/3}\phi_{3_r}^{+5/3}\phi_{5_b}^{-1/3} - \frac{\sqrt{2}}{2}\,H^0\phi_{2_g}^{-4/3}\phi_{3_b}^{+5/3}\phi_{5_r}^{-1/3} + H^0\phi_{2_b}^{-4/3}\phi_{3_r}^{+2/3}\phi_{5_g}^{+2/3}\\
&- H^0\phi_{2_b}^{-4/3}\phi_{3_g}^{+2/3}\phi_{5_r}^{+2/3} - \frac{\sqrt{2}}{2}\,H^0\phi_{2_b}^{-4/3}\phi_{3_r}^{+5/3}\phi_{5_g}^{-1/3} + \frac{\sqrt{2}}{2}\,H^0\phi_{2_b}^{-4/3}\phi_{3_g}^{+5/3}\phi_{5_r}^{-1/3}\\
&- \frac{\sqrt{2}}{2}\,H^+\phi_{2_r}^{-4/3}\phi_{3_g}^{+2/3}\phi_{5_b}^{-1/3} + \frac{\sqrt{2}}{2}\,H^+\phi_{2_r}^{-4/3}\phi_{3_b}^{+2/3}\phi_{5_g}^{-1/3} + H^+\phi_{2_r}^{-4/3}\phi_{3_g}^{+5/3}\phi_{5_b}^{-4/3}\\
&- H^+\phi_{2_r}^{-4/3}\phi_{3_b}^{+5/3}\phi_{5_g}^{-4/3} + \frac{\sqrt{2}}{2}\,H^+\phi_{2_g}^{-4/3}\phi_{3_r}^{+2/3}\phi_{5_b}^{-1/3} - \frac{\sqrt{2}}{2}\,H^+\phi_{2_g}^{-4/3}\phi_{3_b}^{+2/3}\phi_{5_r}^{-1/3}\\
&- H^+\phi_{2_g}^{-4/3}\phi_{3_r}^{+5/3}\phi_{5_b}^{-4/3} + H^+\phi_{2_g}^{-4/3}\phi_{3_b}^{+5/3}\phi_{5_r}^{-4/3} - \frac{\sqrt{2}}{2}\,H^+\phi_{2_b}^{-4/3}\phi_{3_r}^{+2/3}\phi_{5_g}^{-1/3}\\
&+ \frac{\sqrt{2}}{2}\,H^+\phi_{2_b}^{-4/3}\phi_{3_g}^{+2/3}\phi_{5_r}^{-1/3} + H^+\phi_{2_b}^{-4/3}\phi_{3_r}^{+5/3}\phi_{5_g}^{-4/3} - H^+\phi_{2_b}^{-4/3}\phi_{3_g}^{+5/3}\phi_{5_r}^{-4/3}) + \text{h.c.}
\end{aligned}
$$

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\begin{aligned}
&\lambda_{27}(2\sqrt{2}\,H^0{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - 2\sqrt{2}\,H^0{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger}\\
&- 2\sqrt{2}\,H^0{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} + 2\sqrt{2}\,H^0{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger}\\
&+ 2\sqrt{2}\,H^0{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - 2\sqrt{2}\,H^0{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger}\\
&- 2\,H^0{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} + 2\,H^0{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger}\\
&+ 2\,H^0{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - 2\,H^0{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger}\\
&- 2\,H^0{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} + 2\,H^0{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger}\\
&+ 2\,H^+{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - 2\,H^+{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger}\\
&- 2\,H^+{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} + 2\,H^+{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger}\\
&+ 2\,H^+{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - 2\,H^+{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger}\\
&- 2\sqrt{2}\,H^+{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} + 2\sqrt{2}\,H^+{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger}\\
&+ 2\sqrt{2}\,H^+{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - 2\sqrt{2}\,H^+{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger}\\
&- 2\sqrt{2}\,H^+{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} + 2\sqrt{2}\,H^+{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger}) + \text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
&\lambda_{28}(2\sqrt{2}\,H^0\phi_{4_r}^{-1/3}\phi_{5_g}^{-1/3}\phi_{5_b}^{+2/3} - 2\sqrt{2}\,H^0\phi_{4_r}^{-1/3}\phi_{5_b}^{-1/3}\phi_{5_g}^{+2/3}\\
&- 2\sqrt{2}\,H^0\phi_{4_g}^{-1/3}\phi_{5_r}^{-1/3}\phi_{5_b}^{+2/3} + 2\sqrt{2}\,H^0\phi_{4_g}^{-1/3}\phi_{5_b}^{-1/3}\phi_{5_r}^{+2/3} + 2\sqrt{2}\,H^0\phi_{4_b}^{-1/3}\phi_{5_r}^{-1/3}\phi_{5_g}^{+2/3}\\
&- 2\sqrt{2}\,H^0\phi_{4_b}^{-1/3}\phi_{5_g}^{-1/3}\phi_{5_r}^{+2/3} - 2\,H^0\phi_{4_r}^{+2/3}\phi_{5_g}^{-4/3}\phi_{5_b}^{+2/3} + 2\,H^0\phi_{4_r}^{+2/3}\phi_{5_b}^{-4/3}\phi_{5_g}^{+2/3}\\
&+ 2\,H^0\phi_{4_g}^{+2/3}\phi_{5_r}^{-4/3}\phi_{5_b}^{+2/3} - 2\,H^0\phi_{4_g}^{+2/3}\phi_{5_b}^{-4/3}\phi_{5_r}^{+2/3} - 2\,H^0\phi_{4_b}^{+2/3}\phi_{5_r}^{-4/3}\phi_{5_g}^{+2/3}\\
&+ 2\,H^0\phi_{4_b}^{+2/3}\phi_{5_g}^{-4/3}\phi_{5_r}^{+2/3} - 2\,H^+\phi_{4_r}^{-1/3}\phi_{5_g}^{-4/3}\phi_{5_b}^{+2/3} + 2\,H^+\phi_{4_r}^{-1/3}\phi_{5_b}^{-4/3}\phi_{5_g}^{+2/3}\\
&+ 2\,H^+\phi_{4_g}^{-1/3}\phi_{5_r}^{-4/3}\phi_{5_b}^{+2/3} - 2\,H^+\phi_{4_g}^{-1/3}\phi_{5_b}^{-4/3}\phi_{5_r}^{+2/3} - 2\,H^+\phi_{4_b}^{-1/3}\phi_{5_r}^{-4/3}\phi_{5_g}^{+2/3}\\
&+ 2\,H^+\phi_{4_b}^{-1/3}\phi_{5_g}^{-4/3}\phi_{5_r}^{+2/3} + 2\sqrt{2}\,H^+\phi_{4_r}^{+2/3}\phi_{5_g}^{-4/3}\phi_{5_b}^{-1/3} - 2\sqrt{2}\,H^+\phi_{4_r}^{+2/3}\phi_{5_b}^{-4/3}\phi_{5_g}^{-1/3}\\
&- 2\sqrt{2}\,H^+\phi_{4_g}^{+2/3}\phi_{5_r}^{-4/3}\phi_{5_b}^{-1/3} + 2\sqrt{2}\,H^+\phi_{4_g}^{+2/3}\phi_{5_b}^{-4/3}\phi_{5_r}^{-1/3}\\
&+ 2\sqrt{2}\,H^+\phi_{4_b}^{+2/3}\phi_{5_r}^{-4/3}\phi_{5_g}^{-1/3} - 2\sqrt{2}\,H^+\phi_{4_b}^{+2/3}\phi_{5_g}^{-4/3}\phi_{5_r}^{-1/3}) + \text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
&\lambda_{29}(\phi_{1_r}^{-1/3}\phi_{1_r}^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_1}^{-1/3,\bar{r}\dagger} + 2\,\phi_{1_r}^{-1/3}\phi_{1_g}^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_1}^{-1/3,\bar{g}\dagger}\\
&+ 2\,\phi_{1_r}^{-1/3}\phi_{1_b}^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_1}^{-1/3,\bar{b}\dagger} + \phi_{1_g}^{-1/3}\phi_{1_g}^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_1}^{-1/3,\bar{g}\dagger}\\
&+ 2\,\phi_{1_g}^{-1/3}\phi_{1_b}^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_1}^{-1/3,\bar{b}\dagger} + \phi_{1_b}^{-1/3}\phi_{1_b}^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_1}^{-1/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
\lambda_{30}(&4\,{\phi_1}_r^{-1/3}{\phi_1}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - 2\,{\phi_1}_r^{-1/3}{\phi_1}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&+ 4\,{\phi_1}_r^{-1/3}{\phi_1}_g^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} + 4\,{\phi_1}_r^{-1/3}{\phi_1}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} \\
&- 4\,{\phi_1}_r^{-1/3}{\phi_1}_g^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} + 4\,{\phi_1}_r^{-1/3}{\phi_1}_b^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
&+ 4\,{\phi_1}_r^{-1/3}{\phi_1}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - 4\,{\phi_1}_r^{-1/3}{\phi_1}_b^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ 4\,{\phi_1}_g^{-1/3}{\phi_1}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - 2\,{\phi_1}_g^{-1/3}{\phi_1}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
&+ 4\,{\phi_1}_g^{-1/3}{\phi_1}_b^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} + 4\,{\phi_1}_g^{-1/3}{\phi_1}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&- 4\,{\phi_1}_g^{-1/3}{\phi_1}_b^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} + 4\,{\phi_1}_b^{-1/3}{\phi_1}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
&- 2\,{\phi_1}_b^{-1/3}{\phi_1}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger}) + \text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(1)}_{31,c1}(&{\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(2)}_{31,c2}(&{\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(1)}_{33,c1}(&{\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger})
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(2)}_{33,c2}\big(& {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(1)}_{35,c1}\big(& {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(2)}_{35,c2}\big(& {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger} \\
&+ {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} \\
&+ {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger} \\
&+ {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(1)}_{37,c1}\big({\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}^\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}^\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}^\dagger} \\
&\quad + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}^\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}^\dagger} \\
&\quad + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}^\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}^\dagger} \\
&\quad + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}^\dagger} \\
&\quad + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}^\dagger} \\
&\quad + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}^\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}^\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}^\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}^\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}^\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(2)}_{37,c2}\big({\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}^\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}^\dagger} \\
&\quad + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}^\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{r}^\dagger} \\
&\quad + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{r}^\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{r}^\dagger} \\
&\quad + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{r}^\dagger} + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{r}^\dagger} \\
&\quad + {\phi_1}_r^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{r}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{g}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{g}^\dagger} + {\phi_1}_g^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{g}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{b}^\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{b}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{r}^\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{b}^\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{b}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{b}^\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{g}^\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{b}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}^\dagger} + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}^\dagger} \\
&\quad + {\phi_1}_b^{-1/3}{\phi_1}^{-1/3,\bar{b}^\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}^\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(1)}_{39,c1}\big(\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4_r}^{+2/3}+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4_g}^{+2/3}\\
&\quad+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4_b}^{+2/3}+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4_r}^{-1/3}\\
&\quad+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4_g}^{-1/3}+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4_b}^{-1/3}\\
&\quad+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4_r}^{+2/3}+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4_g}^{+2/3}\\
&\quad+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4_b}^{+2/3}+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4_r}^{-1/3}\\
&\quad+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4_g}^{-1/3}+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4_b}^{-1/3}\\
&\quad+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4_r}^{+2/3}+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4_g}^{+2/3}\\
&\quad+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4_b}^{+2/3}+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4_r}^{-1/3}\\
&\quad+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4_g}^{-1/3}+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4_b}^{-1/3}\big)+\text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(2)}_{39,c2}\big(\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4_r}^{+2/3}+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4_r}^{-1/3}\\
&\quad+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4_g}^{+2/3}+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4_g}^{-1/3}\\
&\quad+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4_b}^{+2/3}+\phi_{1_r}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4_b}^{-1/3}\\
&\quad+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4_r}^{+2/3}+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4_r}^{-1/3}\\
&\quad+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4_g}^{+2/3}+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4_g}^{-1/3}\\
&\quad+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4_b}^{+2/3}+\phi_{1_g}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4_b}^{-1/3}\\
&\quad+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4_r}^{+2/3}+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4_r}^{-1/3}\\
&\quad+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4_g}^{+2/3}+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4_g}^{-1/3}\\
&\quad+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4_b}^{+2/3}+\phi_{1_b}^{-1/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4_b}^{-1/3}\big)+\text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(1)}_{41,c1}\Big\{\sqrt{2}\,\phi_{1_r}^{-1/3}\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{r}\dagger}-\phi_{1_r}^{-1/3}\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{r}\dagger}\\
&\quad+\sqrt{2}\,\phi_{1_r}^{-1/3}\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{g}\dagger}-\phi_{1_r}^{-1/3}\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{g}\dagger}\\
&\quad+\sqrt{2}\,\phi_{1_r}^{-1/3}\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{b}\dagger}-\phi_{1_r}^{-1/3}\phi_{3_b}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{b}\dagger}\\
&\quad+\phi_{1_r}^{-1/3}\phi_{3_r}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{r}\dagger}-\sqrt{2}\,\phi_{1_r}^{-1/3}\phi_{3_r}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_5^{+2/3,\bar{r}\dagger}\\
&\quad+\phi_{1_r}^{-1/3}\phi_{3_g}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{g}\dagger}-\sqrt{2}\,\phi_{1_r}^{-1/3}\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_5^{+2/3,\bar{g}\dagger}\\
&\quad+\phi_{1_r}^{-1/3}\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{b}\dagger}-\sqrt{2}\,\phi_{1_r}^{-1/3}\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_5^{+2/3,\bar{b}\dagger}\\
&\quad+\sqrt{2}\,\phi_{1_g}^{-1/3}\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{r}\dagger}-\phi_{1_g}^{-1/3}\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{r}\dagger}\\
&\quad+\sqrt{2}\,\phi_{1_g}^{-1/3}\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{g}\dagger}-\phi_{1_g}^{-1/3}\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{g}\dagger}\\
&\quad+\sqrt{2}\,\phi_{1_g}^{-1/3}\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{b}\dagger}-\phi_{1_g}^{-1/3}\phi_{3_b}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{b}\dagger}\\
&\quad+\phi_{1_g}^{-1/3}\phi_{3_r}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{r}\dagger}-\sqrt{2}\,\phi_{1_g}^{-1/3}\phi_{3_r}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_5^{+2/3,\bar{r}\dagger}\\
&\quad+\phi_{1_g}^{-1/3}\phi_{3_g}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{g}\dagger}-\sqrt{2}\,\phi_{1_g}^{-1/3}\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_5^{+2/3,\bar{g}\dagger}\\
&\quad+\phi_{1_g}^{-1/3}\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{b}\dagger}-\sqrt{2}\,\phi_{1_g}^{-1/3}\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_5^{+2/3,\bar{b}\dagger}\\
&\quad+\sqrt{2}\,\phi_{1_b}^{-1/3}\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{r}\dagger}-\phi_{1_b}^{-1/3}\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_5^{-1/3,\bar{r}\dagger}\\
&\quad+\sqrt{2}\,\phi_{1_b}^{-1/3}\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{g}\dagger}-\phi_{1_b}^{-1/3}\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_5^{-1/3,\bar{g}\dagger}
\end{aligned}
$$

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\begin{aligned}
&+\sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_b}{}^{+2/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-4/3,\bar{b}\dagger} - \phi_{1_b}{}^{-1/3}\phi_{3_b}{}^{+2/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{b}\dagger} \\
&\quad + \phi_{1_b}{}^{-1/3}\phi_{3_r}{}^{+5/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_r}{}^{+5/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{+2/3,\bar{r}\dagger} \\
&\quad + \phi_{1_b}{}^{-1/3}\phi_{3_g}{}^{+5/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_g}{}^{+5/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{+2/3,\bar{g}\dagger} \\
&\quad \left. + \phi_{1_b}{}^{-1/3}\phi_{3_b}{}^{+5/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_b}{}^{+5/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{+2/3,\bar{b}\dagger} \right\} + \text{h.c.}
\end{aligned}
$$

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\begin{aligned}
&\lambda^{(2)}_{41,c2}\Big\{ \sqrt{2}\,\phi_{1_r}{}^{-1/3}\phi_{3_r}{}^{+2/3}\phi_3{}^{+5/3,\bar{r}\dagger}\phi_5{}^{-4/3,\bar{r}\dagger} - \phi_{1_r}{}^{-1/3}\phi_{3_r}{}^{+2/3}\phi_3{}^{+2/3,\bar{r}\dagger}\phi_5{}^{-1/3,\bar{r}\dagger} \\
&\quad + \sqrt{2}\,\phi_{1_r}{}^{-1/3}\phi_{3_g}{}^{+2/3}\phi_3{}^{+5/3,\bar{g}\dagger}\phi_5{}^{-4/3,\bar{r}\dagger} - \phi_{1_r}{}^{-1/3}\phi_{3_g}{}^{+2/3}\phi_3{}^{+2/3,\bar{g}\dagger}\phi_5{}^{-1/3,\bar{r}\dagger} \\
&\quad + \sqrt{2}\,\phi_{1_r}{}^{-1/3}\phi_{3_b}{}^{+2/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-4/3,\bar{r}\dagger} - \phi_{1_r}{}^{-1/3}\phi_{3_b}{}^{+2/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{r}\dagger} \\
&\quad + \phi_{1_r}{}^{-1/3}\phi_{3_r}{}^{+5/3}\phi_3{}^{+5/3,\bar{r}\dagger}\phi_5{}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,\phi_{1_r}{}^{-1/3}\phi_{3_r}{}^{+5/3}\phi_3{}^{+2/3,\bar{r}\dagger}\phi_5{}^{+2/3,\bar{r}\dagger} \\
&\quad + \phi_{1_r}{}^{-1/3}\phi_{3_g}{}^{+5/3}\phi_3{}^{+5/3,\bar{g}\dagger}\phi_5{}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,\phi_{1_r}{}^{-1/3}\phi_{3_g}{}^{+5/3}\phi_3{}^{+2/3,\bar{g}\dagger}\phi_5{}^{+2/3,\bar{r}\dagger} \\
&\quad + \phi_{1_r}{}^{-1/3}\phi_{3_b}{}^{+5/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,\phi_{1_r}{}^{-1/3}\phi_{3_b}{}^{+5/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{+2/3,\bar{r}\dagger} \\
&\quad + \sqrt{2}\,\phi_{1_g}{}^{-1/3}\phi_{3_r}{}^{+2/3}\phi_3{}^{+5/3,\bar{r}\dagger}\phi_5{}^{-4/3,\bar{g}\dagger} - \phi_{1_g}{}^{-1/3}\phi_{3_r}{}^{+2/3}\phi_3{}^{+2/3,\bar{r}\dagger}\phi_5{}^{-1/3,\bar{g}\dagger} \\
&\quad + \sqrt{2}\,\phi_{1_g}{}^{-1/3}\phi_{3_g}{}^{+2/3}\phi_3{}^{+5/3,\bar{g}\dagger}\phi_5{}^{-4/3,\bar{g}\dagger} - \phi_{1_g}{}^{-1/3}\phi_{3_g}{}^{+2/3}\phi_3{}^{+2/3,\bar{g}\dagger}\phi_5{}^{-1/3,\bar{g}\dagger} \\
&\quad + \sqrt{2}\,\phi_{1_g}{}^{-1/3}\phi_{3_b}{}^{+2/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-4/3,\bar{g}\dagger} - \phi_{1_g}{}^{-1/3}\phi_{3_b}{}^{+2/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{g}\dagger} \\
&\quad + \phi_{1_g}{}^{-1/3}\phi_{3_r}{}^{+5/3}\phi_3{}^{+5/3,\bar{r}\dagger}\phi_5{}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,\phi_{1_g}{}^{-1/3}\phi_{3_r}{}^{+5/3}\phi_3{}^{+2/3,\bar{r}\dagger}\phi_5{}^{+2/3,\bar{g}\dagger} \\
&\quad + \phi_{1_g}{}^{-1/3}\phi_{3_g}{}^{+5/3}\phi_3{}^{+5/3,\bar{g}\dagger}\phi_5{}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,\phi_{1_g}{}^{-1/3}\phi_{3_g}{}^{+5/3}\phi_3{}^{+2/3,\bar{g}\dagger}\phi_5{}^{+2/3,\bar{g}\dagger} \\
&\quad + \phi_{1_g}{}^{-1/3}\phi_{3_b}{}^{+5/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,\phi_{1_g}{}^{-1/3}\phi_{3_b}{}^{+5/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{+2/3,\bar{g}\dagger} \\
&\quad + \sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_r}{}^{+2/3}\phi_3{}^{+5/3,\bar{r}\dagger}\phi_5{}^{-4/3,\bar{b}\dagger} - \phi_{1_b}{}^{-1/3}\phi_{3_r}{}^{+2/3}\phi_3{}^{+2/3,\bar{r}\dagger}\phi_5{}^{-1/3,\bar{b}\dagger} \\
&\quad + \sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_g}{}^{+2/3}\phi_3{}^{+5/3,\bar{g}\dagger}\phi_5{}^{-4/3,\bar{b}\dagger} - \phi_{1_b}{}^{-1/3}\phi_{3_g}{}^{+2/3}\phi_3{}^{+2/3,\bar{g}\dagger}\phi_5{}^{-1/3,\bar{b}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
&+\sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_b}{}^{+2/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-4/3,\bar{b}\dagger} - \phi_{1_b}{}^{-1/3}\phi_{3_b}{}^{+2/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{b}\dagger} \\
&\quad + \phi_{1_b}{}^{-1/3}\phi_{3_r}{}^{+5/3}\phi_3{}^{+5/3,\bar{r}\dagger}\phi_5{}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_r}{}^{+5/3}\phi_3{}^{+2/3,\bar{r}\dagger}\phi_5{}^{+2/3,\bar{b}\dagger} \\
&\quad + \phi_{1_b}{}^{-1/3}\phi_{3_g}{}^{+5/3}\phi_3{}^{+5/3,\bar{g}\dagger}\phi_5{}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_g}{}^{+5/3}\phi_3{}^{+2/3,\bar{g}\dagger}\phi_5{}^{+2/3,\bar{b}\dagger} \\
&\quad \left. + \phi_{1_b}{}^{-1/3}\phi_{3_b}{}^{+5/3}\phi_3{}^{+5/3,\bar{b}\dagger}\phi_5{}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{1_b}{}^{-1/3}\phi_{3_b}{}^{+5/3}\phi_3{}^{+2/3,\bar{b}\dagger}\phi_5{}^{+2/3,\bar{b}\dagger} \right\} + \text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(1)}_{43,c1}\Big\{ & \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_1}_r^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_1}_r^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_1}_r^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_1}_r^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_1}_r^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_1}_r^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_1}_g^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_1}_g^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_1}_g^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_1}_g^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_1}_g^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_1}_g^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + \sqrt{2}\,{\phi_1}_b^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_1}_b^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_1}_b^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_1}_b^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
& +\sqrt{2}\,{\phi_1}_b^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_1}_b^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_1}_b^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_1}_b^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_1}_b^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_1}_b^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_1}_b^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_1}_b^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}\Big\} + \text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(2)}_{43,c2}\Big\{ & \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_1}_r^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_1}_r^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_1}_r^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + {\phi_1}_r^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_1}_r^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_1}_r^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_1}_r^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_1}_g^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_1}_g^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_1}_g^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + {\phi_1}_g^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_1}_g^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_1}_g^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_1}_g^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_1}_b^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_1}_b^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + \sqrt{2}\,{\phi_1}_b^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_1}_b^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger}
\end{aligned}
$$

$$+\sqrt{2}\,\phi_{1_b}^{-1/3}\phi_{4_b}^{-1/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{b}\dagger} - \phi_{1_b}^{-1/3}\phi_{4_b}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_5^{-1/3,\bar{b}\dagger}$$
$$+\phi_{1_b}^{-1/3}\phi_{4_r}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{1_b}^{-1/3}\phi_{4_r}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}\phi_5^{+2/3,\bar{b}\dagger}$$
$$+\phi_{1_b}^{-1/3}\phi_{4_g}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{1_b}^{-1/3}\phi_{4_g}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}\phi_5^{+2/3,\bar{b}\dagger}$$
$$\left.+\phi_{1_b}^{-1/3}\phi_{4_b}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_5^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{1_b}^{-1/3}\phi_{4_b}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_5^{+2/3,\bar{b}\dagger}\right\} + \text{h.c.}$$

$$\lambda_{45}\left\{2\,\phi_{1_r}^{-1/3}\phi_{5_g}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{g}\dagger} - 2\,\phi_{1_r}^{-1/3}\phi_{5_g}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{r}\dagger}\right.$$
$$+2\,\phi_{1_r}^{-1/3}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{b}\dagger} - 2\,\phi_{1_r}^{-1/3}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{r}\dagger}$$
$$+2\,\phi_{1_r}^{-1/3}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{g}\dagger} - 2\,\phi_{1_r}^{-1/3}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{r}\dagger}$$
$$+2\,\phi_{1_r}^{-1/3}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{b}\dagger} - 2\,\phi_{1_r}^{-1/3}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{r}\dagger}$$
$$+2\,\phi_{1_r}^{-1/3}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{g}\dagger} - 2\,\phi_{1_r}^{-1/3}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{r}\dagger}$$
$$+2\,\phi_{1_r}^{-1/3}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{b}\dagger} - 2\,\phi_{1_r}^{-1/3}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\phi_5^{-1/3,\bar{r}\dagger}$$
$$-2\,\phi_{1_g}^{-1/3}\phi_{5_r}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{g}\dagger} + 2\,\phi_{1_g}^{-1/3}\phi_{5_r}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{r}\dagger}$$
$$+2\,\phi_{1_g}^{-1/3}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{b}\dagger} - 2\,\phi_{1_g}^{-1/3}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{g}\dagger}$$
$$-2\,\phi_{1_g}^{-1/3}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{g}\dagger} + 2\,\phi_{1_g}^{-1/3}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{r}\dagger}$$
$$+2\,\phi_{1_g}^{-1/3}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{b}\dagger} - 2\,\phi_{1_g}^{-1/3}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{g}\dagger}$$
$$-2\,\phi_{1_g}^{-1/3}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{g}\dagger} + 2\,\phi_{1_g}^{-1/3}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{r}\dagger}$$
$$+2\,\phi_{1_g}^{-1/3}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{b}\dagger} - 2\,\phi_{1_g}^{-1/3}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\phi_5^{-1/3,\bar{g}\dagger}$$
$$-2\,\phi_{1_b}^{-1/3}\phi_{5_r}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{b}\dagger} + 2\,\phi_{1_b}^{-1/3}\phi_{5_r}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{r}\dagger}$$
$$-2\,\phi_{1_b}^{-1/3}\phi_{5_g}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{b}\dagger} + 2\,\phi_{1_b}^{-1/3}\phi_{5_g}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{g}\dagger}$$

$$-2\,\phi_{1_b}^{-1/3}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\phi_5^{-4/3,\bar{b}\dagger} + 2\,\phi_{1_b}^{-1/3}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{r}\dagger}$$
$$-2\,\phi_{1_b}^{-1/3}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}\phi_5^{-4/3,\bar{b}\dagger} + 2\,\phi_{1_b}^{-1/3}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\phi_5^{-4/3,\bar{g}\dagger}$$
$$-2\,\phi_{1_b}^{-1/3}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\phi_5^{-1/3,\bar{b}\dagger} + 2\,\phi_{1_b}^{-1/3}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\phi_5^{-1/3,\bar{r}\dagger}$$
$$\left.-2\,\phi_{1_b}^{-1/3}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\phi_5^{-1/3,\bar{b}\dagger} + 2\,\phi_{1_b}^{-1/3}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\phi_5^{-1/3,\bar{g}\dagger}\right\} + \text{h.c.}$$

$$\lambda_{46}(\phi_{2_r}^{-4/3}\phi_{2_r}^{-4/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_2^{-4/3,\bar{r}\dagger} + 2\,\phi_{2_r}^{-4/3}\phi_{2_g}^{-4/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_2^{-4/3,\bar{g}\dagger}$$
$$+2\,\phi_{2_r}^{-4/3}\phi_{2_b}^{-4/3}\phi_2^{-4/3,\bar{r}\dagger}\phi_2^{-4/3,\bar{b}\dagger} + \phi_{2_g}^{-4/3}\phi_{2_g}^{-4/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_2^{-4/3,\bar{g}\dagger}$$
$$+2\,\phi_{2_g}^{-4/3}\phi_{2_b}^{-4/3}\phi_2^{-4/3,\bar{g}\dagger}\phi_2^{-4/3,\bar{b}\dagger} + \phi_{2_b}^{-4/3}\phi_{2_b}^{-4/3}\phi_2^{-4/3,\bar{b}\dagger}\phi_2^{-4/3,\bar{b}\dagger})$$

$$
\begin{aligned}
&\lambda^{(1)}_{47,c1}\big({\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger} \\
&+ {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger} \\
&+ {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(2)}_{47,c2}\big({\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger} \\
&+ {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger} \\
&+ {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(1)}_{49,c1}\big({\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}^\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}^\dagger} \\
&+ {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}^\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}^\dagger} \\
&+ {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}^\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}^\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}^\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}^\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}^\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}^\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}^\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}^\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}^\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}^\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}^\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}^\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}^\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(2)}_{49,c2}\big({\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_4}_r^{-1/3}{\phi_4^{-1/3,\bar{r}}}^\dagger + {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_4}_r^{+2/3}{\phi_4^{+2/3,\bar{r}}}^\dagger \\
&+ {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_4}_g^{-1/3}{\phi_4^{-1/3,\bar{r}}}^\dagger + {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_4}_g^{+2/3}{\phi_4^{+2/3,\bar{r}}}^\dagger \\
&+ {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_4}_b^{-1/3}{\phi_4^{-1/3,\bar{r}}}^\dagger + {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_4}_b^{+2/3}{\phi_4^{+2/3,\bar{r}}}^\dagger \\
&+ {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_4}_r^{-1/3}{\phi_4^{-1/3,\bar{g}}}^\dagger + {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_4}_r^{+2/3}{\phi_4^{+2/3,\bar{g}}}^\dagger \\
&+ {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_4}_g^{-1/3}{\phi_4^{-1/3,\bar{g}}}^\dagger + {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_4}_g^{+2/3}{\phi_4^{+2/3,\bar{g}}}^\dagger \\
&+ {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_4}_b^{-1/3}{\phi_4^{-1/3,\bar{g}}}^\dagger + {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_4}_b^{+2/3}{\phi_4^{+2/3,\bar{g}}}^\dagger \\
&+ {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_4}_r^{-1/3}{\phi_4^{-1/3,\bar{b}}}^\dagger + {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_4}_r^{+2/3}{\phi_4^{+2/3,\bar{b}}}^\dagger \\
&+ {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_4}_g^{-1/3}{\phi_4^{-1/3,\bar{b}}}^\dagger + {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_4}_g^{+2/3}{\phi_4^{+2/3,\bar{b}}}^\dagger \\
&+ {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_4}_b^{-1/3}{\phi_4^{-1/3,\bar{b}}}^\dagger + {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_4}_b^{+2/3}{\phi_4^{+2/3,\bar{b}}}^\dagger\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(1)}_{51,c1}\big({\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_5}_r^{-4/3}{\phi_5^{-4/3,\bar{r}}}^\dagger + {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_5}_g^{-4/3}{\phi_5^{-4/3,\bar{g}}}^\dagger \\
&+ {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_5}_b^{-4/3}{\phi_5^{-4/3,\bar{b}}}^\dagger + {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_5}_r^{-1/3}{\phi_5^{-1/3,\bar{r}}}^\dagger \\
&+ {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_5}_g^{-1/3}{\phi_5^{-1/3,\bar{g}}}^\dagger + {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_5}_b^{-1/3}{\phi_5^{-1/3,\bar{b}}}^\dagger \\
&+ {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_5}_r^{+2/3}{\phi_5^{+2/3,\bar{r}}}^\dagger + {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_5}_g^{+2/3}{\phi_5^{+2/3,\bar{g}}}^\dagger \\
&+ {\phi_2}_r^{-4/3}{\phi_2^{-4/3,\bar{r}}}^\dagger{\phi_5}_b^{+2/3}{\phi_5^{+2/3,\bar{b}}}^\dagger + {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_5}_r^{-4/3}{\phi_5^{-4/3,\bar{r}}}^\dagger \\
&+ {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_5}_g^{-4/3}{\phi_5^{-4/3,\bar{g}}}^\dagger + {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_5}_b^{-4/3}{\phi_5^{-4/3,\bar{b}}}^\dagger \\
&+ {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_5}_r^{-1/3}{\phi_5^{-1/3,\bar{r}}}^\dagger + {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_5}_g^{-1/3}{\phi_5^{-1/3,\bar{g}}}^\dagger \\
&+ {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_5}_b^{-1/3}{\phi_5^{-1/3,\bar{b}}}^\dagger + {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_5}_r^{+2/3}{\phi_5^{+2/3,\bar{r}}}^\dagger \\
&+ {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_5}_g^{+2/3}{\phi_5^{+2/3,\bar{g}}}^\dagger + {\phi_2}_g^{-4/3}{\phi_2^{-4/3,\bar{g}}}^\dagger{\phi_5}_b^{+2/3}{\phi_5^{+2/3,\bar{b}}}^\dagger \\
&+ {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_5}_r^{-4/3}{\phi_5^{-4/3,\bar{r}}}^\dagger + {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_5}_g^{-4/3}{\phi_5^{-4/3,\bar{g}}}^\dagger \\
&+ {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_5}_b^{-4/3}{\phi_5^{-4/3,\bar{b}}}^\dagger + {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_5}_r^{-1/3}{\phi_5^{-1/3,\bar{r}}}^\dagger \\
&+ {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_5}_g^{-1/3}{\phi_5^{-1/3,\bar{g}}}^\dagger + {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_5}_b^{-1/3}{\phi_5^{-1/3,\bar{b}}}^\dagger \\
&+ {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_5}_r^{+2/3}{\phi_5^{+2/3,\bar{r}}}^\dagger + {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_5}_g^{+2/3}{\phi_5^{+2/3,\bar{g}}}^\dagger \\
&+ {\phi_2}_b^{-4/3}{\phi_2^{-4/3,\bar{b}}}^\dagger{\phi_5}_b^{+2/3}{\phi_5^{+2/3,\bar{b}}}^\dagger\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(2)}_{51,c2}\big({\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&\quad + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} \\
&\quad + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&\quad + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&\quad + {\phi_2}_r^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&\quad + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
&\quad + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
&\quad + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&\quad + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} + {\phi_2}_g^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
&\quad + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&\quad + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} \\
&\quad + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&\quad + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&\quad + {\phi_2}_b^{-4/3}{\phi_2}^{-4/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(1)}_{53,c1}\Big\{ \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_2}_r^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&+ \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_2}_r^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
&+ \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_2}_r^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ {\phi_2}_r^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&+ {\phi_2}_r^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
&+ {\phi_2}_r^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&+ \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_2}_g^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&+ \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_2}_g^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
&+ \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_2}_g^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
&+ {\phi_2}_g^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&+ \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_2}_b^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&+ \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_2}_b^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
&+\sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_2}_b^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
&+ {\phi_2}_b^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}\Big\} + \text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(2)}_{53,c2}\Big\{ & \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_2}_r^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_2}_r^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_2}_r^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + {\phi_2}_r^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_2}_r^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_2}_r^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_2}_r^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_2}_g^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_2}_g^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_2}_g^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + {\phi_2}_g^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_2}_g^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_2}_g^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_2}_g^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_2}_b^{-4/3}{\phi_3}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_2}_b^{-4/3}{\phi_3}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
& +\sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_2}_b^{-4/3}{\phi_3}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_2}_b^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_r^{+5/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + {\phi_2}_b^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_g^{+5/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + {\phi_2}_b^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_2}_b^{-4/3}{\phi_3}_b^{+5/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}\Big\} + \text{h.c.}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(1)}_{55,c1}\big(& 2\,{\phi_3}_r^{+2/3}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_3}^{+2/3,\bar{r}\dagger} + 4\,{\phi_3}_r^{+2/3}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_3}^{+2/3,\bar{g}\dagger} \\
& + 4\,{\phi_3}_r^{+2/3}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_3}^{+2/3,\bar{b}\dagger} + 4\,{\phi_3}_r^{+2/3}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_3}^{+2/3,\bar{r}\dagger} \\
& + 4\,{\phi_3}_r^{+2/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_3}^{+2/3,\bar{r}\dagger} + 4\,{\phi_3}_r^{+2/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_3}^{+2/3,\bar{r}\dagger} \\
& + 2\,{\phi_3}_g^{+2/3}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_3}^{+2/3,\bar{g}\dagger} + 4\,{\phi_3}_g^{+2/3}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_3}^{+2/3,\bar{b}\dagger} \\
& + 4\,{\phi_3}_g^{+2/3}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_3}^{+2/3,\bar{g}\dagger} + 4\,{\phi_3}_g^{+2/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_3}^{+2/3,\bar{g}\dagger} \\
& + 4\,{\phi_3}_g^{+2/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_3}^{+2/3,\bar{g}\dagger} + 2\,{\phi_3}_b^{+2/3}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_3}^{+2/3,\bar{b}\dagger} \\
& + 4\,{\phi_3}_b^{+2/3}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_3}^{+2/3,\bar{b}\dagger} + 4\,{\phi_3}_b^{+2/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_3}^{+2/3,\bar{b}\dagger} \\
& + 4\,{\phi_3}_b^{+2/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_3}^{+2/3,\bar{b}\dagger} + 2\,{\phi_3}_r^{+5/3}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_3}^{+5/3,\bar{r}\dagger} \\
& + 4\,{\phi_3}_r^{+5/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_3}^{+5/3,\bar{g}\dagger} + 4\,{\phi_3}_r^{+5/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_3}^{+5/3,\bar{b}\dagger} \\
& + 2\,{\phi_3}_g^{+5/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_3}^{+5/3,\bar{g}\dagger} + 4\,{\phi_3}_g^{+5/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_3}^{+5/3,\bar{b}\dagger} \\
& + 2\,{\phi_3}_b^{+5/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_3}^{+5/3,\bar{b}\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(2)}_{55,c2}\big(2\,{\phi_3}_r^{+2/3}{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger}{\phi_3}^{+2/3,\bar{r}^\dagger} + 4\,{\phi_3}_r^{+2/3}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger}{\phi_3}^{+2/3,\bar{g}^\dagger}\\
&\quad + 4\,{\phi_3}_r^{+2/3}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger}{\phi_3}^{+2/3,\bar{b}^\dagger} + 4\,{\phi_3}_r^{+2/3}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_3}^{+2/3,\bar{r}^\dagger}\\
&\quad + 4\,{\phi_3}_r^{+2/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_3}^{+2/3,\bar{g}^\dagger} + 4\,{\phi_3}_r^{+2/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_3}^{+2/3,\bar{b}^\dagger}\\
&\quad + 2\,{\phi_3}_g^{+2/3}{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger}{\phi_3}^{+2/3,\bar{g}^\dagger} + 4\,{\phi_3}_g^{+2/3}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger}{\phi_3}^{+2/3,\bar{b}^\dagger}\\
&\quad + 4\,{\phi_3}_g^{+2/3}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_3}^{+2/3,\bar{r}^\dagger} + 4\,{\phi_3}_g^{+2/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_3}^{+2/3,\bar{g}^\dagger}\\
&\quad + 4\,{\phi_3}_g^{+2/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_3}^{+2/3,\bar{b}^\dagger} + 2\,{\phi_3}_b^{+2/3}{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger}{\phi_3}^{+2/3,\bar{b}^\dagger}\\
&\quad + 4\,{\phi_3}_b^{+2/3}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_3}^{+2/3,\bar{r}^\dagger} + 4\,{\phi_3}_b^{+2/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_3}^{+2/3,\bar{g}^\dagger}\\
&\quad + 4\,{\phi_3}_b^{+2/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_3}^{+2/3,\bar{b}^\dagger} + 2\,{\phi_3}_r^{+5/3}{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_3}^{+5/3,\bar{r}^\dagger}\\
&\quad + 4\,{\phi_3}_r^{+5/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_3}^{+5/3,\bar{g}^\dagger} + 4\,{\phi_3}_r^{+5/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_3}^{+5/3,\bar{b}^\dagger}\\
&\quad + 2\,{\phi_3}_g^{+5/3}{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_3}^{+5/3,\bar{g}^\dagger} + 4\,{\phi_3}_g^{+5/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_3}^{+5/3,\bar{b}^\dagger}\\
&\quad + 2\,{\phi_3}_b^{+5/3}{\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_3}^{+5/3,\bar{b}^\dagger}\big)
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(1)}_{57,c1}\Big\{{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}^\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}^\dagger}\\
&\quad + {\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}^\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}^\dagger}\\
&\quad + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}^\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}^\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}^\dagger}\\
&\quad + {\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}^\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}^\dagger}\\
&\quad + {\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}^\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}^\dagger}\\
&\quad + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}^\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}^\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}^\dagger}\\
&\quad + {\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}^\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}^\dagger}\\
&\quad + {\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}^\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}^\dagger}\\
&\quad + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}^\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}^\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}^\dagger}\\
&\quad + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}^\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}^\dagger}\\
&\quad + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}^\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}^\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}^\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}^\dagger}\\
&\quad + {\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}^\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}^\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}^\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}^\dagger}\\
&\quad + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}^\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}^\dagger}\\
&\quad + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}^\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}^\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{g}^\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}^\dagger}
\end{aligned}
$$

$$
\begin{aligned}
&+{\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{g}^\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}^\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{g}^\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}^\dagger}\\
&\quad + {\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}^\dagger} + {\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}^\dagger}\\
&\quad + {\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}^\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}^\dagger} + {\phi_3}_b^{+5/3}{\phi_3}^{+2/3,\bar{b}^\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}^\dagger}\\
&\quad + {\phi_3}_b^{+5/3}{\phi_3}^{+2/3,\bar{b}^\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}^\dagger} + {\phi_3}_b^{+5/3}{\phi_3}^{+2/3,\bar{b}^\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}^\dagger}\Big\}
\end{aligned}
$$

$$\lambda^{(2)}_{57,c2}\Big\{ {\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}$$
$$+ {\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}$$
$$+ {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}$$
$$+ {\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}$$
$$+ {\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}$$
$$+ {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}$$
$$+ {\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}$$
$$+ {\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}$$
$$+ {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}$$
$$+ {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}$$
$$+ {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}$$
$$+ {\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}$$
$$+ {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}$$
$$+ {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}$$

$$+{\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}$$
$$+ {\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger} + {\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}$$
$$+ {\phi_3}_b^{+5/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger} + {\phi_3}_b^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}$$
$$+ {\phi_3}_b^{+5/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger} + {\phi_3}_b^{+5/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}\Big\}$$

$$\lambda^{(3)}_{57,c3}\Big\{ -{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger} - {\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}$$
$$- {\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}$$
$$+ {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}$$
$$- {\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger} - {\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}$$
$$- {\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}$$
$$+ {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}$$
$$- {\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger} - {\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}$$
$$- {\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}$$
$$+ {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}$$
$$+ {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}$$
$$+ {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger} - {\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}$$
$$- {\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger} - {\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}$$
$$+ {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}$$
$$+ {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger} - {\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}$$

$$
\begin{aligned}
&-\phi_{3g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4g}^{-1/3}\phi_4^{+2/3,\bar{g}\dagger}-\phi_{3g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4b}^{-1/3}\phi_4^{+2/3,\bar{b}\dagger}\\
&\quad+\phi_{3b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4r}^{-1/3}\phi_4^{-1/3,\bar{r}\dagger}+\phi_{3b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4g}^{-1/3}\phi_4^{-1/3,\bar{g}\dagger}\\
&\quad+\phi_{3b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}-\phi_{3b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4r}^{-1/3}\phi_4^{+2/3,\bar{r}\dagger}\\
&\quad-\phi_{3b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4g}^{-1/3}\phi_4^{+2/3,\bar{g}\dagger}-\phi_{3b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4b}^{-1/3}\phi_4^{+2/3,\bar{b}\dagger}\Bigg\}
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(4)}_{57,c4}\Bigg\{-\phi_{3r}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4r}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}-\phi_{3r}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4g}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}\\
&-\phi_{3r}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4b}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}+\phi_{3r}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4r}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}\\
&+\phi_{3r}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4g}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}+\phi_{3r}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4b}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}\\
&-\phi_{3g}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4r}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}-\phi_{3g}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4g}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}\\
&-\phi_{3g}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4b}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}+\phi_{3g}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4r}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}\\
&+\phi_{3g}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4g}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}+\phi_{3g}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4b}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}\\
&-\phi_{3b}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4r}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}-\phi_{3b}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4g}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}\\
&-\phi_{3b}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4b}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}+\phi_{3b}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4r}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}\\
&+\phi_{3b}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4g}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}+\phi_{3b}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4b}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}\\
&+\phi_{3r}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4r}^{-1/3}\phi_4^{-1/3,\bar{r}\dagger}+\phi_{3r}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4g}^{-1/3}\phi_4^{-1/3,\bar{r}\dagger}\\
&+\phi_{3r}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{r}\dagger}-\phi_{3r}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4r}^{-1/3}\phi_4^{+2/3,\bar{r}\dagger}\\
&-\phi_{3r}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4g}^{-1/3}\phi_4^{+2/3,\bar{r}\dagger}-\phi_{3r}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4b}^{-1/3}\phi_4^{+2/3,\bar{r}\dagger}\\
&+\phi_{3g}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4r}^{-1/3}\phi_4^{-1/3,\bar{g}\dagger}+\phi_{3g}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4g}^{-1/3}\phi_4^{-1/3,\bar{g}\dagger}\\
&+\phi_{3g}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{g}\dagger}-\phi_{3g}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4r}^{-1/3}\phi_4^{+2/3,\bar{g}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
&-\phi_{3g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4g}^{-1/3}\phi_4^{+2/3,\bar{g}\dagger}-\phi_{3g}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4b}^{-1/3}\phi_4^{+2/3,\bar{g}\dagger}\\
&\quad+\phi_{3b}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{4r}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}+\phi_{3b}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{4g}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}\\
&\quad+\phi_{3b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}-\phi_{3b}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{4r}^{-1/3}\phi_4^{+2/3,\bar{b}\dagger}\\
&\quad-\phi_{3b}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{4g}^{-1/3}\phi_4^{+2/3,\bar{b}\dagger}-\phi_{3b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{4b}^{-1/3}\phi_4^{+2/3,\bar{b}\dagger}\Bigg\}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(1)}_{61,c1}\Bigg\{ & \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} - {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} - {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}
\end{aligned}
$$

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\begin{aligned}
& +\sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} - {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& - {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& - {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& - {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& - {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + \sqrt{2}\,{\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger}
\end{aligned}
$$

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\begin{aligned}
&+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&-\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger}-\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger}\\
&-\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger}+\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}+\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger}+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}\\
&+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\Bigg\}
\end{aligned}
$$

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\begin{aligned}
&\lambda^{(2)}_{61,c2}\Bigg\{\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{-4/3,\bar{r}\dagger}+\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{-1/3,\bar{r}\dagger}\\
&+\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{-4/3,\bar{r}\dagger}+\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{-1/3,\bar{r}\dagger}\\
&+\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{-4/3,\bar{r}\dagger}+\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{-1/3,\bar{r}\dagger}\\
&+\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger}-\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger}-\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger}-\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{-4/3,\bar{g}\dagger}+\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{-1/3,\bar{g}\dagger}\\
&+\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{-4/3,\bar{g}\dagger}+\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{-1/3,\bar{g}\dagger}\\
&+\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{-4/3,\bar{g}\dagger}+\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{-1/3,\bar{g}\dagger}\\
&+\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger}-\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&+\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger}-\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&+\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger}-\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&+\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{-4/3,\bar{b}\dagger}+\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{-1/3,\bar{b}\dagger}\\
&+\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{-4/3,\bar{b}\dagger}+\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{-1/3,\bar{b}\dagger}
\end{aligned}
$$

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\begin{aligned}
&+\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{-4/3,\bar{b}\dagger}+\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{-1/3,\bar{b}\dagger}\\
&+\phi_{3_b}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger}-\phi_{3_b}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&+\phi_{3_b}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger}-\phi_{3_b}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&+\phi_{3_b}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger}-\phi_{3_b}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&-\phi_{3_r}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger}+\phi_{3_r}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&-\phi_{3_r}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger}+\phi_{3_r}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&-\phi_{3_r}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger}+\phi_{3_r}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\sqrt{2}\,\phi_{3_r}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger}+\sqrt{2}\,\phi_{3_r}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\sqrt{2}\,\phi_{3_r}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger}+\sqrt{2}\,\phi_{3_r}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\sqrt{2}\,\phi_{3_r}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger}+\sqrt{2}\,\phi_{3_r}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&-\phi_{3_g}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger}+\phi_{3_g}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&-\phi_{3_g}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger}+\phi_{3_g}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&-\phi_{3_g}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger}+\phi_{3_g}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}
\end{aligned}
$$

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\begin{aligned}
&+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}+\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&-\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger}+\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&-\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger}+\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&-\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger}+\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}+\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\Bigg\}
\end{aligned}
$$

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\begin{aligned}
\lambda^{(3)}_{61,c3}\Bigg\{ & -\sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& - \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& - \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_3}_r^{+2/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + {\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} + 2\,{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + 2\,{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} + 2\,{\phi_3}_r^{+2/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& - \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& - \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& - \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_3}_g^{+2/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + {\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} + 2\,{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + 2\,{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} + 2\,{\phi_3}_g^{+2/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& - \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& - \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
& -\sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_3}_b^{+2/3}{\phi_3}^{+5/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + {\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} + 2\,{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + 2\,{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} + 2\,{\phi_3}_b^{+2/3}{\phi_3}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + 2\,{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + 2\,{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 2\,{\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} + {\phi_3}_r^{+5/3}{\phi_3}^{+5/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& - \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& - \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_3}_r^{+5/3}{\phi_3}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + 2\,{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + 2\,{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 2\,{\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} + {\phi_3}_g^{+5/3}{\phi_3}^{+5/3,\bar{g}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& - \sqrt{2}\,{\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_3}_g^{+5/3}{\phi_3}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
&-\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}-\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&\quad-\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}-\sqrt{2}\,\phi_{3_g}^{+5/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\\
&\quad+2\,\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger}+2\,\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger}\\
&\quad+2\,\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger}+\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_r}^{-1/3}\phi_5^{-1/3,\bar{r}\dagger}\\
&\quad+\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_g}^{-1/3}\phi_5^{-1/3,\bar{g}\dagger}+\phi_{3_b}^{+5/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{-1/3,\bar{b}\dagger}\\
&\quad-\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_r}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger}-\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_g}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}\\
&\quad-\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger}-\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&\quad\left.-\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger}-\sqrt{2}\,\phi_{3_b}^{+5/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger}\right\}
\end{aligned}
$$

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\begin{aligned}
&\lambda^{(4)}_{61,c4}\left\{-\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{-4/3,\bar{r}\dagger}-\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{-1/3,\bar{r}\dagger}\right.\\
&-\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{-4/3,\bar{r}\dagger}-\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{-1/3,\bar{r}\dagger}\\
&-\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{-4/3,\bar{r}\dagger}-\sqrt{2}\,\phi_{3_r}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{-1/3,\bar{r}\dagger}\\
&+\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{-1/3,\bar{r}\dagger}+2\,\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{-1/3,\bar{r}\dagger}+2\,\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&+\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{-1/3,\bar{r}\dagger}+2\,\phi_{3_r}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger}\\
&-\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{-4/3,\bar{g}\dagger}-\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{-1/3,\bar{g}\dagger}\\
&-\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{-4/3,\bar{g}\dagger}-\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{-1/3,\bar{g}\dagger}\\
&-\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{-4/3,\bar{g}\dagger}-\sqrt{2}\,\phi_{3_g}^{+2/3}\phi_3^{+5/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{-1/3,\bar{g}\dagger}\\
&+\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{-1/3,\bar{g}\dagger}+2\,\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&+\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{-1/3,\bar{g}\dagger}+2\,\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&+\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{-1/3}\phi_5^{-1/3,\bar{g}\dagger}+2\,\phi_{3_g}^{+2/3}\phi_3^{+2/3,\bar{b}\dagger}\phi_{5_b}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger}\\
&-\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{-1/3}\phi_5^{-4/3,\bar{b}\dagger}-\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{r}\dagger}\phi_{5_r}^{+2/3}\phi_5^{-1/3,\bar{b}\dagger}\\
&-\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{-1/3}\phi_5^{-4/3,\bar{b}\dagger}-\sqrt{2}\,\phi_{3_b}^{+2/3}\phi_3^{+5/3,\bar{g}\dagger}\phi_{5_g}^{+2/3}\phi_5^{-1/3,\bar{b}\dagger}
\end{aligned}
$$

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\begin{aligned}
&-\sqrt{2}\,{\phi_{3}}_{b}^{+2/3}{\phi_{3}}^{+5/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-1/3}{\phi_{5}}^{-4/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_{3}}_{b}^{+2/3}{\phi_{3}}^{+5/3,\bar{b}\dagger}{\phi_{5}}_{b}^{+2/3}{\phi_{5}}^{-1/3,\bar{b}\dagger}\\
&+ {\phi_{3}}_{b}^{+2/3}{\phi_{3}}^{+2/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-1/3}{\phi_{5}}^{-1/3,\bar{b}\dagger} + 2\,{\phi_{3}}_{b}^{+2/3}{\phi_{3}}^{+2/3,\bar{r}\dagger}{\phi_{5}}_{r}^{+2/3}{\phi_{5}}^{+2/3,\bar{b}\dagger}\\
&+ {\phi_{3}}_{b}^{+2/3}{\phi_{3}}^{+2/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-1/3}{\phi_{5}}^{-1/3,\bar{b}\dagger} + 2\,{\phi_{3}}_{b}^{+2/3}{\phi_{3}}^{+2/3,\bar{g}\dagger}{\phi_{5}}_{g}^{+2/3}{\phi_{5}}^{+2/3,\bar{b}\dagger}\\
&+ {\phi_{3}}_{b}^{+2/3}{\phi_{3}}^{+2/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-1/3}{\phi_{5}}^{-1/3,\bar{b}\dagger} + 2\,{\phi_{3}}_{b}^{+2/3}{\phi_{3}}^{+2/3,\bar{b}\dagger}{\phi_{5}}_{b}^{+2/3}{\phi_{5}}^{+2/3,\bar{b}\dagger}\\
&+ 2\,{\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+5/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-4/3}{\phi_{5}}^{-4/3,\bar{r}\dagger} + {\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+5/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-1/3}{\phi_{5}}^{-1/3,\bar{r}\dagger}\\
&+ 2\,{\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+5/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-4/3}{\phi_{5}}^{-4/3,\bar{r}\dagger} + {\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+5/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-1/3}{\phi_{5}}^{-1/3,\bar{r}\dagger}\\
&+ 2\,{\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+5/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-4/3}{\phi_{5}}^{-4/3,\bar{r}\dagger} + {\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+5/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-1/3}{\phi_{5}}^{-1/3,\bar{r}\dagger}\\
&- \sqrt{2}\,{\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+2/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-4/3}{\phi_{5}}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+2/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-1/3}{\phi_{5}}^{+2/3,\bar{r}\dagger}\\
&- \sqrt{2}\,{\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+2/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-4/3}{\phi_{5}}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+2/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-1/3}{\phi_{5}}^{+2/3,\bar{r}\dagger}\\
&- \sqrt{2}\,{\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+2/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-4/3}{\phi_{5}}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_{3}}_{r}^{+5/3}{\phi_{3}}^{+2/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-1/3}{\phi_{5}}^{+2/3,\bar{r}\dagger}\\
&+ 2\,{\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+5/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-4/3}{\phi_{5}}^{-4/3,\bar{g}\dagger} + {\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+5/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-1/3}{\phi_{5}}^{-1/3,\bar{g}\dagger}\\
&+ 2\,{\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+5/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-4/3}{\phi_{5}}^{-4/3,\bar{g}\dagger} + {\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+5/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-1/3}{\phi_{5}}^{-1/3,\bar{g}\dagger}\\
&+ 2\,{\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+5/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-4/3}{\phi_{5}}^{-4/3,\bar{g}\dagger} + {\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+5/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-1/3}{\phi_{5}}^{-1/3,\bar{g}\dagger}\\
&- \sqrt{2}\,{\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+2/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-4/3}{\phi_{5}}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+2/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-1/3}{\phi_{5}}^{+2/3,\bar{g}\dagger}
\end{aligned}
$$

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\begin{aligned}
&-\sqrt{2}\,{\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+2/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-4/3}{\phi_{5}}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+2/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-1/3}{\phi_{5}}^{+2/3,\bar{g}\dagger}\\
&- \sqrt{2}\,{\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+2/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-4/3}{\phi_{5}}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_{3}}_{g}^{+5/3}{\phi_{3}}^{+2/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-1/3}{\phi_{5}}^{+2/3,\bar{g}\dagger}\\
&+ 2\,{\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+5/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-4/3}{\phi_{5}}^{-4/3,\bar{b}\dagger} + {\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+5/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-1/3}{\phi_{5}}^{-1/3,\bar{b}\dagger}\\
&+ 2\,{\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+5/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-4/3}{\phi_{5}}^{-4/3,\bar{b}\dagger} + {\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+5/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-1/3}{\phi_{5}}^{-1/3,\bar{b}\dagger}\\
&+ 2\,{\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+5/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-4/3}{\phi_{5}}^{-4/3,\bar{b}\dagger} + {\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+5/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-1/3}{\phi_{5}}^{-1/3,\bar{b}\dagger}\\
&- \sqrt{2}\,{\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+2/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-4/3}{\phi_{5}}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+2/3,\bar{r}\dagger}{\phi_{5}}_{r}^{-1/3}{\phi_{5}}^{+2/3,\bar{b}\dagger}\\
&- \sqrt{2}\,{\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+2/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-4/3}{\phi_{5}}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+2/3,\bar{g}\dagger}{\phi_{5}}_{g}^{-1/3}{\phi_{5}}^{+2/3,\bar{b}\dagger}\\
&- \sqrt{2}\,{\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+2/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-4/3}{\phi_{5}}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_{3}}_{b}^{+5/3}{\phi_{3}}^{+2/3,\bar{b}\dagger}{\phi_{5}}_{b}^{-1/3}{\phi_{5}}^{+2/3,\bar{b}\dagger}\Bigg\}
\end{aligned}
$$

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\begin{aligned}
&\lambda^{(1)}_{65,c1}\big(2\,{\phi_{4}}_{r}^{-1/3}{\phi_{4}}_{r}^{-1/3}{\phi_{4}}^{-1/3,\bar{r}\dagger}{\phi_{4}}^{-1/3,\bar{r}\dagger} + 4\,{\phi_{4}}_{r}^{-1/3}{\phi_{4}}_{g}^{-1/3}{\phi_{4}}^{-1/3,\bar{r}\dagger}{\phi_{4}}^{-1/3,\bar{g}\dagger}\\
&\quad + 4\,{\phi_{4}}_{r}^{-1/3}{\phi_{4}}_{b}^{-1/3}{\phi_{4}}^{-1/3,\bar{r}\dagger}{\phi_{4}}^{-1/3,\bar{b}\dagger} + 4\,{\phi_{4}}_{r}^{-1/3}{\phi_{4}}_{r}^{+2/3}{\phi_{4}}^{+2/3,\bar{r}\dagger}{\phi_{4}}^{-1/3,\bar{r}\dagger}\\
&\quad + 4\,{\phi_{4}}_{r}^{-1/3}{\phi_{4}}_{g}^{+2/3}{\phi_{4}}^{+2/3,\bar{g}\dagger}{\phi_{4}}^{-1/3,\bar{r}\dagger} + 4\,{\phi_{4}}_{r}^{-1/3}{\phi_{4}}_{b}^{+2/3}{\phi_{4}}^{+2/3,\bar{b}\dagger}{\phi_{4}}^{-1/3,\bar{r}\dagger}\\
&\quad + 2\,{\phi_{4}}_{g}^{-1/3}{\phi_{4}}_{g}^{-1/3}{\phi_{4}}^{-1/3,\bar{g}\dagger}{\phi_{4}}^{-1/3,\bar{g}\dagger} + 4\,{\phi_{4}}_{g}^{-1/3}{\phi_{4}}_{b}^{-1/3}{\phi_{4}}^{-1/3,\bar{g}\dagger}{\phi_{4}}^{-1/3,\bar{b}\dagger}\\
&\quad + 4\,{\phi_{4}}_{g}^{-1/3}{\phi_{4}}_{r}^{+2/3}{\phi_{4}}^{+2/3,\bar{r}\dagger}{\phi_{4}}^{-1/3,\bar{g}\dagger} + 4\,{\phi_{4}}_{g}^{-1/3}{\phi_{4}}_{g}^{+2/3}{\phi_{4}}^{+2/3,\bar{g}\dagger}{\phi_{4}}^{-1/3,\bar{g}\dagger}\\
&\quad + 4\,{\phi_{4}}_{g}^{-1/3}{\phi_{4}}_{b}^{+2/3}{\phi_{4}}^{+2/3,\bar{b}\dagger}{\phi_{4}}^{-1/3,\bar{g}\dagger} + 2\,{\phi_{4}}_{b}^{-1/3}{\phi_{4}}_{b}^{-1/3}{\phi_{4}}^{-1/3,\bar{b}\dagger}{\phi_{4}}^{-1/3,\bar{b}\dagger}\\
&\quad + 4\,{\phi_{4}}_{b}^{-1/3}{\phi_{4}}_{r}^{+2/3}{\phi_{4}}^{+2/3,\bar{r}\dagger}{\phi_{4}}^{-1/3,\bar{b}\dagger} + 4\,{\phi_{4}}_{b}^{-1/3}{\phi_{4}}_{g}^{+2/3}{\phi_{4}}^{+2/3,\bar{g}\dagger}{\phi_{4}}^{-1/3,\bar{b}\dagger}\\
&\quad + 4\,{\phi_{4}}_{b}^{-1/3}{\phi_{4}}_{b}^{+2/3}{\phi_{4}}^{+2/3,\bar{b}\dagger}{\phi_{4}}^{-1/3,\bar{b}\dagger} + 2\,{\phi_{4}}_{r}^{+2/3}{\phi_{4}}_{r}^{+2/3}{\phi_{4}}^{+2/3,\bar{r}\dagger}{\phi_{4}}^{+2/3,\bar{r}\dagger}\\
&\quad + 4\,{\phi_{4}}_{r}^{+2/3}{\phi_{4}}_{g}^{+2/3}{\phi_{4}}^{+2/3,\bar{r}\dagger}{\phi_{4}}^{+2/3,\bar{g}\dagger} + 4\,{\phi_{4}}_{r}^{+2/3}{\phi_{4}}_{b}^{+2/3}{\phi_{4}}^{+2/3,\bar{r}\dagger}{\phi_{4}}^{+2/3,\bar{b}\dagger}\\
&\quad + 2\,{\phi_{4}}_{g}^{+2/3}{\phi_{4}}_{g}^{+2/3}{\phi_{4}}^{+2/3,\bar{g}\dagger}{\phi_{4}}^{+2/3,\bar{g}\dagger} + 4\,{\phi_{4}}_{g}^{+2/3}{\phi_{4}}_{b}^{+2/3}{\phi_{4}}^{+2/3,\bar{g}\dagger}{\phi_{4}}^{+2/3,\bar{b}\dagger}\\
&\quad + 2\,{\phi_{4}}_{b}^{+2/3}{\phi_{4}}_{b}^{+2/3}{\phi_{4}}^{+2/3,\bar{b}\dagger}{\phi_{4}}^{+2/3,\bar{b}\dagger}\big)
\end{aligned}
$$

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\begin{aligned}
&\lambda^{(2)}_{65,c2}(2\,{\phi_4}_r^{-1/3}{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_4}^{-1/3,\bar{r}\dagger} + 4\,{\phi_4}_r^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_4}^{-1/3,\bar{g}\dagger} \\
&\quad + 4\,{\phi_4}_r^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_4}^{-1/3,\bar{b}\dagger} + 4\,{\phi_4}_r^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_4}^{-1/3,\bar{r}\dagger} \\
&\quad + 4\,{\phi_4}_r^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_4}^{-1/3,\bar{g}\dagger} + 4\,{\phi_4}_r^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_4}^{-1/3,\bar{b}\dagger} \\
&\quad + 2\,{\phi_4}_g^{-1/3}{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_4}^{-1/3,\bar{g}\dagger} + 4\,{\phi_4}_g^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_4}^{-1/3,\bar{b}\dagger} \\
&\quad + 4\,{\phi_4}_g^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_4}^{-1/3,\bar{r}\dagger} + 4\,{\phi_4}_g^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_4}^{-1/3,\bar{g}\dagger} \\
&\quad + 4\,{\phi_4}_g^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_4}^{-1/3,\bar{b}\dagger} + 2\,{\phi_4}_b^{-1/3}{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_4}^{-1/3,\bar{b}\dagger} \\
&\quad + 4\,{\phi_4}_b^{-1/3}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_4}^{-1/3,\bar{r}\dagger} + 4\,{\phi_4}_b^{-1/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_4}^{-1/3,\bar{g}\dagger} \\
&\quad + 4\,{\phi_4}_b^{-1/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_4}^{-1/3,\bar{b}\dagger} + 2\,{\phi_4}_r^{+2/3}{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_4}^{+2/3,\bar{r}\dagger} \\
&\quad + 4\,{\phi_4}_r^{+2/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_4}^{+2/3,\bar{g}\dagger} + 4\,{\phi_4}_r^{+2/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_4}^{+2/3,\bar{b}\dagger} \\
&\quad + 2\,{\phi_4}_g^{+2/3}{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_4}^{+2/3,\bar{g}\dagger} + 4\,{\phi_4}_g^{+2/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_4}^{+2/3,\bar{b}\dagger} \\
&\quad + 2\,{\phi_4}_b^{+2/3}{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_4}^{+2/3,\bar{b}\dagger})
\end{aligned}
$$

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\begin{aligned}
&\lambda^{(1)}_{67,c1}\Bigg\{\sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&+ \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&+ \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&+ {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&- {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} - {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&+ \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&+ \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&+ \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&+ {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&- {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} - {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&+ \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&+ \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}
\end{aligned}
$$

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\begin{aligned}
&+\sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&\quad + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&\quad + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&\quad - {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} - {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&\quad - {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&\quad - {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&\quad + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&\quad + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
&\quad + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&\quad + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&\quad - {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&\quad - {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&\quad + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&\quad + \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger}
\end{aligned}
$$

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\begin{aligned}
&+\sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&\quad + \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&\quad - {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&\quad - {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&\quad + {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} + {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&\quad + \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
&\quad + \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
&\quad + \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}\Bigg\}
\end{aligned}
$$

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\begin{aligned}
\lambda^{(2)}_{67,c2}\Big\{ & \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} - {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} - {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger}
\end{aligned}
$$

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\begin{aligned}
& +\sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} + \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} - {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& - {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} + \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& - {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& - {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger} + \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
&+\sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger}+\sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}\\
&\quad+\sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger}+\sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}\\
&\quad-{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger}+{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}\\
&\quad-{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger}+{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}\\
&\quad-{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger}+{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}\\
&\quad+\sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger}+\sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}\\
&\quad+\sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger}+\sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}\\
&\quad+\sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger}+\sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}\Bigg\}
\end{aligned}
$$

$$
\begin{aligned}
&\lambda^{(3)}_{67,c3}\Bigg\{-\sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger}-\sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger}\\
&-\sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger}-\sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}\\
&-\sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger}-\sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger}\\
&+{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}+{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger}\\
&+{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger}+2\,{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}\\
&+2\,{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}+2\,{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}\\
&-\sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger}-\sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger}\\
&-\sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger}-\sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}\\
&-\sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger}-\sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger}\\
&+{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}+{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger}\\
&+{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger}+2\,{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}\\
&+2\,{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}+2\,{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}\\
&-\sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger}-\sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger}\\
&-\sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger}-\sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
&-\sqrt{2}\,\phi_{4b}^{-1/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_{5g}^{+2/3}\phi_5^{-1/3,\bar{g}\dagger} - \sqrt{2}\,\phi_{4b}^{-1/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_{5b}^{+2/3}\phi_5^{-1/3,\bar{b}\dagger} \\
&+ \phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5r}^{-1/3}\phi_5^{-1/3,\bar{r}\dagger} + \phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5g}^{-1/3}\phi_5^{-1/3,\bar{g}\dagger} \\
&+ \phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5b}^{-1/3}\phi_5^{-1/3,\bar{b}\dagger} + 2\,\phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5r}^{+2/3}\phi_5^{+2/3,\bar{r}\dagger} \\
&+ 2\,\phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5g}^{+2/3}\phi_5^{+2/3,\bar{g}\dagger} + 2\,\phi_{4b}^{-1/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5b}^{+2/3}\phi_5^{+2/3,\bar{b}\dagger} \\
&+ 2\,\phi_{4r}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}\phi_{5r}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger} + 2\,\phi_{4r}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}\phi_{5g}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger} \\
&+ 2\,\phi_{4r}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}\phi_{5b}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger} + \phi_{4r}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}\phi_{5r}^{-1/3}\phi_5^{-1/3,\bar{r}\dagger} \\
&+ \phi_{4r}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}\phi_{5g}^{-1/3}\phi_5^{-1/3,\bar{g}\dagger} + \phi_{4r}^{+2/3}\phi_4^{+2/3,\bar{r}\dagger}\phi_{5b}^{-1/3}\phi_5^{-1/3,\bar{b}\dagger} \\
&- \sqrt{2}\,\phi_{4r}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}\phi_{5r}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger} - \sqrt{2}\,\phi_{4r}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}\phi_{5g}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger} \\
&- \sqrt{2}\,\phi_{4r}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}\phi_{5b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{4r}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}\phi_{5r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger} \\
&- \sqrt{2}\,\phi_{4r}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}\phi_{5g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger} - \sqrt{2}\,\phi_{4r}^{+2/3}\phi_4^{-1/3,\bar{r}\dagger}\phi_{5b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger} \\
&+ 2\,\phi_{4g}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}\phi_{5r}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger} + 2\,\phi_{4g}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}\phi_{5g}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger} \\
&+ 2\,\phi_{4g}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}\phi_{5b}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger} + \phi_{4g}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}\phi_{5r}^{-1/3}\phi_5^{-1/3,\bar{r}\dagger} \\
&+ \phi_{4g}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}\phi_{5g}^{-1/3}\phi_5^{-1/3,\bar{g}\dagger} + \phi_{4g}^{+2/3}\phi_4^{+2/3,\bar{g}\dagger}\phi_{5b}^{-1/3}\phi_5^{-1/3,\bar{b}\dagger} \\
&- \sqrt{2}\,\phi_{4g}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}\phi_{5r}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger} - \sqrt{2}\,\phi_{4g}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}\phi_{5g}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger}
\end{aligned}
$$

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\begin{aligned}
&-\sqrt{2}\,\phi_{4g}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}\phi_{5b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{4g}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}\phi_{5r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger} \\
&- \sqrt{2}\,\phi_{4g}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}\phi_{5g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger} - \sqrt{2}\,\phi_{4g}^{+2/3}\phi_4^{-1/3,\bar{g}\dagger}\phi_{5b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger} \\
&+ 2\,\phi_{4b}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_{5r}^{-4/3}\phi_5^{-4/3,\bar{r}\dagger} + 2\,\phi_{4b}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_{5g}^{-4/3}\phi_5^{-4/3,\bar{g}\dagger} \\
&+ 2\,\phi_{4b}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_{5b}^{-4/3}\phi_5^{-4/3,\bar{b}\dagger} + \phi_{4b}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_{5r}^{-1/3}\phi_5^{-1/3,\bar{r}\dagger} \\
&+ \phi_{4b}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_{5g}^{-1/3}\phi_5^{-1/3,\bar{g}\dagger} + \phi_{4b}^{+2/3}\phi_4^{+2/3,\bar{b}\dagger}\phi_{5b}^{-1/3}\phi_5^{-1/3,\bar{b}\dagger} \\
&- \sqrt{2}\,\phi_{4b}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5r}^{-4/3}\phi_5^{-1/3,\bar{r}\dagger} - \sqrt{2}\,\phi_{4b}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5g}^{-4/3}\phi_5^{-1/3,\bar{g}\dagger} \\
&- \sqrt{2}\,\phi_{4b}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5b}^{-4/3}\phi_5^{-1/3,\bar{b}\dagger} - \sqrt{2}\,\phi_{4b}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5r}^{-1/3}\phi_5^{+2/3,\bar{r}\dagger} \\
&- \sqrt{2}\,\phi_{4b}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5g}^{-1/3}\phi_5^{+2/3,\bar{g}\dagger} - \sqrt{2}\,\phi_{4b}^{+2/3}\phi_4^{-1/3,\bar{b}\dagger}\phi_{5b}^{-1/3}\phi_5^{+2/3,\bar{b}\dagger} \Bigg\}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(4)}_{67,c4}\Big\{ & -\sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& - \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& - \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_4}_r^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} + 2\,{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} + 2\,{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + {\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} + 2\,{\phi_4}_r^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& - \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& - \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_4}_g^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} + 2\,{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} + 2\,{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& + {\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} + 2\,{\phi_4}_g^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
& - \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& - \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger}
\end{aligned}
$$

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\begin{aligned}
& -\sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-4/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_4}_b^{-1/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} + 2\,{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} + 2\,{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + {\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} + 2\,{\phi_4}_b^{-1/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + 2\,{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + 2\,{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& + 2\,{\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger} + {\phi_4}_r^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& - \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& - \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{r}\dagger} - \sqrt{2}\,{\phi_4}_r^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + 2\,{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + 2\,{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + 2\,{\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger} + {\phi_4}_g^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& - \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}
\end{aligned}
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$$
\begin{aligned}
&-\sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
&- \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{g}\dagger} - \sqrt{2}\,{\phi_4}_g^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger} \\
&+ 2\,{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ 2\,{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ 2\,{\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger} + {\phi_4}_b^{+2/3}{\phi_4}^{+2/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&- \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{r}\dagger}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&- \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{g}\dagger}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger} \\
&- \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-4/3}{\phi_5}^{-1/3,\bar{b}\dagger} - \sqrt{2}\,{\phi_4}_b^{+2/3}{\phi_4}^{-1/3,\bar{b}\dagger}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger} \Bigg\}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(1)}_{71,c1}\Bigg\{ & 8\,{\phi_5}_r^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - 4\,{\phi_5}_r^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&+ 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} \\
&- 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
&+ 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} - 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ 4\,{\phi_5}_g^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} + 4\,{\phi_5}_g^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} \\
&- 4\,{\phi_5}_g^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} + 8\,{\phi_5}_g^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&- 4\,{\phi_5}_g^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} + 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
&+ 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ 4\,{\phi_5}_b^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} + 4\,{\phi_5}_b^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} \\
&- 4\,{\phi_5}_b^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} + 4\,{\phi_5}_b^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
&+ 4\,{\phi_5}_b^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - 4\,{\phi_5}_b^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&+ 8\,{\phi_5}_b^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - 4\,{\phi_5}_b^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&- 4\,{\phi_5}_r^{-1/3}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} + 2\,{\phi_5}_r^{-1/3}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
&- 4\,{\phi_5}_r^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} - 4\,{\phi_5}_r^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger}
\end{aligned}
$$

$$
\begin{aligned}
&+4\,{\phi_5}_r^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - 4\,{\phi_5}_r^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
&- 4\,{\phi_5}_r^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} + 4\,{\phi_5}_r^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
&- 4\,{\phi_5}_g^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} + 2\,{\phi_5}_g^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
&- 4\,{\phi_5}_g^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - 4\,{\phi_5}_g^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
&+ 4\,{\phi_5}_g^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - 4\,{\phi_5}_b^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
&+ 2\,{\phi_5}_b^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \Bigg\}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(2)}_{71,c2}\Big\{ & 2\,{\phi_5}_r^{-4/3}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\,\dagger}{\phi_5}^{-4/3,\bar{r}\,\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{r}\,\dagger}{\phi_5}^{-4/3,\bar{g}\,\dagger} \\
& + 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{r}\,\dagger}{\phi_5}^{-4/3,\bar{b}\,\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\,\dagger}{\phi_5}^{-4/3,\bar{r}\,\dagger} \\
& + 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\,\dagger}{\phi_5}^{-4/3,\bar{r}\,\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\,\dagger}{\phi_5}^{-4/3,\bar{r}\,\dagger} \\
& + 4\,{\phi_5}_r^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\,\dagger}{\phi_5}^{-4/3,\bar{r}\,\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\,\dagger}{\phi_5}^{-4/3,\bar{r}\,\dagger} \\
& + 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\,\dagger}{\phi_5}^{-4/3,\bar{r}\,\dagger} + 2\,{\phi_5}_g^{-4/3}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\,\dagger}{\phi_5}^{-4/3,\bar{g}\,\dagger} \\
& + 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{g}\,\dagger}{\phi_5}^{-4/3,\bar{b}\,\dagger} + 4\,{\phi_5}_g^{-4/3}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\,\dagger}{\phi_5}^{-4/3,\bar{g}\,\dagger} \\
& + 4\,{\phi_5}_g^{-4/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\,\dagger}{\phi_5}^{-4/3,\bar{g}\,\dagger} + 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\,\dagger}{\phi_5}^{-4/3,\bar{g}\,\dagger} \\
& + 4\,{\phi_5}_g^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\,\dagger}{\phi_5}^{-4/3,\bar{g}\,\dagger} + 4\,{\phi_5}_g^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\,\dagger}{\phi_5}^{-4/3,\bar{g}\,\dagger} \\
& + 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\,\dagger}{\phi_5}^{-4/3,\bar{g}\,\dagger} + 2\,{\phi_5}_b^{-4/3}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\,\dagger}{\phi_5}^{-4/3,\bar{b}\,\dagger} \\
& + 4\,{\phi_5}_b^{-4/3}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\,\dagger}{\phi_5}^{-4/3,\bar{b}\,\dagger} + 4\,{\phi_5}_b^{-4/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\,\dagger}{\phi_5}^{-4/3,\bar{b}\,\dagger} \\
& + 4\,{\phi_5}_b^{-4/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\,\dagger}{\phi_5}^{-4/3,\bar{b}\,\dagger} + 4\,{\phi_5}_b^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\,\dagger}{\phi_5}^{-4/3,\bar{b}\,\dagger} \\
& + 4\,{\phi_5}_b^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\,\dagger}{\phi_5}^{-4/3,\bar{b}\,\dagger} + 4\,{\phi_5}_b^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\,\dagger}{\phi_5}^{-4/3,\bar{b}\,\dagger} \\
& + 2\,{\phi_5}_r^{-1/3}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\,\dagger}{\phi_5}^{-1/3,\bar{r}\,\dagger} + 4\,{\phi_5}_r^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{r}\,\dagger}{\phi_5}^{-1/3,\bar{g}\,\dagger} \\
& + 4\,{\phi_5}_r^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{r}\,\dagger}{\phi_5}^{-1/3,\bar{b}\,\dagger} + 4\,{\phi_5}_r^{-1/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\,\dagger}{\phi_5}^{-1/3,\bar{r}\,\dagger}
\end{aligned}
$$

$$
\begin{aligned}
& +4\,{\phi_5}_r^{-1/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\,\dagger}{\phi_5}^{-1/3,\bar{r}\,\dagger} + 4\,{\phi_5}_r^{-1/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\,\dagger}{\phi_5}^{-1/3,\bar{r}\,\dagger} \\
& + 2\,{\phi_5}_g^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\,\dagger}{\phi_5}^{-1/3,\bar{g}\,\dagger} + 4\,{\phi_5}_g^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{g}\,\dagger}{\phi_5}^{-1/3,\bar{b}\,\dagger} \\
& + 4\,{\phi_5}_g^{-1/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\,\dagger}{\phi_5}^{-1/3,\bar{g}\,\dagger} + 4\,{\phi_5}_g^{-1/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\,\dagger}{\phi_5}^{-1/3,\bar{g}\,\dagger} \\
& + 4\,{\phi_5}_g^{-1/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\,\dagger}{\phi_5}^{-1/3,\bar{g}\,\dagger} + 2\,{\phi_5}_b^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\,\dagger}{\phi_5}^{-1/3,\bar{b}\,\dagger} \\
& + 4\,{\phi_5}_b^{-1/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\,\dagger}{\phi_5}^{-1/3,\bar{b}\,\dagger} + 4\,{\phi_5}_b^{-1/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\,\dagger}{\phi_5}^{-1/3,\bar{b}\,\dagger} \\
& + 4\,{\phi_5}_b^{-1/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\,\dagger}{\phi_5}^{-1/3,\bar{b}\,\dagger} + 2\,{\phi_5}_r^{+2/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\,\dagger}{\phi_5}^{+2/3,\bar{r}\,\dagger} \\
& + 4\,{\phi_5}_r^{+2/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{r}\,\dagger}{\phi_5}^{+2/3,\bar{g}\,\dagger} + 4\,{\phi_5}_r^{+2/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{r}\,\dagger}{\phi_5}^{+2/3,\bar{b}\,\dagger} \\
& + 2\,{\phi_5}_g^{+2/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\,\dagger}{\phi_5}^{+2/3,\bar{g}\,\dagger} + 4\,{\phi_5}_g^{+2/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{g}\,\dagger}{\phi_5}^{+2/3,\bar{b}\,\dagger} \\
& + 2\,{\phi_5}_b^{+2/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\,\dagger}{\phi_5}^{+2/3,\bar{b}\,\dagger}\Big\}
\end{aligned}
$$

$$
\begin{aligned}
\lambda^{(3)}_{71,c3}\Big\{ & 2\,{\phi_5}_r^{-4/3}{\phi_5}_r^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} \\
& + 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
& - 4\,{\phi_5}_r^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} \\
& - 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& - 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} + 4\,{\phi_5}_r^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} \\
& + 2\,{\phi_5}_g^{-4/3}{\phi_5}_g^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} + 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
& + 4\,{\phi_5}_g^{-4/3}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} + 4\,{\phi_5}_g^{-4/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - 4\,{\phi_5}_g^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_g^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - 4\,{\phi_5}_g^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_g^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} - 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_g^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} + 2\,{\phi_5}_b^{-4/3}{\phi_5}_b^{-4/3}{\phi_5}^{-4/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
& + 4\,{\phi_5}_b^{-4/3}{\phi_5}_r^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} + 4\,{\phi_5}_b^{-4/3}{\phi_5}_g^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_b^{-4/3}{\phi_5}_b^{-1/3}{\phi_5}^{-1/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} - 4\,{\phi_5}_b^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger}
\end{aligned}
$$

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\begin{aligned}
& +4\,{\phi_5}_b^{-4/3}{\phi_5}_r^{+2/3}{\phi_5}^{-1/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - 4\,{\phi_5}_b^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
& + 4\,{\phi_5}_b^{-4/3}{\phi_5}_g^{+2/3}{\phi_5}^{-1/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} - 4\,{\phi_5}_b^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
& + 4\,{\phi_5}_b^{-4/3}{\phi_5}_b^{+2/3}{\phi_5}^{-1/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} + 4\,{\phi_5}_r^{-1/3}{\phi_5}_r^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} \\
& + 4\,{\phi_5}_r^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} + 4\,{\phi_5}_r^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} \\
& + 4\,{\phi_5}_r^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} + 4\,{\phi_5}_r^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{r}\dagger} \\
& + 4\,{\phi_5}_r^{-1/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} + 4\,{\phi_5}_r^{-1/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_r^{-1/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} + 4\,{\phi_5}_g^{-1/3}{\phi_5}_g^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_g^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} + 4\,{\phi_5}_g^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_g^{-1/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} + 4\,{\phi_5}_g^{-1/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_g^{-1/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} + 4\,{\phi_5}_b^{-1/3}{\phi_5}_b^{-1/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-4/3,\bar{b}\dagger} \\
& + 4\,{\phi_5}_b^{-1/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{r}\dagger} + 4\,{\phi_5}_b^{-1/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{g}\dagger} \\
& + 4\,{\phi_5}_b^{-1/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{-1/3,\bar{b}\dagger} + 2\,{\phi_5}_r^{+2/3}{\phi_5}_r^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{r}\dagger} \\
& + 4\,{\phi_5}_r^{+2/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} + 4\,{\phi_5}_r^{+2/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{r}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger} \\
& + 2\,{\phi_5}_g^{+2/3}{\phi_5}_g^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{g}\dagger} + 4\,{\phi_5}_g^{+2/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{g}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}
\end{aligned}
$$

$$
+2\,{\phi_5}_b^{+2/3}{\phi_5}_b^{+2/3}{\phi_5}^{+2/3,\bar{b}\dagger}{\phi_5}^{+2/3,\bar{b}\dagger}\Big\}
$$

3.3 Gauge and Kinetic Terms

$$-\frac{1}{4}B_{\mu\nu}B^{\mu\nu} + -\frac{1}{4}W^{a}_{\mu\nu}W^{a,\mu\nu} + -\frac{1}{4}G^{A}_{\mu\nu}G^{A,\mu\nu}$$

$$\frac{1}{6}\,g_1\,B_\mu\,q^{+2/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{+2/3} + \frac{1}{6}\,g_1\,B_\mu\,q^{+2/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{+2/3} + \frac{1}{6}\,g_1\,B_\mu\,q^{+2/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_b^{+2/3}$$

$$\frac{1}{6}\,g_1\,B_\mu\,q^{-1/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{-1/3} + \frac{1}{6}\,g_1\,B_\mu\,q^{-1/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{-1/3} + \frac{1}{6}\,g_1\,B_\mu\,q^{-1/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_b^{-1/3}$$

$$\frac{1}{2}\,g_2\,W^1_\mu\,q^{+2/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{-1/3} + \frac{1}{2}\,g_2\,W^1_\mu\,q^{+2/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{-1/3} + \frac{1}{2}\,g_2\,W^1_\mu\,q^{+2/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_b^{-1/3}$$

$$\frac{1}{2}\,g_2\,W^1_\mu\,q^{-1/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{+2/3} + \frac{1}{2}\,g_2\,W^1_\mu\,q^{-1/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{+2/3} + \frac{1}{2}\,g_2\,W^1_\mu\,q^{-1/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_b^{+2/3}$$

$$\left(-\frac{i}{2}\right)\,g_2\,W^2_\mu\,q^{+2/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{-1/3} + \left(-\frac{i}{2}\right)\,g_2\,W^2_\mu\,q^{+2/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{-1/3} + \left(-\frac{i}{2}\right)\,g_2\,W^2_\mu\,q^{+2/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_b^{-1/3}$$

$$\frac{i}{2}\,g_2\,W^2_\mu\,q^{-1/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{+2/3} + \frac{i}{2}\,g_2\,W^2_\mu\,q^{-1/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{+2/3} + \frac{i}{2}\,g_2\,W^2_\mu\,q^{-1/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_b^{+2/3}$$

$$\frac{1}{2}\,g_2\,W^3_\mu\,q^{+2/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{+2/3} + \frac{1}{2}\,g_2\,W^3_\mu\,q^{+2/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{+2/3} + \frac{1}{2}\,g_2\,W^3_\mu\,q^{+2/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_b^{+2/3}$$

$$\left(-\frac{1}{2}\right)\,g_2\,W^3_\mu\,q^{-1/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{-1/3} + \left(-\frac{1}{2}\right)\,g_2\,W^3_\mu\,q^{-1/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{-1/3} + \left(-\frac{1}{2}\right)\,g_2\,W^3_\mu\,q^{-1/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_b^{-1/3}$$

$$\frac{1}{2}\,g_3\,G^1_\mu\,q^{+2/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_g^{+2/3} + \frac{1}{2}\,g_3\,G^1_\mu\,q^{-1/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_g^{-1/3} + \frac{1}{2}\,g_3\,G^1_\mu\,q^{+2/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_r^{+2/3}$$

$$\frac{1}{2}\,g_3\,G^1_\mu\,q^{-1/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_r^{-1/3} + \left(-\frac{i}{2}\right)\,g_3\,G^2_\mu\,q^{+2/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_g^{+2/3} + \left(-\frac{i}{2}\right)\,g_3\,G^2_\mu\,q^{-1/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_g^{-1/3}$$

$$\frac{i}{2}\,g_3\,G^2_\mu\,q^{+2/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_r^{+2/3} + \frac{i}{2}\,g_3\,G^2_\mu\,q^{-1/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_r^{-1/3} + \frac{1}{2}\,g_3\,G^3_\mu\,q^{+2/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{+2/3}$$

$$\frac{1}{2}\,g_3\,G^3_\mu\,q^{-1/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_r^{-1/3} + \left(-\frac{1}{2}\right)\,g_3\,G^3_\mu\,q^{+2/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{+2/3} + \left(-\frac{1}{2}\right)\,g_3\,G^3_\mu\,q^{-1/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_g^{-1/3}$$

$$\frac{1}{2}\,g_3\,G^4_\mu\,q^{+2/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_b^{+2/3} + \frac{1}{2}\,g_3\,G^4_\mu\,q^{-1/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_b^{-1/3} + \frac{1}{2}\,g_3\,G^4_\mu\,q^{+2/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_r^{+2/3}$$

$$\frac{1}{2}\,g_3\,G^4_\mu\,q^{-1/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_r^{-1/3} + \left(-\frac{i}{2}\right)\,g_3\,G^5_\mu\,q^{+2/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_b^{+2/3} + \left(-\frac{i}{2}\right)\,g_3\,G^5_\mu\,q^{-1/3,\bar{r}\dagger}\,\bar{\sigma}^\mu\,q_b^{-1/3}$$

$$\frac{i}{2}\,g_3\,G^5_\mu\,q^{+2/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_r^{+2/3} + \frac{i}{2}\,g_3\,G^5_\mu\,q^{-1/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_r^{-1/3} + \frac{1}{2}\,g_3\,G^6_\mu\,q^{+2/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_b^{+2/3}$$

$$\frac{1}{2}\,g_3\,G^6_\mu\,q^{-1/3,\bar{g}\dagger}\,\bar{\sigma}^\mu\,q_b^{-1/3} + \frac{1}{2}\,g_3\,G^6_\mu\,q^{+2/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_g^{+2/3} + \frac{1}{2}\,g_3\,G^6_\mu\,q^{-1/3,\bar{b}\dagger}\,\bar{\sigma}^\mu\,q_g^{-1/3}$$

$$\left(-\frac{i}{2}\right) g_3\, G^7_\mu\, q^{+2/3,\bar{g}\,\dagger}\, \bar\sigma^\mu\, q^{+2/3}_b + \left(-\frac{i}{2}\right) g_3\, G^7_\mu\, q^{-1/3,\bar{g}\,\dagger}\, \bar\sigma^\mu\, q^{-1/3}_b + \frac{i}{2}\, g_3\, G^7_\mu\, q^{+2/3,\bar{b}\,\dagger}\, \bar\sigma^\mu\, q^{+2/3}_g$$

$$\frac{i}{2}\, g_3\, G^7_\mu\, q^{-1/3,\bar{b}\,\dagger}\, \bar\sigma^\mu\, q^{-1/3}_g + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, q^{+2/3,\bar{r}\,\dagger}\, \bar\sigma^\mu\, q^{+2/3}_r + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, q^{-1/3,\bar{r}\,\dagger}\, \bar\sigma^\mu\, q^{-1/3}_r$$

$$\frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, q^{+2/3,\bar{g}\,\dagger}\, \bar\sigma^\mu\, q^{+2/3}_g + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, q^{-1/3,\bar{g}\,\dagger}\, \bar\sigma^\mu\, q^{-1/3}_g + \left(-\frac{\sqrt{3}}{3}\right) g_3\, G^8_\mu\, q^{+2/3,\bar{b}\,\dagger}\, \bar\sigma^\mu\, q^{+2/3}_b$$

$$\left(-\frac{\sqrt{3}}{3}\right) g_3\, G^8_\mu\, q^{-1/3,\bar{b}\,\dagger}\, \bar\sigma^\mu\, q^{-1/3}_b + \left(-\frac{1}{2}\right) g_1\, B_\mu\, l^{0\,\dagger}\, \bar\sigma^\mu\, l^0 + \left(-\frac{1}{2}\right) g_1\, B_\mu\, l^{-\,\dagger}\, \bar\sigma^\mu\, l^-$$

$$\frac{1}{2}\, g_2\, W^1_\mu\, l^{0\,\dagger}\, \bar\sigma^\mu\, l^- + \frac{1}{2}\, g_2\, W^1_\mu\, l^{-\,\dagger}\, \bar\sigma^\mu\, l^0 + \left(-\frac{i}{2}\right) g_2\, W^2_\mu\, l^{0\,\dagger}\, \bar\sigma^\mu\, l^-$$

$$\frac{i}{2}\, g_2\, W^2_\mu\, l^{-\,\dagger}\, \bar\sigma^\mu\, l^0 + \frac{1}{2}\, g_2\, W^3_\mu\, l^{0\,\dagger}\, \bar\sigma^\mu\, l^0 + \left(-\frac{1}{2}\right) g_2\, W^3_\mu\, l^{-\,\dagger}\, \bar\sigma^\mu\, l^-$$

$$\frac{1}{3}\, g_1\, B_\mu\, {d^C}^{+1/3\,\dagger}_r\, \bar\sigma^\mu\, d^{C+1/3,\bar{r}} + \frac{1}{3}\, g_1\, B_\mu\, {d^C}^{+1/3\,\dagger}_g\, \bar\sigma^\mu\, d^{C+1/3,\bar{g}} + \frac{1}{3}\, g_1\, B_\mu\, {d^C}^{+1/3\,\dagger}_b\, \bar\sigma^\mu\, d^{C+1/3,\bar{b}}$$

$$\left(-\frac{1}{2}\right) g_3\, G^1_\mu\, {d^C}^{+1/3\,\dagger}_r\, \bar\sigma^\mu\, d^{C+1/3,\bar{g}} + \left(-\frac{1}{2}\right) g_3\, G^1_\mu\, {d^C}^{+1/3\,\dagger}_g\, \bar\sigma^\mu\, d^{C+1/3,\bar{r}} + \left(-\frac{i}{2}\right) g_3\, G^2_\mu\, {d^C}^{+1/3\,\dagger}_r\, \bar\sigma^\mu\, d^{C+1/3,\bar{g}}$$

$$\frac{i}{2}\, g_3\, G^2_\mu\, {d^C}^{+1/3\,\dagger}_g\, \bar\sigma^\mu\, d^{C+1/3,\bar{r}} + \left(-\frac{1}{2}\right) g_3\, G^3_\mu\, {d^C}^{+1/3\,\dagger}_r\, \bar\sigma^\mu\, d^{C+1/3,\bar{r}} + \frac{1}{2}\, g_3\, G^3_\mu\, {d^C}^{+1/3\,\dagger}_g\, \bar\sigma^\mu\, d^{C+1/3,\bar{g}}$$

$$\left(-\frac{1}{2}\right) g_3\, G^4_\mu\, {d^C}^{+1/3\,\dagger}_r\, \bar\sigma^\mu\, d^{C+1/3,\bar{b}} + \left(-\frac{1}{2}\right) g_3\, G^4_\mu\, {d^C}^{+1/3\,\dagger}_b\, \bar\sigma^\mu\, d^{C+1/3,\bar{r}} + \left(-\frac{i}{2}\right) g_3\, G^5_\mu\, {d^C}^{+1/3\,\dagger}_r\, \bar\sigma^\mu\, d^{C+1/3,\bar{b}}$$

$$\frac{i}{2}\, g_3\, G^5_\mu\, {d^C}^{+1/3\,\dagger}_b\, \bar\sigma^\mu\, d^{C+1/3,\bar{r}} + \left(-\frac{1}{2}\right) g_3\, G^6_\mu\, {d^C}^{+1/3\,\dagger}_g\, \bar\sigma^\mu\, d^{C+1/3,\bar{b}} + \left(-\frac{1}{2}\right) g_3\, G^6_\mu\, {d^C}^{+1/3\,\dagger}_b\, \bar\sigma^\mu\, d^{C+1/3,\bar{g}}$$

$$\left(-\frac{i}{2}\right) g_3\, G^7_\mu\, {d^C}^{+1/3\,\dagger}_g\, \bar\sigma^\mu\, d^{C+1/3,\bar{b}} + \frac{i}{2}\, g_3\, G^7_\mu\, {d^C}^{+1/3\,\dagger}_b\, \bar\sigma^\mu\, d^{C+1/3,\bar{g}} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, G^8_\mu\, {d^C}^{+1/3\,\dagger}_r\, \bar\sigma^\mu\, d^{C+1/3,\bar{r}}$$

$$\left(-\frac{\sqrt{3}}{6}\right) g_3\, G^8_\mu\, {d^C}^{+1/3\,\dagger}_g\, \bar\sigma^\mu\, d^{C+1/3,\bar{g}} + \frac{\sqrt{3}}{3}\, g_3\, G^8_\mu\, {d^C}^{+1/3\,\dagger}_b\, \bar\sigma^\mu\, d^{C+1/3,\bar{b}} + \left(-\frac{2}{3}\right) g_1\, B_\mu\, {u^C}^{-2/3\,\dagger}_r\, \bar\sigma^\mu\, u^{C-2/3,\bar{r}}$$

$$\left(-\frac{2}{3}\right) g_1\, B_\mu\, {u^C}^{-2/3\,\dagger}_g\, \bar\sigma^\mu\, u^{C-2/3,\bar{g}} + \left(-\frac{2}{3}\right) g_1\, B_\mu\, {u^C}^{-2/3\,\dagger}_b\, \bar\sigma^\mu\, u^{C-2/3,\bar{b}} + \left(-\frac{1}{2}\right) g_3\, G^1_\mu\, {u^C}^{-2/3\,\dagger}_r\, \bar\sigma^\mu\, u^{C-2/3,\bar{g}}$$

$$\left(-\frac{1}{2}\right) g_3\, G^1_\mu\, {u^C}^{-2/3\,\dagger}_g\, \bar\sigma^\mu\, u^{C-2/3,\bar{r}} + \left(-\frac{i}{2}\right) g_3\, G^2_\mu\, {u^C}^{-2/3\,\dagger}_r\, \bar\sigma^\mu\, u^{C-2/3,\bar{g}} + \frac{i}{2}\, g_3\, G^2_\mu\, {u^C}^{-2/3\,\dagger}_g\, \bar\sigma^\mu\, u^{C-2/3,\bar{r}}$$

$$\left(-\frac{1}{2}\right) g_3\, G^3_\mu\, {u^C}^{-2/3\,\dagger}_r\, \bar\sigma^\mu\, u^{C-2/3,\bar{r}} + \frac{1}{2}\, g_3\, G^3_\mu\, {u^C}^{-2/3\,\dagger}_g\, \bar\sigma^\mu\, u^{C-2/3,\bar{g}} + \left(-\frac{1}{2}\right) g_3\, G^4_\mu\, {u^C}^{-2/3\,\dagger}_r\, \bar\sigma^\mu\, u^{C-2/3,\bar{b}}$$

$$\left(-\frac{1}{2}\right)\, g_3\, G^4_\mu\, u^{C_b^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{r}}} + \left(-\frac{i}{2}\right)\, g_3\, G^5_\mu\, u^{C_r^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{b}}} + \frac{i}{2}\, g_3\, G^5_\mu\, u^{C_b^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{r}}}$$

$$\left(-\frac{1}{2}\right)\, g_3\, G^6_\mu\, u^{C_g^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{b}}} + \left(-\frac{1}{2}\right)\, g_3\, G^6_\mu\, u^{C_b^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{g}}} + \left(-\frac{i}{2}\right)\, g_3\, G^7_\mu\, u^{C_g^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{b}}}$$

$$\frac{i}{2}\, g_3\, G^7_\mu\, u^{C_b^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{g}}} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, G^8_\mu\, u^{C_r^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{r}}} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, G^8_\mu\, u^{C_g^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{g}}}$$

$$\frac{\sqrt{3}}{3}\, g_3\, G^8_\mu\, u^{C_b^{-2/3}\dagger}\, \bar{\sigma}^\mu\, u^{C^{-2/3,\bar{b}}} + g_1\, B_\mu\, e^{C^{+}\dagger}\, \bar{\sigma}^\mu\, e^{C^{+}} + \frac{1}{2}\, g_1\, B_\mu\, H^{+\dagger}\, i\overleftrightarrow{\partial^\mu}\, H^{+}$$

$$\frac{1}{2}\, g_1\, B_\mu\, H^{0\dagger}\, i\overleftrightarrow{\partial^\mu}\, H^{0} + \frac{1}{2}\, g_2\, W^1_\mu\, H^{+\dagger}\, i\overleftrightarrow{\partial^\mu}\, H^{0} + \frac{1}{2}\, g_2\, W^1_\mu\, H^{0\dagger}\, i\overleftrightarrow{\partial^\mu}\, H^{+}$$

$$\left(-\frac{i}{2}\right)\, g_2\, W^2_\mu\, H^{+\dagger}\, i\overleftrightarrow{\partial^\mu}\, H^{0} + \frac{i}{2}\, g_2\, W^2_\mu\, H^{0\dagger}\, i\overleftrightarrow{\partial^\mu}\, H^{+} + \frac{1}{2}\, g_2\, W^3_\mu\, H^{+\dagger}\, i\overleftrightarrow{\partial^\mu}\, H^{+}$$

$$\left(-\frac{1}{2}\right)\, g_2\, W^3_\mu\, H^{0\dagger}\, i\overleftrightarrow{\partial^\mu}\, H^{0} + \left(-\frac{1}{3}\right)\, g_1\, B_\mu\, \phi_1{}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_r^{-1/3} + \left(-\frac{1}{3}\right)\, g_1\, B_\mu\, \phi_1{}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_g^{-1/3}$$

$$\left(-\frac{1}{3}\right)\, g_1\, B_\mu\, \phi_1{}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_b^{-1/3} + \frac{1}{2}\, g_3\, G^1_\mu\, \phi_1{}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_g^{-1/3} + \frac{1}{2}\, g_3\, G^1_\mu\, \phi_1{}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_r^{-1/3}$$

$$\left(-\frac{i}{2}\right)\, g_3\, G^2_\mu\, \phi_1{}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_g^{-1/3} + \frac{i}{2}\, g_3\, G^2_\mu\, \phi_1{}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_r^{-1/3} + \frac{1}{2}\, g_3\, G^3_\mu\, \phi_1{}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_r^{-1/3}$$

$$\left(-\frac{1}{2}\right)\, g_3\, G^3_\mu\, \phi_1{}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_g^{-1/3} + \frac{1}{2}\, g_3\, G^4_\mu\, \phi_1{}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_b^{-1/3} + \frac{1}{2}\, g_3\, G^4_\mu\, \phi_1{}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_r^{-1/3}$$

$$\left(-\frac{i}{2}\right)\, g_3\, G^5_\mu\, \phi_1{}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_b^{-1/3} + \frac{i}{2}\, g_3\, G^5_\mu\, \phi_1{}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_r^{-1/3} + \frac{1}{2}\, g_3\, G^6_\mu\, \phi_1{}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_b^{-1/3}$$

$$\frac{1}{2}\, g_3\, G^6_\mu\, \phi_1{}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_g^{-1/3} + \left(-\frac{i}{2}\right)\, g_3\, G^7_\mu\, \phi_1{}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_b^{-1/3} + \frac{i}{2}\, g_3\, G^7_\mu\, \phi_1{}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_g^{-1/3}$$

$$\frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, \phi_1{}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_r^{-1/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, \phi_1{}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_g^{-1/3} + \left(-\frac{\sqrt{3}}{3}\right)\, g_3\, G^8_\mu\, \phi_1{}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_1}_b^{-1/3}$$

$$\left(-\frac{4}{3}\right)\, g_1\, B_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_r^{-4/3} + \left(-\frac{4}{3}\right)\, g_1\, B_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_g^{-4/3} + \left(-\frac{4}{3}\right)\, g_1\, B_\mu\, \phi_2{}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_b^{-4/3}$$

$$\frac{1}{2}\, g_3\, G^1_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_g^{-4/3} + \frac{1}{2}\, g_3\, G^1_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_r^{-4/3} + \left(-\frac{i}{2}\right)\, g_3\, G^2_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_g^{-4/3}$$

$$\frac{i}{2}\, g_3\, G^2_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_r^{-4/3} + \frac{1}{2}\, g_3\, G^3_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_r^{-4/3} + \left(-\frac{1}{2}\right)\, g_3\, G^3_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_g^{-4/3}$$

$$\frac{1}{2}\, g_3\, G^4_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_b^{-4/3} + \frac{1}{2}\, g_3\, G^4_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_r^{-4/3} + \left(-\frac{i}{2}\right)\, g_3\, G^5_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_b^{-4/3}$$

$$\frac{i}{2}\, g_3\, G^5_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_r^{-4/3} + \frac{1}{2}\, g_3\, G^6_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_b^{-4/3} + \frac{1}{2}\, g_3\, G^6_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_g^{-4/3}$$

$$\left(-\frac{i}{2}\right)\, g_3\, G^7_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_b^{-4/3} + \frac{i}{2}\, g_3\, G^7_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_g^{-4/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_r^{-4/3}$$

$$\frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_g^{-4/3} + \left(-\frac{\sqrt{3}}{3}\right)\, g_3\, G^8_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_2}_b^{-4/3} + \frac{7}{6}\, g_1\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3}$$

$$\frac{7}{6}\, g_1\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3} + \frac{7}{6}\, g_1\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+5/3} + \frac{7}{6}\, g_1\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3}$$

$$\frac{7}{6}\, g_1\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3} + \frac{7}{6}\, g_1\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+2/3} + \frac{1}{2}\, g_2\, W^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3}$$

$$\frac{1}{2}\, g_2\, W^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3} + \frac{1}{2}\, g_2\, W^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+2/3} + \frac{1}{2}\, g_2\, W^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3}$$

$$\frac{1}{2}\, g_2\, W^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3} + \frac{1}{2}\, g_2\, W^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+5/3} + \left(-\frac{i}{2}\right)\, g_2\, W^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{i}{2}\right)\, g_2\, W^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3} + \left(-\frac{i}{2}\right)\, g_2\, W^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+2/3} + \frac{i}{2}\, g_2\, W^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3}$$

$$\frac{i}{2}\, g_2\, W^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3} + \frac{i}{2}\, g_2\, W^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+5/3} + \frac{1}{2}\, g_2\, W^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3}$$

$$\frac{1}{2}\, g_2\, W^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3} + \frac{1}{2}\, g_2\, W^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+5/3} + \left(-\frac{1}{2}\right)\, g_2\, W^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{1}{2}\right)\, g_2\, W^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3} + \left(-\frac{1}{2}\right)\, g_2\, W^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+2/3} + \frac{1}{2}\, g_3\, G^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3}$$

$$\frac{1}{2}\, g_3\, G^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3} + \frac{1}{2}\, g_3\, G^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3} + \frac{1}{2}\, g_3\, G^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{i}{2}\right)\, g_3\, G^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3} + \left(-\frac{i}{2}\right)\, g_3\, G^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3} + \frac{i}{2}\, g_3\, G^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3}$$

$$\frac{i}{2}\, g_3\, G^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3} + \frac{1}{2}\, g_3\, G^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3} + \frac{1}{2}\, g_3\, G^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{1}{2}\right)\, g_3\, G^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3} + \left(-\frac{1}{2}\right)\, g_3\, G^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3} + \frac{1}{2}\, g_3\, G^4_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+5/3}$$

$$\frac{1}{2}\, g_3\, G^4_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+2/3} + \frac{1}{2}\, g_3\, G^4_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3} + \frac{1}{2}\, g_3\, G^4_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{i}{2}\right) g_3\, G^5_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+5/3} + \left(-\frac{i}{2}\right) g_3\, G^5_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+2/3} + \frac{i}{2}\, g_3\, G^5_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3}$$

$$\frac{i}{2}\, g_3\, G^5_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3} + \frac{1}{2}\, g_3\, G^6_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+5/3} + \frac{1}{2}\, g_3\, G^6_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+2/3}$$

$$\frac{1}{2}\, g_3\, G^6_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3} + \frac{1}{2}\, g_3\, G^6_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3} + \left(-\frac{i}{2}\right) g_3\, G^7_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+5/3}$$

$$\left(-\frac{i}{2}\right) g_3\, G^7_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+2/3} + \frac{i}{2}\, g_3\, G^7_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3} + \frac{i}{2}\, g_3\, G^7_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3}$$

$$\frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+5/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_r^{+2/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+5/3}$$

$$\frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_g^{+2/3} + \left(-\frac{\sqrt{3}}{3}\right) g_3\, G^8_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+5/3} + \left(-\frac{\sqrt{3}}{3}\right) g_3\, G^8_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_3}_b^{+2/3}$$

$$\frac{1}{6}\, g_1\, B_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3} + \frac{1}{6}\, g_1\, B_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3} + \frac{1}{6}\, g_1\, B_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{+2/3}$$

$$\frac{1}{6}\, g_1\, B_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3} + \frac{1}{6}\, g_1\, B_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3} + \frac{1}{6}\, g_1\, B_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{-1/3}$$

$$\frac{1}{2}\, g_2\, W^1_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3} + \frac{1}{2}\, g_2\, W^1_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3} + \frac{1}{2}\, g_2\, W^1_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{-1/3}$$

$$\frac{1}{2}\, g_2\, W^1_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3} + \frac{1}{2}\, g_2\, W^1_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3} + \frac{1}{2}\, g_2\, W^1_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{+2/3}$$

$$\left(-\frac{i}{2}\right) g_2\, W^2_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{2}\right) g_2\, W^2_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3} + \left(-\frac{i}{2}\right) g_2\, W^2_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{-1/3}$$

$$\frac{i}{2}\, g_2\, W^2_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3} + \frac{i}{2}\, g_2\, W^2_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3} + \frac{i}{2}\, g_2\, W^2_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{+2/3}$$

$$\frac{1}{2}\, g_2\, W^3_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3} + \frac{1}{2}\, g_2\, W^3_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3} + \frac{1}{2}\, g_2\, W^3_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{+2/3}$$

$$\left(-\frac{1}{2}\right) g_2\, W^3_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3} + \left(-\frac{1}{2}\right) g_2\, W^3_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3} + \left(-\frac{1}{2}\right) g_2\, W^3_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{-1/3}$$

$$\frac{1}{2}\, g_3\, G^1_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3} + \frac{1}{2}\, g_3\, G^1_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3} + \frac{1}{2}\, g_3\, G^1_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{2}\, g_3\, G^1_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{2}\right) g_3\, G^2_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{2}\right) g_3\, G^2_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3}$$

$$\frac{i}{2}\, g_3\, G^2_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3} + \frac{i}{2}\, g_3\, G^2_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3} + \frac{1}{2}\, g_3\, G^3_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{2}\, g_3\, G^3_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3} + \left(-\frac{1}{2}\right)\, g_3\, G^3_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3} + \left(-\frac{1}{2}\right)\, g_3\, G^3_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{2}\, g_3\, G^4_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{+2/3} + \frac{1}{2}\, g_3\, G^4_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{-1/3} + \frac{1}{2}\, g_3\, G^4_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{2}\, g_3\, G^4_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{2}\right)\, g_3\, G^5_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{+2/3} + \left(-\frac{i}{2}\right)\, g_3\, G^5_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{-1/3}$$

$$\frac{i}{2}\, g_3\, G^5_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3} + \frac{i}{2}\, g_3\, G^5_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3} + \frac{1}{2}\, g_3\, G^6_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{+2/3}$$

$$\frac{1}{2}\, g_3\, G^6_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{-1/3} + \frac{1}{2}\, g_3\, G^6_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3} + \frac{1}{2}\, g_3\, G^6_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3}$$

$$\left(-\frac{i}{2}\right)\, g_3\, G^7_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{+2/3} + \left(-\frac{i}{2}\right)\, g_3\, G^7_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{-1/3} + \frac{i}{2}\, g_3\, G^7_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3}$$

$$\frac{i}{2}\, g_3\, G^7_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{+2/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_r^{-1/3}$$

$$\frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{+2/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_g^{-1/3} + \left(-\frac{\sqrt{3}}{3}\right)\, g_3\, G^8_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{+2/3}$$

$$\left(-\frac{\sqrt{3}}{3}\right)\, g_3\, G^8_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_4}_b^{-1/3} + \left(-\frac{1}{3}\right)\, g_1\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{+2/3} + \left(-\frac{1}{3}\right)\, g_1\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{1}{3}\right)\, g_1\, B_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{+2/3} + \left(-\frac{1}{3}\right)\, g_1\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3} + \left(-\frac{1}{3}\right)\, g_1\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{1}{3}\right)\, g_1\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-1/3} + \left(-\frac{1}{3}\right)\, g_1\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-4/3} + \left(-\frac{1}{3}\right)\, g_1\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{1}{3}\right)\, g_1\, B_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{+2/3}$$

$$\frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{2}}{2}\, g_2\, W^1_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}i}{2}\right)\, g_2\, W^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{2}i}{2}\right)\, g_2\, W^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{2}\right)\, g_2\, W_\mu^2\, {\phi_5}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{2}\, g_2\, W_\mu^2\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}i}{2}\, g_2\, W_\mu^2\, {\phi_5}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{+2/3}$$

$$\frac{\sqrt{2}i}{2}\, g_2\, W_\mu^2\, {\phi_5}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{2}i}{2}\right)\, g_2\, W_\mu^2\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-4/3} + \left(-\frac{\sqrt{2}i}{2}\right)\, g_2\, W_\mu^2\, {\phi_5}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{\sqrt{2}i}{2}\right)\, g_2\, W_\mu^2\, {\phi_5}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}i}{2}\, g_2\, W_\mu^2\, {\phi_5}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}i}{2}\, g_2\, W_\mu^2\, {\phi_5}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{2}i}{2}\, g_2\, W_\mu^2\, {\phi_5}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-1/3} + g_2\, W_\mu^3\, {\phi_5}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{+2/3} + g_2\, W_\mu^3\, {\phi_5}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{+2/3}$$

$$g_2\, W_\mu^3\, {\phi_5}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{+2/3} + -\, g_2\, W_\mu^3\, {\phi_5}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-4/3} + -\, g_2\, W_\mu^3\, {\phi_5}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-4/3}$$

$$-\, g_2\, W_\mu^3\, {\phi_5}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-4/3} + \frac{1}{2}\, g_3\, G_\mu^1\, {\phi_5}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{+2/3} + \frac{1}{2}\, g_3\, G_\mu^1\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-1/3}$$

$$\frac{1}{2}\, g_3\, G_\mu^1\, {\phi_5}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-4/3} + \frac{1}{2}\, g_3\, G_\mu^1\, {\phi_5}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{+2/3} + \frac{1}{2}\, g_3\, G_\mu^1\, {\phi_5}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3}$$

$$\frac{1}{2}\, g_3\, G_\mu^1\, {\phi_5}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-4/3} + \left(-\frac{i}{2}\right)\, g_3\, G_\mu^2\, {\phi_5}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{+2/3} + \left(-\frac{i}{2}\right)\, g_3\, G_\mu^2\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{i}{2}\right)\, g_3\, G_\mu^2\, {\phi_5}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-4/3} + \frac{i}{2}\, g_3\, G_\mu^2\, {\phi_5}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{+2/3} + \frac{i}{2}\, g_3\, G_\mu^2\, {\phi_5}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3}$$

$$\frac{i}{2}\, g_3\, G_\mu^2\, {\phi_5}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-4/3} + \frac{1}{2}\, g_3\, G_\mu^3\, {\phi_5}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{+2/3} + \frac{1}{2}\, g_3\, G_\mu^3\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3}$$

$$\frac{1}{2}\, g_3\, G_\mu^3\, {\phi_5}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-4/3} + \left(-\frac{1}{2}\right)\, g_3\, G_\mu^3\, {\phi_5}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{2}\right)\, g_3\, G_\mu^3\, {\phi_5}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{1}{2}\right)\, g_3\, G_\mu^3\, {\phi_5}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_g^{-4/3} + \frac{1}{2}\, g_3\, G_\mu^4\, {\phi_5}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{+2/3} + \frac{1}{2}\, g_3\, G_\mu^4\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-1/3}$$

$$\frac{1}{2}\, g_3\, G_\mu^4\, {\phi_5}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-4/3} + \frac{1}{2}\, g_3\, G_\mu^4\, {\phi_5}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{+2/3} + \frac{1}{2}\, g_3\, G_\mu^4\, {\phi_5}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3}$$

$$\frac{1}{2}\, g_3\, G_\mu^4\, {\phi_5}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-4/3} + \left(-\frac{i}{2}\right)\, g_3\, G_\mu^5\, {\phi_5}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{+2/3} + \left(-\frac{i}{2}\right)\, g_3\, G_\mu^5\, {\phi_5}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-1/3}$$

$$\left(-\frac{i}{2}\right)\, g_3\, G_\mu^5\, {\phi_5}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-4/3} + \frac{i}{2}\, g_3\, G_\mu^5\, {\phi_5}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{+2/3} + \frac{i}{2}\, g_3\, G_\mu^5\, {\phi_5}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-1/3}$$

$$\frac{i}{2}\, g_3\, G_\mu^5\, {\phi_5}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_r^{-4/3} + \frac{1}{2}\, g_3\, G_\mu^6\, {\phi_5}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{+2/3} + \frac{1}{2}\, g_3\, G_\mu^6\, {\phi_5}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, {\phi_5}_b^{-1/3}$$

$$\frac{1}{2}\, g_3\, G^6_\mu\, \phi_5{}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5b}{}^{-4/3} + \frac{1}{2}\, g_3\, G^6_\mu\, \phi_5{}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5g}{}^{+2/3} + \frac{1}{2}\, g_3\, G^6_\mu\, \phi_5{}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5g}{}^{-1/3}$$

$$\frac{1}{2}\, g_3\, G^6_\mu\, \phi_5{}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5g}{}^{-4/3} + \left(-\frac{i}{2}\right)\, g_3\, G^7_\mu\, \phi_5{}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5b}{}^{+2/3} + \left(-\frac{i}{2}\right)\, g_3\, G^7_\mu\, \phi_5{}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5b}{}^{-1/3}$$

$$\left(-\frac{i}{2}\right)\, g_3\, G^7_\mu\, \phi_5{}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5b}{}^{-4/3} + \frac{i}{2}\, g_3\, G^7_\mu\, \phi_5{}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5g}{}^{+2/3} + \frac{i}{2}\, g_3\, G^7_\mu\, \phi_5{}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5g}{}^{-1/3}$$

$$\frac{i}{2}\, g_3\, G^7_\mu\, \phi_5{}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5g}{}^{-4/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, \phi_5{}^{+2/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5r}{}^{+2/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, \phi_5{}^{-1/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5r}{}^{-1/3}$$

$$\frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, \phi_5{}^{-4/3,\bar{r}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5r}{}^{-4/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, \phi_5{}^{+2/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5g}{}^{+2/3} + \frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, \phi_5{}^{-1/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5g}{}^{-1/3}$$

$$\frac{\sqrt{3}}{6}\, g_3\, G^8_\mu\, \phi_5{}^{-4/3,\bar{g}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5g}{}^{-4/3} + \left(-\frac{\sqrt{3}}{3}\right)\, g_3\, G^8_\mu\, \phi_5{}^{+2/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5b}{}^{+2/3} + \left(-\frac{\sqrt{3}}{3}\right)\, g_3\, G^8_\mu\, \phi_5{}^{-1/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5b}{}^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{3}\right)\, g_3\, G^8_\mu\, \phi_5{}^{-4/3,\bar{b}\dagger}\, i\overleftrightarrow{\partial^\mu}\, \phi_{5b}{}^{-4/3}$$

$$\frac{1}{4}\, g_1\, g_1\, B_\mu\, B_\mu\, H^{+\dagger}\, H^+ + \frac{1}{4}\, g_1\, g_1\, B_\mu\, B_\mu\, H^{0\dagger}\, H^0 + \frac{1}{4}\, g_1\, g_2\, B_\mu\, W^1_\mu\, H^{+\dagger}\, H^0$$

$$\frac{1}{4}\, g_1\, g_2\, B_\mu\, W^1_\mu\, H^{0\dagger}\, H^+ + \left(-\frac{i}{4}\right)\, g_1\, g_2\, B_\mu\, W^2_\mu\, H^{+\dagger}\, H^0 + \frac{i}{4}\, g_1\, g_2\, B_\mu\, W^2_\mu\, H^{0\dagger}\, H^+$$

$$\frac{1}{4}\, g_1\, g_2\, B_\mu\, W^3_\mu\, H^{+\dagger}\, H^+ + \left(-\frac{1}{4}\right)\, g_1\, g_2\, B_\mu\, W^3_\mu\, H^{0\dagger}\, H^0 + \frac{1}{4}\, g_2\, g_1\, W^1_\mu\, B_\mu\, H^{+\dagger}\, H^0$$

$$\frac{1}{4}\, g_2\, g_1\, W^1_\mu\, B_\mu\, H^{0\dagger}\, H^+ + \left(-\frac{i}{4}\right)\, g_2\, g_1\, W^2_\mu\, B_\mu\, H^{+\dagger}\, H^0 + \frac{i}{4}\, g_2\, g_1\, W^2_\mu\, B_\mu\, H^{0\dagger}\, H^+$$

$$\frac{1}{4}\, g_2\, g_1\, W^3_\mu\, B_\mu\, H^{+\dagger}\, H^+ + \left(-\frac{1}{4}\right)\, g_2\, g_1\, W^3_\mu\, B_\mu\, H^{0\dagger}\, H^0 + \frac{1}{4}\, g_2\, g_2\, W^1_\mu\, W^1_\mu\, H^{+\dagger}\, H^+$$

$$\frac{1}{4}\, g_2\, g_2\, W^1_\mu\, W^1_\mu\, H^{0\dagger}\, H^0 + \frac{i}{4}\, g_2\, g_2\, W^1_\mu\, W^2_\mu\, H^{+\dagger}\, H^+ + \left(-\frac{i}{4}\right)\, g_2\, g_2\, W^1_\mu\, W^2_\mu\, H^{0\dagger}\, H^0$$

$$\left(-\frac{1}{4}\right)\, g_2\, g_2\, W^1_\mu\, W^3_\mu\, H^{+\dagger}\, H^0 + \frac{1}{4}\, g_2\, g_2\, W^1_\mu\, W^3_\mu\, H^{0\dagger}\, H^+ + \left(-\frac{i}{4}\right)\, g_2\, g_2\, W^2_\mu\, W^1_\mu\, H^{+\dagger}\, H^+$$

$$\frac{i}{4}\, g_2\, g_2\, W^2_\mu\, W^1_\mu\, H^{0\dagger}\, H^0 + \frac{1}{4}\, g_2\, g_2\, W^2_\mu\, W^2_\mu\, H^{+\dagger}\, H^+ + \frac{1}{4}\, g_2\, g_2\, W^2_\mu\, W^2_\mu\, H^{0\dagger}\, H^0$$

$$\frac{i}{4}\, g_2\, g_2\, W^2_\mu\, W^3_\mu\, H^{+\dagger}\, H^0 + \frac{i}{4}\, g_2\, g_2\, W^2_\mu\, W^3_\mu\, H^{0\dagger}\, H^+ + \frac{1}{4}\, g_2\, g_2\, W^3_\mu\, W^1_\mu\, H^{+\dagger}\, H^0$$

$$\left(-\frac{1}{4}\right) g_2\, g_2\, W^3_\mu\, W^1_\mu\, H^{0\dagger}\, H^+ + \left(-\frac{i}{4}\right) g_2\, g_2\, W^3_\mu\, W^2_\mu\, H^{+\dagger}\, H^0 + \left(-\frac{i}{4}\right) g_2\, g_2\, W^3_\mu\, W^2_\mu\, H^{0\dagger}\, H^+$$

$$\frac{1}{4} g_2\, g_2\, W^3_\mu\, W^3_\mu\, H^{+\dagger}\, H^+ + \frac{1}{4} g_2\, g_2\, W^3_\mu\, W^3_\mu\, H^{0\dagger}\, H^0 + \frac{1}{9} g_1\, g_1\, B_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\frac{1}{9} g_1\, g_1\, B_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3} + \frac{1}{9} g_1\, g_1\, B_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3}$$

$$\left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3} + \frac{i}{6} g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{i}{6}\right) g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3} + \frac{1}{6} g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3}$$

$$\left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_r^{-1/3} + \frac{i}{6} g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{i}{6}\right) g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_g^{-1/3} + \frac{i}{6} g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3}$$

$$\left(-\frac{i}{6}\right) g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{\sqrt{3}}{18}\right) g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3} + \left(-\frac{\sqrt{3}}{18}\right) g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3}$$

$$\frac{\sqrt{3}}{9} g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\frac{i}{6} g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{i}{6}\right) g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\frac{1}{6} g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\frac{i}{6} g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{i}{6}\right) g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_r^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3}$$

$$\left(-\frac{1}{6}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_g^{-1/3} + \frac{i}{6} g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{i}{6}\right) g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_g^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{18}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3} + \left(-\frac{\sqrt{3}}{18}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3} + \frac{\sqrt{3}}{9} g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_b^{-1/3}$$

$$\frac{1}{4} g_3\, g_3\, G^1_\mu\, G^1_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3} + \frac{1}{4} g_3\, g_3\, G^1_\mu\, G^1_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3} + \frac{i}{4} g_3\, g_3\, G^1_\mu\, G^2_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^1\, G_\mu^2\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G_\mu^1\, G_\mu^3\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G_\mu^1\, G_\mu^3\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G_\mu^1\, G_\mu^4\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^1\, G_\mu^5\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G_\mu^1\, G_\mu^6\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^1\, G_\mu^7\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G_\mu^1\, G_\mu^8\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G_\mu^1\, G_\mu^8\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^2\, G_\mu^1\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G_\mu^2\, G_\mu^1\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G_\mu^2\, G_\mu^2\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G_\mu^2\, G_\mu^2\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G_\mu^2\, G_\mu^3\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G_\mu^2\, G_\mu^3\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G_\mu^2\, G_\mu^4\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G_\mu^2\, G_\mu^5\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^2\, G_\mu^6\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3}$$

$$\begin{aligned}&\left(-\frac{1}{4}\right) g_3\, g_3\, G_\mu^2\, G_\mu^7\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3}\\ &+ \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G_\mu^2\, G_\mu^8\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G_\mu^2\, G_\mu^8\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3}\end{aligned}$$

$$\frac{1}{4}\, g_3\, g_3\, G_\mu^3\, G_\mu^1\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G_\mu^3\, G_\mu^1\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^3\, G_\mu^2\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^3\, G_\mu^2\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G_\mu^3\, G_\mu^3\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G_\mu^3\, G_\mu^3\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G_\mu^3\, G_\mu^4\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^3\, G_\mu^5\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G_\mu^3\, G_\mu^6\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G_\mu^3\, G_\mu^7\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G_\mu^3\, G_\mu^8\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3} + \left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G_\mu^3\, G_\mu^8\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G_\mu^4\, G_\mu^1\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^4\, G_\mu^2\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G_\mu^4\, G_\mu^3\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G_\mu^4\, G_\mu^4\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G_\mu^4\, G_\mu^4\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_b^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G_\mu^4\, G_\mu^5\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G_\mu^4\, G_\mu^5\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G_\mu^4\, G_\mu^6\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G_\mu^4\, G_\mu^7\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_g^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G_\mu^4\, G_\mu^8\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_b^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G_\mu^4\, G_\mu^8\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G_\mu^5\, G_\mu^1\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_g^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^2_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_g^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^3_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_r^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^5_\mu\,G^4_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_r^{-1/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^4_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_b^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^5_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_r^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^5_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_b^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^5_\mu\,G^6_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_g^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^7_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_g^{-1/3}+\frac{\sqrt{3}i}{6}\,g_3\,g_3\,G^5_\mu\,G^8_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_b^{-1/3}$$

$$\frac{\sqrt{3}i}{12}\,g_3\,g_3\,G^5_\mu\,G^8_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_r^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^6_\mu\,G^1_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_r^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^6_\mu\,G^2_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_r^{-1/3}$$

$$\left(-\frac{1}{4}\right)\,g_3\,g_3\,G^6_\mu\,G^3_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_g^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^6_\mu\,G^4_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_r^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^6_\mu\,G^5_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_r^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^6_\mu\,G^6_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_g^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^6_\mu\,G^6_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_b^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^6_\mu\,G^7_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_g^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^6_\mu\,G^7_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_b^{-1/3}$$
$$+\left(-\frac{\sqrt{3}}{6}\right)\,g_3\,g_3\,G^6_\mu\,G^8_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_b^{-1/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^6_\mu\,G^8_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_g^{-1/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^7_\mu\,G^1_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_r^{-1/3}+\left(-\frac{1}{4}\right)\,g_3\,g_3\,G^7_\mu\,G^2_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_r^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^7_\mu\,G^3_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_g^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^7_\mu\,G^4_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_r^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^7_\mu\,G^5_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_r^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^7_\mu\,G^6_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_g^{-1/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^7_\mu\,G^6_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_b^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^7_\mu\,G^7_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_g^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^7_\mu\,G^7_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_b^{-1/3}$$

$$\frac{\sqrt{3}i}{6}\,g_3\,g_3\,G^7_\mu\,G^8_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_b^{-1/3}+\frac{\sqrt{3}i}{12}\,g_3\,g_3\,G^7_\mu\,G^8_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_g^{-1/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^8_\mu\,G^1_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_g^{-1/3}$$

$$\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^8_\mu\,G^1_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_r^{-1/3}+\left(-\frac{\sqrt{3}i}{12}\right)\,g_3\,g_3\,G^8_\mu\,G^2_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_g^{-1/3}+\frac{\sqrt{3}i}{12}\,g_3\,g_3\,G^8_\mu\,G^2_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_r^{-1/3}$$

$$\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^8_\mu\,G^3_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_r^{-1/3}+\left(-\frac{\sqrt{3}}{12}\right)\,g_3\,g_3\,G^8_\mu\,G^3_\mu\,{\phi_1}^{-1/3,\bar{g}\dagger}\,{\phi_1}_g^{-1/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^8_\mu\,G^4_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_b^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right)\,g_3\,g_3\,G^8_\mu\,G^4_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_r^{-1/3}$$
$$+\left(-\frac{\sqrt{3}i}{12}\right)\,g_3\,g_3\,G^8_\mu\,G^5_\mu\,{\phi_1}^{-1/3,\bar{r}\dagger}\,{\phi_1}_b^{-1/3}+\left(-\frac{\sqrt{3}i}{6}\right)\,g_3\,g_3\,G^8_\mu\,G^5_\mu\,{\phi_1}^{-1/3,\bar{b}\dagger}\,{\phi_1}_r^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_g^{-1/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_b^{-1/3}$$

$$\left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_g^{-1/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_1}^{-1/3,\bar{r}\dagger}\, {\phi_1}_r^{-1/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_1}^{-1/3,\bar{g}\dagger}\, {\phi_1}_g^{-1/3}$$

$$\frac{1}{3}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_1}^{-1/3,\bar{b}\dagger}\, {\phi_1}_b^{-1/3} + \frac{16}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_r^{-4/3} + \frac{16}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_g^{-4/3}$$

$$\frac{16}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{2}{3}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3} + \left(-\frac{2}{3}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\frac{2i}{3}\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3} + \left(-\frac{2i}{3}\right) g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3} + \left(-\frac{2}{3}\right) g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\frac{2}{3}\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_g^{-4/3} + \left(-\frac{2}{3}\right) g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{2}{3}\right) g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\frac{2i}{3}\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{2i}{3}\right) g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3} + \left(-\frac{2}{3}\right) g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_b^{-4/3}$$

$$\left(-\frac{2}{3}\right) g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3} + \frac{2i}{3}\, g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{2i}{3}\right) g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3}$$

$$\left(-\frac{2\sqrt{3}}{9}\right) g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_r^{-4/3} + \left(-\frac{2\sqrt{3}}{9}\right) g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_g^{-4/3} + \frac{4\sqrt{3}}{9}\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_b^{-4/3}$$

$$\left(-\frac{2}{3}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3} + \left(-\frac{2}{3}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3} + \frac{2i}{3}\, g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3}$$

$$\left(-\frac{2i}{3}\right) g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3} + \left(-\frac{2}{3}\right) g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_r^{-4/3} + \frac{2}{3}\, g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_g^{-4/3}$$

$$\left(-\frac{2}{3}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{2}{3}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3} + \frac{2i}{3}\, g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_b^{-4/3}$$

$$\left(-\frac{2i}{3}\right) g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3} + \left(-\frac{2}{3}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{2}{3}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3}$$

$$\frac{2i}{3}\, g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{2i}{3}\right) g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3} + \left(-\frac{2\sqrt{3}}{9}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\left(-\frac{2\sqrt{3}}{9}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{4\sqrt{3}}{9}\, g_3\, g_1\, G^8_\mu\, B_\mu\, \phi_2{}^{-4/3,\bar{b}\dagger}\, \phi_{2b}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^1_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2r}{}^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^1_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^1_\mu\, G^2_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2r}{}^{-4/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^1_\mu\, G^2_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2g}{}^{-4/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_3\, G^1_\mu\, G^3_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^3_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2r}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^4_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2b}{}^{-4/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^1_\mu\, G^5_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2b}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^6_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2b}{}^{-4/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^1_\mu\, G^7_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2b}{}^{-4/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^1_\mu\, G^8_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^1_\mu\, G^8_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2r}{}^{-4/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^2_\mu\, G^1_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2r}{}^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^1_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^2_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2r}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^2_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2g}{}^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^3_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^3_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2r}{}^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^4_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2b}{}^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^5_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2b}{}^{-4/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^2_\mu\, G^6_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2b}{}^{-4/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^2_\mu\, G^7_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2b}{}^{-4/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^2_\mu\, G^8_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^2_\mu\, G^8_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2r}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^1_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2g}{}^{-4/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^1_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2r}{}^{-4/3}$$
$$+ \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2g}{}^{-4/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2r}{}^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2r}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^4_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2b}{}^{-4/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^5_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2b}{}^{-4/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^6_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2b}{}^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^3_\mu\, G^7_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2b}{}^{-4/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^3_\mu\, G^8_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2r}{}^{-4/3} + \left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^3_\mu\, G^8_\mu\, \phi_2{}^{-4/3,\bar{g}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^1_\mu\, \phi_2{}^{-4/3,\bar{b}\dagger}\, \phi_{2g}{}^{-4/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^2_\mu\, \phi_2{}^{-4/3,\bar{b}\dagger}\, \phi_{2g}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^3_\mu\, \phi_2{}^{-4/3,\bar{b}\dagger}\, \phi_{2r}{}^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2r}{}^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, \phi_2{}^{-4/3,\bar{b}\dagger}\, \phi_{2b}{}^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^5_\mu\, \phi_2{}^{-4/3,\bar{r}\dagger}\, \phi_{2r}{}^{-4/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^5_\mu\, \phi_2{}^{-4/3,\bar{b}\dagger}\, \phi_{2b}{}^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_b^{-4/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^1_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^2_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^3_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_r^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_b^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_r^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^5_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_b^{-4/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^1_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^2_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^3_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^4_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^5_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_g^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_b^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_g^{-4/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_b^{-4/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_b^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^1_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^2_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3}$$
$$+ \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^3_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^4_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^5_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_g^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_b^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_g^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_b^{-4/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_b^{-4/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_g^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_g^{-4/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_r^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\left(-\frac{\sqrt{3}}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_2}^{-4/3,\bar{g}\dagger}\, {\phi_2}_g^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_2}^{-4/3,\bar{r}\dagger}\, {\phi_2}_b^{-4/3}$$
$$+ \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_2}^{-4/3,\bar{b}\dagger}\, {\phi_2}_r^{-4/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_2}^{-4/3,\bar{r}\,\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_2}^{-4/3,\bar{b}\,\dagger}\, {\phi_2}_r^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{g}\,\dagger}\, {\phi_2}_b^{-4/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_2}^{-4/3,\bar{b}\,\dagger}\, {\phi_2}_g^{-4/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{g}\,\dagger}\, {\phi_2}_b^{-4/3} + \left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_2}^{-4/3,\bar{b}\,\dagger}\, {\phi_2}_g^{-4/3}$$

$$\frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{r}\,\dagger}\, {\phi_2}_r^{-4/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{g}\,\dagger}\, {\phi_2}_g^{-4/3} + \frac{1}{3}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_2}^{-4/3,\bar{b}\,\dagger}\, {\phi_2}_b^{-4/3}$$

$$\frac{49}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\,\dagger}\, {\phi_3}_r^{+5/3} + \frac{49}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\,\dagger}\, {\phi_3}_g^{+5/3} + \frac{49}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\,\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{49}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\,\dagger}\, {\phi_3}_r^{+2/3} + \frac{49}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\,\dagger}\, {\phi_3}_g^{+2/3} + \frac{49}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\,\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{7}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{r}\,\dagger}\, {\phi_3}_r^{+2/3} + \frac{7}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{g}\,\dagger}\, {\phi_3}_g^{+2/3} + \frac{7}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{b}\,\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{7}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{r}\,\dagger}\, {\phi_3}_r^{+5/3} + \frac{7}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{g}\,\dagger}\, {\phi_3}_g^{+5/3} + \frac{7}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{b}\,\dagger}\, {\phi_3}_b^{+5/3}$$

$$\left(-\frac{7i}{12}\right) g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{r}\,\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{7i}{12}\right) g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{g}\,\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{7i}{12}\right) g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{b}\,\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{7i}{12}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{r}\,\dagger}\, {\phi_3}_r^{+5/3} + \frac{7i}{12}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{g}\,\dagger}\, {\phi_3}_g^{+5/3} + \frac{7i}{12}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{b}\,\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{7}{12}\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{r}\,\dagger}\, {\phi_3}_r^{+5/3} + \frac{7}{12}\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{g}\,\dagger}\, {\phi_3}_g^{+5/3} + \frac{7}{12}\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{b}\,\dagger}\, {\phi_3}_b^{+5/3}$$

$$\left(-\frac{7}{12}\right) g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{r}\,\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{7}{12}\right) g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{g}\,\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{7}{12}\right) g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{b}\,\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{7}{12}\, g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{r}\,\dagger}\, {\phi_3}_g^{+5/3} + \frac{7}{12}\, g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{g}\,\dagger}\, {\phi_3}_r^{+5/3} + \frac{7}{12}\, g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{r}\,\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{7}{12}\, g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{g}\,\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{7i}{12}\right) g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{r}\,\dagger}\, {\phi_3}_g^{+5/3} + \frac{7i}{12}\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{g}\,\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{7i}{12}\right) g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{r}\,\dagger}\, {\phi_3}_g^{+2/3} + \frac{7i}{12}\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{g}\,\dagger}\, {\phi_3}_r^{+2/3} + \frac{7}{12}\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{r}\,\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{7}{12}\right) g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \frac{7}{12}\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{7}{12}\right) g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{7}{12}\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{7}{12}\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \frac{7}{12}\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{7}{12}\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{7i}{12}\right) g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{7i}{12}\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{7i}{12}\right) g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \frac{7i}{12}\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{7}{12}\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{7}{12}\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \frac{7}{12}\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \frac{7}{12}\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\left(-\frac{7i}{12}\right) g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{7i}{12}\, g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{7i}{12}\right) g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{7i}{12}\, g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{7\sqrt{3}}{36}\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{7\sqrt{3}}{36}\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{7\sqrt{3}}{18}\right) g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{7\sqrt{3}}{36}\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{7\sqrt{3}}{36}\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\left(-\frac{7\sqrt{3}}{18}\right) g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{7}{12}\, g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{7}{12}\, g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{7}{12}\, g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{7}{12}\, g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{7}{12}\, g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{7}{12}\, g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{7i}{12}\right) g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{7i}{12}\right) g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\left(-\frac{7i}{12}\right) g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{7i}{12}\, g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{7i}{12}\, g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{7i}{12}\, g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{7}{12}\, g_2\, g_1\, W^3_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{7}{12}\, g_2\, g_1\, W^3_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{7}{12}\, g_2\, g_1\, W^3_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{7}{12}\right) g_2\, g_1\, W^3_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{7}{12}\right) g_2\, g_1\, W^3_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\left(-\frac{7}{12}\right) g_2\, g_1\, W^3_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_2\, g_2\, W^1_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{4}\, g_2\, g_2\, W^1_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_3}^{+5/3,\bar{b}\dagger}\,{\phi_3}_b^{+5/3}+\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_3}^{+2/3,\bar{r}\dagger}\,{\phi_3}_r^{+2/3}+\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_3}^{+2/3,\bar{g}\dagger}\,{\phi_3}_g^{+2/3}$$

$$\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_3}^{+2/3,\bar{b}\dagger}\,{\phi_3}_b^{+2/3}+\frac{i}{4}\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_3}^{+5/3,\bar{r}\dagger}\,{\phi_3}_r^{+5/3}+\frac{i}{4}\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_3}^{+5/3,\bar{g}\dagger}\,{\phi_3}_g^{+5/3}$$

$$\frac{i}{4}\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_3}^{+5/3,\bar{b}\dagger}\,{\phi_3}_b^{+5/3}+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_3}^{+2/3,\bar{r}\dagger}\,{\phi_3}_r^{+2/3}+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_3}^{+2/3,\bar{g}\dagger}\,{\phi_3}_g^{+2/3}$$

$$\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_3}^{+2/3,\bar{b}\dagger}\,{\phi_3}_b^{+2/3}$$
$$+\left(-\frac{1}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^3_\mu\,{\phi_3}^{+5/3,\bar{r}\dagger}\,{\phi_3}_r^{+2/3}+\left(-\frac{1}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^3_\mu\,{\phi_3}^{+5/3,\bar{g}\dagger}\,{\phi_3}_g^{+2/3}$$

$$\left(-\frac{1}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^3_\mu\,{\phi_3}^{+5/3,\bar{b}\dagger}\,{\phi_3}_b^{+2/3}+\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^3_\mu\,{\phi_3}^{+2/3,\bar{r}\dagger}\,{\phi_3}_r^{+5/3}+\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^3_\mu\,{\phi_3}^{+2/3,\bar{g}\dagger}\,{\phi_3}_g^{+5/3}$$

$$\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^3_\mu\,{\phi_3}^{+2/3,\bar{b}\dagger}\,{\phi_3}_b^{+5/3}+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^2_\mu\,W^1_\mu\,{\phi_3}^{+5/3,\bar{r}\dagger}\,{\phi_3}_r^{+5/3}+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^2_\mu\,W^1_\mu\,{\phi_3}^{+5/3,\bar{g}\dagger}\,{\phi_3}_g^{+5/3}$$

$$\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^2_\mu\,W^1_\mu\,{\phi_3}^{+5/3,\bar{b}\dagger}\,{\phi_3}_b^{+5/3}+\frac{i}{4}\,g_2\,g_2\,W^2_\mu\,W^1_\mu\,{\phi_3}^{+2/3,\bar{r}\dagger}\,{\phi_3}_r^{+2/3}+\frac{i}{4}\,g_2\,g_2\,W^2_\mu\,W^1_\mu\,{\phi_3}^{+2/3,\bar{g}\dagger}\,{\phi_3}_g^{+2/3}$$

$$\frac{i}{4}\,g_2\,g_2\,W^2_\mu\,W^1_\mu\,{\phi_3}^{+2/3,\bar{b}\dagger}\,{\phi_3}_b^{+2/3}+\frac{1}{4}\,g_2\,g_2\,W^2_\mu\,W^2_\mu\,{\phi_3}^{+5/3,\bar{r}\dagger}\,{\phi_3}_r^{+5/3}+\frac{1}{4}\,g_2\,g_2\,W^2_\mu\,W^2_\mu\,{\phi_3}^{+5/3,\bar{g}\dagger}\,{\phi_3}_g^{+5/3}$$

$$\frac{1}{4}\,g_2\,g_2\,W^2_\mu\,W^2_\mu\,{\phi_3}^{+5/3,\bar{b}\dagger}\,{\phi_3}_b^{+5/3}+\frac{1}{4}\,g_2\,g_2\,W^2_\mu\,W^2_\mu\,{\phi_3}^{+2/3,\bar{r}\dagger}\,{\phi_3}_r^{+2/3}+\frac{1}{4}\,g_2\,g_2\,W^2_\mu\,W^2_\mu\,{\phi_3}^{+2/3,\bar{g}\dagger}\,{\phi_3}_g^{+2/3}$$

$$\frac{1}{4}\,g_2\,g_2\,W^2_\mu\,W^2_\mu\,{\phi_3}^{+2/3,\bar{b}\dagger}\,{\phi_3}_b^{+2/3}+\frac{i}{4}\,g_2\,g_2\,W^2_\mu\,W^3_\mu\,{\phi_3}^{+5/3,\bar{r}\dagger}\,{\phi_3}_r^{+2/3}+\frac{i}{4}\,g_2\,g_2\,W^2_\mu\,W^3_\mu\,{\phi_3}^{+5/3,\bar{g}\dagger}\,{\phi_3}_g^{+2/3}$$

$$\frac{i}{4}\,g_2\,g_2\,W^2_\mu\,W^3_\mu\,{\phi_3}^{+5/3,\bar{b}\dagger}\,{\phi_3}_b^{+2/3}+\frac{i}{4}\,g_2\,g_2\,W^2_\mu\,W^3_\mu\,{\phi_3}^{+2/3,\bar{r}\dagger}\,{\phi_3}_r^{+5/3}+\frac{i}{4}\,g_2\,g_2\,W^2_\mu\,W^3_\mu\,{\phi_3}^{+2/3,\bar{g}\dagger}\,{\phi_3}_g^{+5/3}$$

$$\frac{i}{4}\,g_2\,g_2\,W^2_\mu\,W^3_\mu\,{\phi_3}^{+2/3,\bar{b}\dagger}\,{\phi_3}_b^{+5/3}+\frac{1}{4}\,g_2\,g_2\,W^3_\mu\,W^1_\mu\,{\phi_3}^{+5/3,\bar{r}\dagger}\,{\phi_3}_r^{+2/3}+\frac{1}{4}\,g_2\,g_2\,W^3_\mu\,W^1_\mu\,{\phi_3}^{+5/3,\bar{g}\dagger}\,{\phi_3}_g^{+2/3}$$

$$\frac{1}{4}\,g_2\,g_2\,W^3_\mu\,W^1_\mu\,{\phi_3}^{+5/3,\bar{b}\dagger}\,{\phi_3}_b^{+2/3}+\left(-\frac{1}{4}\right)\,g_2\,g_2\,W^3_\mu\,W^1_\mu\,{\phi_3}^{+2/3,\bar{r}\dagger}\,{\phi_3}_r^{+5/3}+\left(-\frac{1}{4}\right)\,g_2\,g_2\,W^3_\mu\,W^1_\mu\,{\phi_3}^{+2/3,\bar{g}\dagger}\,{\phi_3}_g^{+5/3}$$

$$\left(-\frac{1}{4}\right)\,g_2\,g_2\,W^3_\mu\,W^1_\mu\,{\phi_3}^{+2/3,\bar{b}\dagger}\,{\phi_3}_b^{+5/3}$$
$$+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^3_\mu\,W^2_\mu\,{\phi_3}^{+5/3,\bar{r}\dagger}\,{\phi_3}_r^{+2/3}+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^3_\mu\,W^2_\mu\,{\phi_3}^{+5/3,\bar{g}\dagger}\,{\phi_3}_g^{+2/3}$$

$$\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^3_\mu\,W^2_\mu\,{\phi_3}^{+5/3,\bar{b}\dagger}\,{\phi_3}_b^{+2/3}$$
$$+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^3_\mu\,W^2_\mu\,{\phi_3}^{+2/3,\bar{r}\dagger}\,{\phi_3}_r^{+5/3}+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^3_\mu\,W^2_\mu\,{\phi_3}^{+2/3,\bar{g}\dagger}\,{\phi_3}_g^{+5/3}$$

$$\left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^1\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^1\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^1\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^1\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^2\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^2\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^2\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^2\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^3\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{1}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^3\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^3\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^3\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^4\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^4\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^4\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^4\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^5\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^5\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^5\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^5\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^6\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^6\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^6\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^6\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^7\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^7\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^7\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^7\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right) g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\begin{aligned} &\frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} \\ &+ \left(-\frac{\sqrt{3}}{6}\right) g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^1\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} \end{aligned}$$

$$\left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^1\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^1\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^1\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^2\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^2\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^2\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_3\, W^2_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{i}{4}\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{i}{4}\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_3\, W^2_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W^2_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W^2_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_3\, W^2_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W^2_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W^2_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W^2_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W^2_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{1}{4}\right) g_2\, g_3\, W^2_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W^2_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W^2_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{1}{4}\right) g_2\, g_3\, W^2_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right) g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{\sqrt{3}i}{6}\, g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{\sqrt{3}i}{12}\, g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{\sqrt{3}i}{12}\, g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{\sqrt{3}i}{6}\right) g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W^3_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_2\, g_3\, W^3_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{1}{4}\right) g_2\, g_3\, W^3_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_3\, W^3_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W^3_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W^3_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W^3_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W^3_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W^3_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_3\, W^3_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{1}{4}\right) g_2\, g_3\, W^3_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W^3_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W^3_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_2\, g_3\, W^3_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{1}{4}\right) g_2\, g_3\, W^3_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_3\, W_\mu^3\, G_\mu^4\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^3\, G_\mu^5\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^3\, G_\mu^5\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W_\mu^3\, G_\mu^5\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^3\, G_\mu^5\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^3\, G_\mu^6\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^3\, G_\mu^6\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{1}{4}\right) g_2\, g_3\, W_\mu^3\, G_\mu^6\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{1}{4}\right) g_2\, g_3\, W_\mu^3\, G_\mu^6\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^3\, G_\mu^7\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^3\, G_\mu^7\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^3\, G_\mu^7\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^3\, G_\mu^7\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^3\, G_\mu^8\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^3\, G_\mu^8\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\begin{aligned}&\left(-\frac{\sqrt{3}}{6}\right) g_2\, g_3\, W_\mu^3\, G_\mu^8\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3}\\ &+ \left(-\frac{\sqrt{3}}{12}\right) g_2\, g_3\, W_\mu^3\, G_\mu^8\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{\sqrt{3}}{12}\right) g_2\, g_3\, W_\mu^3\, G_\mu^8\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}\end{aligned}$$

$$\frac{\sqrt{3}}{6}\, g_2\, g_3\, W_\mu^3\, G_\mu^8\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{7}{12}\, g_3\, g_1\, G_\mu^1\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{7}{12}\, g_3\, g_1\, G_\mu^1\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{7}{12}\, g_3\, g_1\, G_\mu^1\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \frac{7}{12}\, g_3\, g_1\, G_\mu^1\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{7i}{12}\right) g_3\, g_1\, G_\mu^2\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{7i}{12}\, g_3\, g_1\, G_\mu^2\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{7i}{12}\right) g_3\, g_1\, G_\mu^2\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \frac{7i}{12}\, g_3\, g_1\, G_\mu^2\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{7}{12}\, g_3\, g_1\, G_\mu^3\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{7}{12}\right) g_3\, g_1\, G_\mu^3\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \frac{7}{12}\, g_3\, g_1\, G_\mu^3\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{7}{12}\right) g_3\, g_1\, G_\mu^3\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{7}{12}\, g_3\, g_1\, G_\mu^4\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{7}{12}\, g_3\, g_1\, G_\mu^4\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{7}{12}\, g_3\, g_1\, G_\mu^4\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \frac{7}{12}\, g_3\, g_1\, G_\mu^4\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{7i}{12}\right) g_3\, g_1\, G_\mu^5\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{7i}{12}\, g_3\, g_1\, G_\mu^5\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{7i}{12}\right) g_3\, g_1\, G_\mu^5\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \frac{7i}{12}\, g_3\, g_1\, G_\mu^5\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{7}{12}\, g_3\, g_1\, G_\mu^6\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{7}{12}\, g_3\, g_1\, G_\mu^6\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \frac{7}{12}\, g_3\, g_1\, G_\mu^6\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{7}{12}\, g_3\, g_1\, G_\mu^6\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{7i}{12}\right) g_3\, g_1\, G_\mu^7\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{7i}{12}\, g_3\, g_1\, G_\mu^7\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{7i}{12}\right)\, g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \frac{7i}{12}\, g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{7\sqrt{3}}{36}\, g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{7\sqrt{3}}{36}\, g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{7\sqrt{3}}{18}\right)\, g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{7\sqrt{3}}{36}\, g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{7\sqrt{3}}{36}\, g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{7\sqrt{3}}{18}\right)\, g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{i}{4}\, g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^1_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^1_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\begin{aligned}&\left(-\frac{1}{4}\right)\, g_3\, g_2\, G^1_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3}\\ &\quad + \left(-\frac{1}{4}\right)\, g_3\, g_2\, G^1_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3}\end{aligned}$$

$$\frac{i}{4}\, g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^2_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_3\, g_2\, G^2_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^2_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^2_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{1}{4}\right)\, g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{i}{4}\, g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{i}{4}\, g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^3_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{1}{4}\right)\, g_3\, g_2\, G^3_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_2\, G^3_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^3_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{i}{4}\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^4_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^4_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_2\, G^4_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{1}{4}\right) g_3\, g_2\, G^4_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{i}{4}\right) g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{i}{4}\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{1}{4}\right) g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{i}{4}\right) g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{\sqrt{3}i}{6}\right)\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_2\, G^8_\mu\, W^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{\sqrt{3}}{12}\right)\, g_3\, g_2\, G^8_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{\sqrt{3}}{12}\right)\, g_3\, g_2\, G^8_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{\sqrt{3}}{6}\, g_3\, g_2\, G^8_\mu\, W^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^1_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^1_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^1_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^1_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^1_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_3\, G^1_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^1_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^1_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^1_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^1_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^1_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^1_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^1_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{\sqrt{3}}{12} g_3 g_3 G^1_\mu G^8_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3g}{}^{+2/3} + \frac{\sqrt{3}}{12} g_3 g_3 G^1_\mu G^8_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3r}{}^{+2/3} + \left(-\frac{i}{4}\right) g_3 g_3 G^2_\mu G^1_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3r}{}^{+5/3}$$

$$\frac{i}{4} g_3 g_3 G^2_\mu G^1_\mu \phi_3{}^{+5/3,\bar{g}\dagger} \phi_{3g}{}^{+5/3} + \left(-\frac{i}{4}\right) g_3 g_3 G^2_\mu G^1_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3r}{}^{+2/3} + \frac{i}{4} g_3 g_3 G^2_\mu G^1_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3g}{}^{+2/3}$$

$$\frac{1}{4} g_3 g_3 G^2_\mu G^2_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3r}{}^{+5/3} + \frac{1}{4} g_3 g_3 G^2_\mu G^2_\mu \phi_3{}^{+5/3,\bar{g}\dagger} \phi_{3g}{}^{+5/3} + \frac{1}{4} g_3 g_3 G^2_\mu G^2_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3r}{}^{+2/3}$$

$$\frac{1}{4} g_3 g_3 G^2_\mu G^2_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3g}{}^{+2/3} + \frac{i}{4} g_3 g_3 G^2_\mu G^3_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3g}{}^{+5/3} + \frac{i}{4} g_3 g_3 G^2_\mu G^3_\mu \phi_3{}^{+5/3,\bar{g}\dagger} \phi_{3r}{}^{+5/3}$$

$$\frac{i}{4} g_3 g_3 G^2_\mu G^3_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3g}{}^{+2/3} + \frac{i}{4} g_3 g_3 G^2_\mu G^3_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3r}{}^{+2/3} + \frac{i}{4} g_3 g_3 G^2_\mu G^4_\mu \phi_3{}^{+5/3,\bar{g}\dagger} \phi_{3b}{}^{+5/3}$$

$$\frac{i}{4} g_3 g_3 G^2_\mu G^4_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3b}{}^{+2/3} + \frac{1}{4} g_3 g_3 G^2_\mu G^5_\mu \phi_3{}^{+5/3,\bar{g}\dagger} \phi_{3b}{}^{+5/3} + \frac{1}{4} g_3 g_3 G^2_\mu G^5_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3b}{}^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_3 g_3 G^2_\mu G^6_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3b}{}^{+5/3}$$
$$+ \left(-\frac{i}{4}\right) g_3 g_3 G^2_\mu G^6_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3b}{}^{+2/3} + \left(-\frac{1}{4}\right) g_3 g_3 G^2_\mu G^7_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3b}{}^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_3 g_3 G^2_\mu G^7_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3b}{}^{+2/3}$$
$$+ \left(-\frac{\sqrt{3}i}{12}\right) g_3 g_3 G^2_\mu G^8_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3g}{}^{+5/3} + \frac{\sqrt{3}i}{12} g_3 g_3 G^2_\mu G^8_\mu \phi_3{}^{+5/3,\bar{g}\dagger} \phi_{3r}{}^{+5/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right) g_3 g_3 G^2_\mu G^8_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3g}{}^{+2/3} + \frac{\sqrt{3}i}{12} g_3 g_3 G^2_\mu G^8_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3r}{}^{+2/3} + \frac{1}{4} g_3 g_3 G^3_\mu G^1_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3g}{}^{+5/3}$$

$$\left(-\frac{1}{4}\right) g_3 g_3 G^3_\mu G^1_\mu \phi_3{}^{+5/3,\bar{g}\dagger} \phi_{3r}{}^{+5/3} + \frac{1}{4} g_3 g_3 G^3_\mu G^1_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3g}{}^{+2/3} + \left(-\frac{1}{4}\right) g_3 g_3 G^3_\mu G^1_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3r}{}^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_3 g_3 G^3_\mu G^2_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3g}{}^{+5/3}$$
$$+ \left(-\frac{i}{4}\right) g_3 g_3 G^3_\mu G^2_\mu \phi_3{}^{+5/3,\bar{g}\dagger} \phi_{3r}{}^{+5/3} + \left(-\frac{i}{4}\right) g_3 g_3 G^3_\mu G^2_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3g}{}^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_3 g_3 G^3_\mu G^2_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3r}{}^{+2/3} + \frac{1}{4} g_3 g_3 G^3_\mu G^3_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3r}{}^{+5/3} + \frac{1}{4} g_3 g_3 G^3_\mu G^3_\mu \phi_3{}^{+5/3,\bar{g}\dagger} \phi_{3g}{}^{+5/3}$$

$$\frac{1}{4} g_3 g_3 G^3_\mu G^3_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3r}{}^{+2/3} + \frac{1}{4} g_3 g_3 G^3_\mu G^3_\mu \phi_3{}^{+2/3,\bar{g}\dagger} \phi_{3g}{}^{+2/3} + \frac{1}{4} g_3 g_3 G^3_\mu G^4_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3b}{}^{+5/3}$$

$$\frac{1}{4} g_3 g_3 G^3_\mu G^4_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3b}{}^{+2/3} + \left(-\frac{i}{4}\right) g_3 g_3 G^3_\mu G^5_\mu \phi_3{}^{+5/3,\bar{r}\dagger} \phi_{3b}{}^{+5/3} + \left(-\frac{i}{4}\right) g_3 g_3 G^3_\mu G^5_\mu \phi_3{}^{+2/3,\bar{r}\dagger} \phi_{3b}{}^{+2/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^3_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^3_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^5_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^5_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_g^{+5/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_g^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_r^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_r^{+5/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_r^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_b^{+5/3} + \left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_g^{+5/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_b^{+2/3}$$

$$\left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_g^{+2/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{r}\dagger}\, {\phi_3}_r^{+5/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{g}\dagger}\, {\phi_3}_g^{+5/3}$$

$$\frac{1}{3}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_3}^{+5/3,\bar{b}\dagger}\, {\phi_3}_b^{+5/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{r}\dagger}\, {\phi_3}_r^{+2/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{g}\dagger}\, {\phi_3}_g^{+2/3}$$

$$\frac{1}{3}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_3}^{+2/3,\bar{b}\dagger}\, {\phi_3}_b^{+2/3} + \frac{1}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \frac{1}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{1}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \frac{1}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{36}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \frac{1}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \frac{1}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{1}{12}\, g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{i}{12}\right)\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{12}\right)\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\left(-\frac{i}{12}\right)\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{i}{12}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \frac{i}{12}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{i}{12}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{12}\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \frac{1}{12}\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{1}{12}\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{1}{12}\right)\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{1}{12}\right)\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\left(-\frac{1}{12}\right)\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{12}\, g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{12}\, g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{12}\, g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{12}\, g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{12}\right)\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{i}{12}\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{i}{12}\right)\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \frac{i}{12}\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{12}\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{1}{12}\right)\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{12}\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\left(-\frac{1}{12}\right)\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{12}\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{12}\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{12}\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{12}\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{12}\right)\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{i}{12}\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{i}{12}\right)\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3} + \frac{i}{12}\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{12}\,g_1\,g_3\,B_\mu\,G^6_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}+\frac{1}{12}\,g_1\,g_3\,B_\mu\,G^6_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_g^{+2/3}+\frac{1}{12}\,g_1\,g_3\,B_\mu\,G^6_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{-1/3}$$

$$\frac{1}{12}\,g_1\,g_3\,B_\mu\,G^6_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_g^{-1/3}+\left(-\frac{i}{12}\right)\,g_1\,g_3\,B_\mu\,G^7_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}+\frac{i}{12}\,g_1\,g_3\,B_\mu\,G^7_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\left(-\frac{i}{12}\right)\,g_1\,g_3\,B_\mu\,G^7_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{-1/3}+\frac{i}{12}\,g_1\,g_3\,B_\mu\,G^7_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_g^{-1/3}+\frac{\sqrt{3}}{36}\,g_1\,g_3\,B_\mu\,G^8_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\frac{\sqrt{3}}{36}\,g_1\,g_3\,B_\mu\,G^8_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\left(-\frac{\sqrt{3}}{18}\right)\,g_1\,g_3\,B_\mu\,G^8_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_b^{+2/3}+\frac{\sqrt{3}}{36}\,g_1\,g_3\,B_\mu\,G^8_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{\sqrt{3}}{36}\,g_1\,g_3\,B_\mu\,G^8_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\left(-\frac{\sqrt{3}}{18}\right)\,g_1\,g_3\,B_\mu\,G^8_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_b^{-1/3}+\frac{1}{12}\,g_2\,g_1\,W^1_\mu\,B_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{12}\,g_2\,g_1\,W^1_\mu\,B_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\frac{1}{12}\,g_2\,g_1\,W^1_\mu\,B_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_b^{-1/3}+\frac{1}{12}\,g_2\,g_1\,W^1_\mu\,B_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\frac{1}{12}\,g_2\,g_1\,W^1_\mu\,B_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\frac{1}{12}\,g_2\,g_1\,W^1_\mu\,B_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_b^{+2/3}+\left(-\frac{i}{12}\right)\,g_2\,g_1\,W^2_\mu\,B_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\left(-\frac{i}{12}\right)\,g_2\,g_1\,W^2_\mu\,B_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}$$
$$+\left(-\frac{i}{12}\right)\,g_2\,g_1\,W^2_\mu\,B_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_b^{-1/3}+\frac{i}{12}\,g_2\,g_1\,W^2_\mu\,B_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\frac{i}{12}\,g_2\,g_1\,W^2_\mu\,B_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\frac{i}{12}\,g_2\,g_1\,W^2_\mu\,B_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_b^{+2/3}+\frac{1}{12}\,g_2\,g_1\,W^3_\mu\,B_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\frac{1}{12}\,g_2\,g_1\,W^3_\mu\,B_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\frac{1}{12}\,g_2\,g_1\,W^3_\mu\,B_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_b^{+2/3}+\left(-\frac{1}{12}\right)\,g_2\,g_1\,W^3_\mu\,B_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\left(-\frac{1}{12}\right)\,g_2\,g_1\,W^3_\mu\,B_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}$$
$$+\left(-\frac{1}{12}\right)\,g_2\,g_1\,W^3_\mu\,B_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_b^{-1/3}+\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_b^{+2/3}+\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\frac{1}{4}\,g_2\,g_2\,W^1_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_b^{-1/3}+\frac{i}{4}\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\frac{i}{4}\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\frac{i}{4}\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_b^{+2/3}+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}$$
$$+\left(-\frac{i}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_b^{-1/3}+\left(-\frac{1}{4}\right)\,g_2\,g_2\,W^1_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_2\, W_\mu^1\, W_\mu^3\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{1}{4}\right) g_2\, g_2\, W_\mu^1\, W_\mu^3\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^1\, W_\mu^3\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{4}\, g_2\, g_2\, W_\mu^1\, W_\mu^3\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^1\, W_\mu^3\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^2\, W_\mu^1\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^2\, W_\mu^1\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^2\, W_\mu^1\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{i}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^1\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{i}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^1\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{i}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^1\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{i}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{i}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{i}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{i}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{i}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{i}{4}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^1\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^1\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^1\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{1}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^1\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\left(-\frac{1}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^1\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{1}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^1\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{4}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{4}\, g_2\, g_2\, W_\mu^3\, W_\mu^3\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^1\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^1\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^1\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^1\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^2\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^2\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{4}\right) g_2\, g_3\, W_\mu^1\, G_\mu^2\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^2\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^3\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{1}{4}\right)\, g_2\, g_3\, W_\mu^1\, G_\mu^3\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^3\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{1}{4}\right)\, g_2\, g_3\, W_\mu^1\, G_\mu^3\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^4\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^4\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^4\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^4\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^1\, G_\mu^5\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^5\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^1\, G_\mu^5\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^5\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^6\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^6\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^6\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^6\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^1\, G_\mu^7\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^7\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^1\, G_\mu^7\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^1\, G_\mu^7\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \frac{\sqrt{3}}{12}\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_2\, g_3\, W_\mu^1\, G_\mu^8\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^1\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^1\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^1\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^1\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{1}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^2\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^2\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^2\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{1}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^2\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^3\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^3\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^3\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^3\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^4\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^4\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \frac{i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^4\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^4\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{1}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{1}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{i}{4}\right)\, g_2\, g_3\, W_\mu^2\, G_\mu^6\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_2\,g_3\,W_\mu^2\,G_\mu^6\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_g^{-1/3}+\frac{i}{4}\,g_2\,g_3\,W_\mu^2\,G_\mu^6\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}+\frac{i}{4}\,g_2\,g_3\,W_\mu^2\,G_\mu^6\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^2\,G_\mu^7\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_b^{-1/3}+\frac{1}{4}\,g_2\,g_3\,W_\mu^2\,G_\mu^7\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_g^{-1/3}+\frac{1}{4}\,g_2\,g_3\,W_\mu^2\,G_\mu^7\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}$$

$$\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^2\,G_\mu^7\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_g^{+2/3}$$
$$+\left(-\frac{\sqrt{3}i}{12}\right)\,g_2\,g_3\,W_\mu^2\,G_\mu^8\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}+\left(-\frac{\sqrt{3}i}{12}\right)\,g_2\,g_3\,W_\mu^2\,G_\mu^8\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}$$

$$\frac{\sqrt{3}i}{6}\,g_2\,g_3\,W_\mu^2\,G_\mu^8\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_b^{-1/3}+\frac{\sqrt{3}i}{12}\,g_2\,g_3\,W_\mu^2\,G_\mu^8\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}+\frac{\sqrt{3}i}{12}\,g_2\,g_3\,W_\mu^2\,G_\mu^8\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\left(-\frac{\sqrt{3}i}{6}\right)\,g_2\,g_3\,W_\mu^2\,G_\mu^8\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_b^{+2/3}+\frac{1}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^1\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}+\frac{1}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^1\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^1\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{-1/3}$$
$$+\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^1\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{-1/3}+\left(-\frac{i}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^2\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{i}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^2\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}+\frac{i}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^2\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{-1/3}+\left(-\frac{i}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^2\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^3\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^3\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^3\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^3\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\frac{1}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^4\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{+2/3}+\frac{1}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^4\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^4\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}$$
$$+\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^4\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_r^{-1/3}+\left(-\frac{i}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^5\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{+2/3}$$

$$\frac{i}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^5\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_r^{+2/3}+\frac{i}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^5\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}+\left(-\frac{i}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^5\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^6\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}+\frac{1}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^6\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_g^{+2/3}+\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^6\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{-1/3}$$

$$\left(-\frac{1}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^6\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_g^{-1/3}+\left(-\frac{i}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^7\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}+\frac{i}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^7\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{i}{4}\,g_2\,g_3\,W_\mu^3\,G_\mu^7\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{-1/3}+\left(-\frac{i}{4}\right)\,g_2\,g_3\,W_\mu^3\,G_\mu^7\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_g^{-1/3}+\frac{\sqrt{3}}{12}\,g_2\,g_3\,W_\mu^3\,G_\mu^8\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\frac{\sqrt{3}}{12}\,g_2\,g_3\,W^3_\mu\,G^8_\mu\,\phi_4{}^{+2/3,\bar{g}\dagger}\,\phi_{4g}^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right)\,g_2\,g_3\,W^3_\mu\,G^8_\mu\,\phi_4{}^{+2/3,\bar{b}\dagger}\,\phi_{4b}^{+2/3} + \left(-\frac{\sqrt{3}}{12}\right)\,g_2\,g_3\,W^3_\mu\,G^8_\mu\,\phi_4{}^{-1/3,\bar{r}\dagger}\,\phi_{4r}^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{12}\right)\,g_2\,g_3\,W^3_\mu\,G^8_\mu\,\phi_4{}^{-1/3,\bar{g}\dagger}\,\phi_{4g}^{-1/3} + \frac{\sqrt{3}}{6}\,g_2\,g_3\,W^3_\mu\,G^8_\mu\,\phi_4{}^{-1/3,\bar{b}\dagger}\,\phi_{4b}^{-1/3} + \frac{1}{12}\,g_3\,g_1\,G^1_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{r}\dagger}\,\phi_{4g}^{+2/3}$$

$$\frac{1}{12}\,g_3\,g_1\,G^1_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{g}\dagger}\,\phi_{4r}^{+2/3} + \frac{1}{12}\,g_3\,g_1\,G^1_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{r}\dagger}\,\phi_{4g}^{-1/3} + \frac{1}{12}\,g_3\,g_1\,G^1_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{g}\dagger}\,\phi_{4r}^{-1/3}$$

$$\left(-\frac{i}{12}\right)\,g_3\,g_1\,G^2_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{r}\dagger}\,\phi_{4g}^{+2/3} + \frac{i}{12}\,g_3\,g_1\,G^2_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{g}\dagger}\,\phi_{4r}^{+2/3} + \left(-\frac{i}{12}\right)\,g_3\,g_1\,G^2_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{r}\dagger}\,\phi_{4g}^{-1/3}$$

$$\frac{i}{12}\,g_3\,g_1\,G^2_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{g}\dagger}\,\phi_{4r}^{-1/3} + \frac{1}{12}\,g_3\,g_1\,G^3_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{r}\dagger}\,\phi_{4r}^{+2/3} + \left(-\frac{1}{12}\right)\,g_3\,g_1\,G^3_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{g}\dagger}\,\phi_{4g}^{+2/3}$$

$$\frac{1}{12}\,g_3\,g_1\,G^3_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{r}\dagger}\,\phi_{4r}^{-1/3} + \left(-\frac{1}{12}\right)\,g_3\,g_1\,G^3_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{g}\dagger}\,\phi_{4g}^{-1/3} + \frac{1}{12}\,g_3\,g_1\,G^4_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{r}\dagger}\,\phi_{4b}^{+2/3}$$

$$\frac{1}{12}\,g_3\,g_1\,G^4_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{b}\dagger}\,\phi_{4r}^{+2/3} + \frac{1}{12}\,g_3\,g_1\,G^4_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{r}\dagger}\,\phi_{4b}^{-1/3} + \frac{1}{12}\,g_3\,g_1\,G^4_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{b}\dagger}\,\phi_{4r}^{-1/3}$$

$$\left(-\frac{i}{12}\right)\,g_3\,g_1\,G^5_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{r}\dagger}\,\phi_{4b}^{+2/3} + \frac{i}{12}\,g_3\,g_1\,G^5_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{b}\dagger}\,\phi_{4r}^{+2/3} + \left(-\frac{i}{12}\right)\,g_3\,g_1\,G^5_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{r}\dagger}\,\phi_{4b}^{-1/3}$$

$$\frac{i}{12}\,g_3\,g_1\,G^5_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{b}\dagger}\,\phi_{4r}^{-1/3} + \frac{1}{12}\,g_3\,g_1\,G^6_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{g}\dagger}\,\phi_{4b}^{+2/3} + \frac{1}{12}\,g_3\,g_1\,G^6_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{b}\dagger}\,\phi_{4g}^{+2/3}$$

$$\frac{1}{12}\,g_3\,g_1\,G^6_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{g}\dagger}\,\phi_{4b}^{-1/3} + \frac{1}{12}\,g_3\,g_1\,G^6_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{b}\dagger}\,\phi_{4g}^{-1/3} + \left(-\frac{i}{12}\right)\,g_3\,g_1\,G^7_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{g}\dagger}\,\phi_{4b}^{+2/3}$$

$$\frac{i}{12}\,g_3\,g_1\,G^7_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{b}\dagger}\,\phi_{4g}^{+2/3} + \left(-\frac{i}{12}\right)\,g_3\,g_1\,G^7_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{g}\dagger}\,\phi_{4b}^{-1/3} + \frac{i}{12}\,g_3\,g_1\,G^7_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{b}\dagger}\,\phi_{4g}^{-1/3}$$

$$\frac{\sqrt{3}}{36}\,g_3\,g_1\,G^8_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{r}\dagger}\,\phi_{4r}^{+2/3} + \frac{\sqrt{3}}{36}\,g_3\,g_1\,G^8_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{g}\dagger}\,\phi_{4g}^{+2/3} + \left(-\frac{\sqrt{3}}{18}\right)\,g_3\,g_1\,G^8_\mu\,B_\mu\,\phi_4{}^{+2/3,\bar{b}\dagger}\,\phi_{4b}^{+2/3}$$

$$\frac{\sqrt{3}}{36}\,g_3\,g_1\,G^8_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{r}\dagger}\,\phi_{4r}^{-1/3} + \frac{\sqrt{3}}{36}\,g_3\,g_1\,G^8_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{g}\dagger}\,\phi_{4g}^{-1/3} + \left(-\frac{\sqrt{3}}{18}\right)\,g_3\,g_1\,G^8_\mu\,B_\mu\,\phi_4{}^{-1/3,\bar{b}\dagger}\,\phi_{4b}^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_2\,G^1_\mu\,W^1_\mu\,\phi_4{}^{+2/3,\bar{r}\dagger}\,\phi_{4g}^{-1/3} + \frac{1}{4}\,g_3\,g_2\,G^1_\mu\,W^1_\mu\,\phi_4{}^{+2/3,\bar{g}\dagger}\,\phi_{4r}^{-1/3} + \frac{1}{4}\,g_3\,g_2\,G^1_\mu\,W^1_\mu\,\phi_4{}^{-1/3,\bar{r}\dagger}\,\phi_{4g}^{+2/3}$$

$$\frac{1}{4}\,g_3\,g_2\,G^1_\mu\,W^1_\mu\,\phi_4{}^{-1/3,\bar{g}\dagger}\,\phi_{4r}^{+2/3} + \left(-\frac{i}{4}\right)\,g_3\,g_2\,G^1_\mu\,W^2_\mu\,\phi_4{}^{+2/3,\bar{r}\dagger}\,\phi_{4g}^{-1/3} + \left(-\frac{i}{4}\right)\,g_3\,g_2\,G^1_\mu\,W^2_\mu\,\phi_4{}^{+2/3,\bar{g}\dagger}\,\phi_{4r}^{-1/3}$$

$$\frac{i}{4}\,g_3\,g_2\,G^1_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}+\frac{i}{4}\,g_3\,g_2\,G^1_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}+\frac{1}{4}\,g_3\,g_2\,G^1_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{1}{4}\,g_3\,g_2\,G^1_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^1_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{-1/3}+\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^1_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^2_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_g^{-1/3}+\frac{i}{4}\,g_3\,g_2\,G^2_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_r^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^2_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{i}{4}\,g_3\,g_2\,G^2_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^2_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_g^{-1/3}+\frac{1}{4}\,g_3\,g_2\,G^2_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_2\,G^2_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}+\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^2_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^2_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{i}{4}\,g_3\,g_2\,G^2_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}+\frac{i}{4}\,g_3\,g_2\,G^2_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^2_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_2\,G^3_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}+\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^3_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\frac{1}{4}\,g_3\,g_2\,G^3_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^3_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^3_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}+\frac{i}{4}\,g_3\,g_2\,G^3_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}$$

$$\frac{i}{4}\,g_3\,g_2\,G^3_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^3_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\frac{1}{4}\,g_3\,g_2\,G^3_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}$$

$$\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^3_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^3_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}+\frac{1}{4}\,g_3\,g_2\,G^3_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_2\,G^4_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}+\frac{1}{4}\,g_3\,g_2\,G^4_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_r^{-1/3}+\frac{1}{4}\,g_3\,g_2\,G^4_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_b^{+2/3}$$

$$\frac{1}{4}\,g_3\,g_2\,G^4_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^4_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^4_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{i}{4}\,g_3\,g_2\,G^4_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_b^{+2/3}+\frac{i}{4}\,g_3\,g_2\,G^4_\mu\,W^2_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_r^{+2/3}+\frac{1}{4}\,g_3\,g_2\,G^4_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{+2/3}$$

$$\frac{1}{4}\,g_3\,g_2\,G^4_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^4_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}+\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^4_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^5_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}+\frac{i}{4}\,g_3\,g_2\,G^5_\mu\,W^1_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_r^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_2\,G^5_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_b^{+2/3}$$

$$\frac{i}{4}\,g_3\,g_2\,G^5_\mu\,W^1_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{1}{4}\right)\,g_3\,g_2\,G^5_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}+\frac{1}{4}\,g_3\,g_2\,G^5_\mu\,W^2_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} + \frac{i}{4}\, g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3} + \frac{i}{4}\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{1}{4}\right)\, g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3} + \frac{i}{4}\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \frac{i}{4}\, g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{\sqrt{3}i}{6}\right)\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{\sqrt{3}}{12}\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}+\frac{\sqrt{3}}{12}\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\left(-\frac{\sqrt{3}}{6}\right)\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_4}^{+2/3,\bar{b}\dagger}\,{\phi_4}_b^{+2/3}$$

$$\left(-\frac{\sqrt{3}}{12}\right)\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$
$$+\left(-\frac{\sqrt{3}}{12}\right)\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\frac{\sqrt{3}}{6}\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_4}^{-1/3,\bar{b}\dagger}\,{\phi_4}_b^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\left(-\frac{1}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}+\left(-\frac{1}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^4_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^4_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^5_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^5_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^6_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^6_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^7_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^7_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^1_\mu\,G^8_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^1_\mu\,G^8_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^1_\mu\,G^8_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{-1/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^1_\mu\,G^8_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^2_\mu\,G^1_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}+\frac{i}{4}\,g_3\,g_3\,G^2_\mu\,G^1_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^2_\mu\,G^1_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^2_\mu\,G^1_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^2_\mu\,G^2_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_r^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^2_\mu\,G^2_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^2_\mu\,G^2_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_r^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^2_\mu\,G^2_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_g^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^2_\mu\,G^3_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_g^{+2/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^2_\mu\,G^3_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_r^{+2/3}+\frac{i}{4}\,g_3\,g_3\,G^2_\mu\,G^3_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_g^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^2_\mu\,G^3_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_r^{-1/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^2_\mu\,G^4_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}+\frac{i}{4}\,g_3\,g_3\,G^2_\mu\,G^4_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^2_\mu\,G^5_\mu\,{\phi_4}^{+2/3,\bar{g}\dagger}\,{\phi_4}_b^{+2/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^2_\mu\,G^5_\mu\,{\phi_4}^{-1/3,\bar{g}\dagger}\,{\phi_4}_b^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^2_\mu\,G^6_\mu\,{\phi_4}^{+2/3,\bar{r}\dagger}\,{\phi_4}_b^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^2_\mu\,G^6_\mu\,{\phi_4}^{-1/3,\bar{r}\dagger}\,{\phi_4}_b^{-1/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_3\, G^2_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{r}\,\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^2_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{r}\,\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{r}\,\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{g}\,\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{r}\,\dagger}\, {\phi_4}_g^{-1/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{g}\,\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{r}\,\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{g}\,\dagger}\, {\phi_4}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{r}\,\dagger}\, {\phi_4}_g^{-1/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{g}\,\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{r}\,\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{g}\,\dagger}\, {\phi_4}_r^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{r}\,\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{g}\,\dagger}\, {\phi_4}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{r}\,\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{g}\,\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{r}\,\dagger}\, {\phi_4}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{g}\,\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{r}\,\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{r}\,\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{r}\,\dagger}\, {\phi_4}_b^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{r}\,\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^6_\mu\, {\phi_4}^{+2/3,\bar{g}\,\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^6_\mu\, {\phi_4}^{-1/3,\bar{g}\,\dagger}\, {\phi_4}_b^{-1/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^3_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{g}\,\dagger}\, {\phi_4}_b^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^3_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{g}\,\dagger}\, {\phi_4}_b^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{r}\,\dagger}\, {\phi_4}_r^{+2/3}$$

$$\left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{g}\,\dagger}\, {\phi_4}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{r}\,\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{g}\,\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{b}\,\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{b}\,\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{b}\,\dagger}\, {\phi_4}_g^{+2/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{b}\,\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{b}\,\dagger}\, {\phi_4}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{b}\,\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{r}\,\dagger}\, {\phi_4}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{b}\,\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{r}\,\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^4_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^4_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^6_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^6_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^4_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} \\ + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^4_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^5_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^5_\mu\, G^6_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^5_\mu\, G^6_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_g^{-1/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{\sqrt{3}}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_b^{+2/3}$$

$$\left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_r^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3} + \left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_b^{+2/3} + \left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_g^{+2/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_b^{-1/3} + \left(-\frac{\sqrt{3}i}{6}\right) g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{r}\dagger}\, {\phi_4}_r^{+2/3}$$

$$\frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{g}\dagger}\, {\phi_4}_g^{+2/3} + \frac{1}{3}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_4}^{+2/3,\bar{b}\dagger}\, {\phi_4}_b^{+2/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{r}\dagger}\, {\phi_4}_r^{-1/3}$$

$$\frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{g}\dagger}\, {\phi_4}_g^{-1/3} + \frac{1}{3}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_4}^{-1/3,\bar{b}\dagger}\, {\phi_4}_b^{-1/3} + \frac{1}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{1}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{1}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \frac{1}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{1}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{1}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \frac{1}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{1}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \frac{1}{9}\, g_1\, g_1\, B_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3}$$
$$+\left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$
$$+\left(-\frac{\sqrt{2}}{6}\right) g_1\, g_2\, B_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{6}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{\sqrt{2}i}{6}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}i}{6}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$
$$+\left(-\frac{\sqrt{2}i}{6}\right) g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{\sqrt{2}i}{6}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$
$$+\left(-\frac{\sqrt{2}i}{6}\right) g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{2}i}{6}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{\sqrt{2}i}{6}\right) g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{2}i}{6}\, g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$
$$+\left(-\frac{\sqrt{2}i}{6}\right) g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{6}\right) g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$
$$+\left(-\frac{\sqrt{2}i}{6}\right) g_1\, g_2\, B_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{1}{3}\right) g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{1}{3}\right) g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{3}\right) g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \frac{1}{3}\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{1}{3}\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \frac{1}{3}\, g_1\, g_2\, B_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3}$$
$$+\left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{1}{6}\right) g_1\, g_3\, B_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \frac{i}{6}\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{i}{6}\right) g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{i}{6}\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{i}{6}\right) g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{i}{6}\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{i}{6}\right)\, g_1\, g_3\, B_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{1}{6}\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \frac{1}{6}\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \frac{1}{6}\, g_1\, g_3\, B_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3}$$
$$+ \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \frac{i}{6}\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\left(-\frac{i}{6}\right)\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \frac{i}{6}\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{i}{6}\right)\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{i}{6}\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{i}{6}\right)\, g_1\, g_3\, B_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3}$$
$$+ \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{1}{6}\right)\, g_1\, g_3\, B_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \frac{i}{6}\, g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\left(-\frac{i}{6}\right)\, g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{i}{6}\, g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{i}{6}\right)\, g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{i}{6}\, g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{i}{6}\right)\, g_1\, g_3\, B_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{\sqrt{3}}{18}\right)\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{\sqrt{3}}{18}\right)\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{3}}{9}\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3}$$
$$+ \left(-\frac{\sqrt{3}}{18}\right)\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{18}\right)\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{3}}{9}\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$
$$+ \left(-\frac{\sqrt{3}}{18}\right)\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{\sqrt{3}}{18}\right) g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{3}}{9}\, g_1\, g_3\, B_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$
$$+ \left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$
$$+ \left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3}$$
$$+ \left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}}{6}\right) g_2\, g_1\, W^1_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{6}\, g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{\sqrt{2}i}{6}\, g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}i}{6}\, g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}i}{6}\right) g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{\sqrt{2}i}{6}\, g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$
$$+ \left(-\frac{\sqrt{2}i}{6}\right) g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{2}i}{6}\, g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{\sqrt{2}i}{6}\right) g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{2}i}{6}\, g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$
$$+ \left(-\frac{\sqrt{2}i}{6}\right) g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{6}\right) g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}i}{6}\right) g_2\, g_1\, W^2_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{1}{3}\right) g_2\, g_1\, W^3_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{1}{3}\right)\,g_2\,g_1\,W_\mu^3\,B_\mu\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}+\left(-\frac{1}{3}\right)\,g_2\,g_1\,W_\mu^3\,B_\mu\,{\phi_5}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}+\frac{1}{3}\,g_2\,g_1\,W_\mu^3\,B_\mu\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}$$

$$\frac{1}{3}\,g_2\,g_1\,W_\mu^3\,B_\mu\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}+\frac{1}{3}\,g_2\,g_1\,W_\mu^3\,B_\mu\,{\phi_5}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}+\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}$$

$$\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}+\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}+\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}+g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{-1/3,\bar{r}\dagger}\,{\phi_5}_r^{-1/3}$$

$$g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{-1/3,\bar{g}\dagger}\,{\phi_5}_g^{-1/3}+g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{-1/3,\bar{b}\dagger}\,{\phi_5}_b^{-1/3}+\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}$$

$$\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}+\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}+\frac{1}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^1\,{\phi_5}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}+\frac{i}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}$$

$$\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\frac{i}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}+\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\frac{i}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}+\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}+\frac{i}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}$$

$$\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\frac{i}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}+\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\frac{i}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}+\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^1\,W_\mu^2\,{\phi_5}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}+\frac{\sqrt{2}}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^3\,{\phi_5}^{-1/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}$$

$$\left(-\frac{\sqrt{2}}{2}\right)\,g_2\,g_2\,W_\mu^1\,W_\mu^3\,{\phi_5}^{-1/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\frac{\sqrt{2}}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^3\,{\phi_5}^{-1/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}$$
$$+\left(-\frac{\sqrt{2}}{2}\right)\,g_2\,g_2\,W_\mu^1\,W_\mu^3\,{\phi_5}^{-1/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{2}}{2}\,g_2\,g_2\,W_\mu^1\,W_\mu^3\,{\phi_5}^{-1/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}$$
$$+\left(-\frac{\sqrt{2}}{2}\right)\,g_2\,g_2\,W_\mu^1\,W_\mu^3\,{\phi_5}^{-1/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}+\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^2\,W_\mu^1\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}$$

$$\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^2\,W_\mu^1\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}$$
$$+\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^2\,W_\mu^1\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}+\left(-\frac{i}{2}\right)\,g_2\,g_2\,W_\mu^2\,W_\mu^1\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\left(-\frac{i}{2}\right) g_2\, g_2\, W_\mu^2\, W_\mu^1\, \phi_5{}^{+2/3,\bar{b}\dagger}\, \phi_{5b}{}^{+2/3} + \left(-\frac{i}{2}\right) g_2\, g_2\, W_\mu^2\, W_\mu^1\, \phi_5{}^{+2/3,\bar{b}\dagger}\, \phi_{5b}{}^{-4/3} + \frac{i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^1\, \phi_5{}^{-4/3,\bar{r}\dagger}\, \phi_{5r}{}^{+2/3}$$

$$\frac{i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^1\, \phi_5{}^{-4/3,\bar{r}\dagger}\, \phi_{5r}{}^{-4/3} + \frac{i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^1\, \phi_5{}^{-4/3,\bar{g}\dagger}\, \phi_{5g}{}^{+2/3} + \frac{i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^1\, \phi_5{}^{-4/3,\bar{g}\dagger}\, \phi_{5g}{}^{-4/3}$$

$$\frac{i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^1\, \phi_5{}^{-4/3,\bar{b}\dagger}\, \phi_{5b}{}^{+2/3} + \frac{i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^1\, \phi_5{}^{-4/3,\bar{b}\dagger}\, \phi_{5b}{}^{-4/3} + \frac{1}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{+2/3,\bar{r}\dagger}\, \phi_{5r}{}^{+2/3}$$

$$\left(-\frac{1}{2}\right) g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{+2/3,\bar{r}\dagger}\, \phi_{5r}{}^{-4/3} + \frac{1}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{+2/3,\bar{g}\dagger}\, \phi_{5g}{}^{+2/3} + \left(-\frac{1}{2}\right) g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{+2/3,\bar{g}\dagger}\, \phi_{5g}{}^{-4/3}$$

$$\frac{1}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{+2/3,\bar{b}\dagger}\, \phi_{5b}{}^{+2/3} + \left(-\frac{1}{2}\right) g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{+2/3,\bar{b}\dagger}\, \phi_{5b}{}^{-4/3} + g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{-1/3,\bar{r}\dagger}\, \phi_{5r}{}^{-1/3}$$

$$g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{-1/3,\bar{g}\dagger}\, \phi_{5g}{}^{-1/3} + g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{-1/3,\bar{b}\dagger}\, \phi_{5b}{}^{-1/3} + \left(-\frac{1}{2}\right) g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{-4/3,\bar{r}\dagger}\, \phi_{5r}{}^{+2/3}$$

$$\frac{1}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{-4/3,\bar{r}\dagger}\, \phi_{5r}{}^{-4/3} + \left(-\frac{1}{2}\right) g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{-4/3,\bar{g}\dagger}\, \phi_{5g}{}^{+2/3} + \frac{1}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{-4/3,\bar{g}\dagger}\, \phi_{5g}{}^{-4/3}$$

$$\left(-\frac{1}{2}\right) g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{-4/3,\bar{b}\dagger}\, \phi_{5b}{}^{+2/3} + \frac{1}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^2\, \phi_5{}^{-4/3,\bar{b}\dagger}\, \phi_{5b}{}^{-4/3} + \frac{\sqrt{2}i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, \phi_5{}^{-1/3,\bar{r}\dagger}\, \phi_{5r}{}^{+2/3}$$

$$\frac{\sqrt{2}i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, \phi_5{}^{-1/3,\bar{r}\dagger}\, \phi_{5r}{}^{-4/3} + \frac{\sqrt{2}i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, \phi_5{}^{-1/3,\bar{g}\dagger}\, \phi_{5g}{}^{+2/3} + \frac{\sqrt{2}i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, \phi_5{}^{-1/3,\bar{g}\dagger}\, \phi_{5g}{}^{-4/3}$$

$$\frac{\sqrt{2}i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, \phi_5{}^{-1/3,\bar{b}\dagger}\, \phi_{5b}{}^{+2/3} + \frac{\sqrt{2}i}{2}\, g_2\, g_2\, W_\mu^2\, W_\mu^3\, \phi_5{}^{-1/3,\bar{b}\dagger}\, \phi_{5b}{}^{-4/3} + \frac{\sqrt{2}}{2}\, g_2\, g_2\, W_\mu^3\, W_\mu^1\, \phi_5{}^{+2/3,\bar{r}\dagger}\, \phi_{5r}{}^{-1/3}$$

$$\frac{\sqrt{2}}{2}\, g_2\, g_2\, W_\mu^3\, W_\mu^1\, \phi_5{}^{+2/3,\bar{g}\dagger}\, \phi_{5g}{}^{-1/3} + \frac{\sqrt{2}}{2}\, g_2\, g_2\, W_\mu^3\, W_\mu^1\, \phi_5{}^{+2/3,\bar{b}\dagger}\, \phi_{5b}{}^{-1/3} + \left(-\frac{\sqrt{2}}{2}\right) g_2\, g_2\, W_\mu^3\, W_\mu^1\, \phi_5{}^{-4/3,\bar{r}\dagger}\, \phi_{5r}{}^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{2}\right) g_2\, g_2\, W_\mu^3\, W_\mu^1\, \phi_5{}^{-4/3,\bar{g}\dagger}\, \phi_{5g}{}^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}}{2}\right) g_2\, g_2\, W_\mu^3\, W_\mu^1\, \phi_5{}^{-4/3,\bar{b}\dagger}\, \phi_{5b}{}^{-1/3} + \left(-\frac{\sqrt{2}i}{2}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, \phi_5{}^{+2/3,\bar{r}\dagger}\, \phi_{5r}{}^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{2}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, \phi_5{}^{+2/3,\bar{g}\dagger}\, \phi_{5g}{}^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}i}{2}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, \phi_5{}^{+2/3,\bar{b}\dagger}\, \phi_{5b}{}^{-1/3} + \left(-\frac{\sqrt{2}i}{2}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, \phi_5{}^{-4/3,\bar{r}\dagger}\, \phi_{5r}{}^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{2}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, \phi_5{}^{-4/3,\bar{g}\dagger}\, \phi_{5g}{}^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}i}{2}\right) g_2\, g_2\, W_\mu^3\, W_\mu^2\, \phi_5{}^{-4/3,\bar{b}\dagger}\, \phi_{5b}{}^{-1/3} + g_2\, g_2\, W_\mu^3\, W_\mu^3\, \phi_5{}^{+2/3,\bar{r}\dagger}\, \phi_{5r}{}^{+2/3}$$

$$g_2\, g_2\, W^3_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + g_2\, g_2\, W^3_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + g_2\, g_2\, W^3_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$

$$g_2\, g_2\, W^3_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + g_2\, g_2\, W^3_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^1_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^1_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^1_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^1_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{2}}{4}\right)\, g_2\, g_3\, W^1_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{\sqrt{2}}{4}\right)\, g_2\, g_3\, W^1_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{\sqrt{2}}{4}\right)\, g_2\, g_3\, W^1_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{2}}{4}\right)\, g_2\, g_3\, W^1_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^1_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W^1_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W^1_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W^1_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^1_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W^1_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W^1_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W^1_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W^1_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^1_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{6}}{12}\, g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{6}}{12}\, g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{\sqrt{6}}{6}\right) g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{6}}{12}\, g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{6}}{12}\, g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{\sqrt{6}}{12}\, g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{6}}{12}\, g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{\sqrt{6}}{6}\right) g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\left(-\frac{\sqrt{6}}{6}\right) g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{6}}{12}\, g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{6}}{12}\, g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{\sqrt{6}}{6}\right) g_2\, g_3\, W^1_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W^2_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W^2_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3}$$
$$+\left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$
$$+\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^2_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^2_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3}$$
$$+\left(-\frac{\sqrt{2}}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W^2_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W^2_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$
$$+\left(-\frac{\sqrt{2}}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$
$$+\left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$
$$+\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$
$$+\left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3}$$
$$+\left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3}$$
$$+\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W^2_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{\sqrt{2}}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{\sqrt{2}}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^5\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^6\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^6\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^6\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^6\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^6\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^6\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^6\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{2}i}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^6\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{2}}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^7\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^7\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^7\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{2}}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^7\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{\sqrt{2}}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^7\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\frac{\sqrt{2}}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^7\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}}{4}\, g_2\, g_3\, W_\mu^2\, G_\mu^7\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{\sqrt{2}}{4}\right) g_2\, g_3\, W_\mu^2\, G_\mu^7\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{\sqrt{6}i}{12}\right) g_2\, g_3\, W_\mu^2\, G_\mu^8\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{6}i}{12}\right) g_2\, g_3\, W_\mu^2\, G_\mu^8\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{6}i}{6}\, g_2\, g_3\, W_\mu^2\, G_\mu^8\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{6}i}{12}\, g_2\, g_3\, W_\mu^2\, G_\mu^8\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{\sqrt{6}i}{12}\right) g_2\, g_3\, W_\mu^2\, G_\mu^8\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{6}i}{12}\, g_2\, g_3\, W_\mu^2\, G_\mu^8\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{\sqrt{6}i}{12}\right) g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{\sqrt{6}i}{6}\right) g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{6}i}{6} g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{6}i}{12} g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{6}i}{12} g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{6}i}{6}\right) g_2\, g_3\, W^2_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{1}{2} g_2\, g_3\, W^3_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{1}{2} g_2\, g_3\, W^3_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{1}{2}\right) g_2\, g_3\, W^3_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{1}{2}\right) g_2\, g_3\, W^3_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{i}{2}\right) g_2\, g_3\, W^3_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{i}{2} g_2\, g_3\, W^3_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{i}{2} g_2\, g_3\, W^3_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{i}{2}\right) g_2\, g_3\, W^3_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \frac{1}{2} g_2\, g_3\, W^3_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{1}{2}\right) g_2\, g_3\, W^3_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{2}\right) g_2\, g_3\, W^3_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \frac{1}{2} g_2\, g_3\, W^3_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{1}{2} g_2\, g_3\, W^3_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \frac{1}{2} g_2\, g_3\, W^3_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{1}{2}\right) g_2\, g_3\, W^3_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{1}{2}\right) g_2\, g_3\, W^3_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{i}{2}\right) g_2\, g_3\, W^3_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \frac{i}{2} g_2\, g_3\, W^3_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{i}{2} g_2\, g_3\, W^3_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{i}{2}\right) g_2\, g_3\, W^3_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \frac{1}{2} g_2\, g_3\, W^3_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\frac{1}{2} g_2\, g_3\, W^3_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{2}\right) g_2\, g_3\, W^3_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{1}{2}\right) g_2\, g_3\, W^3_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{i}{2}\right) g_2\, g_3\, W^3_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \frac{i}{2} g_2\, g_3\, W^3_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{i}{2} g_2\, g_3\, W^3_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{i}{2}\right) g_2\, g_3\, W^3_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{3}}{6} g_2\, g_3\, W^3_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{3}}{6} g_2\, g_3\, W^3_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{\sqrt{3}}{3}\right) g_2\, g_3\, W^3_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right) g_2\, g_3\, W^3_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{\sqrt{3}}{6}\right) g_2\, g_3\, W^3_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{3}}{3} g_2\, g_3\, W^3_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{1}{6}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{1}{6}\right) g_3\, g_1\, G^1_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \frac{i}{6}\, g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{i}{6}\right) g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{i}{6}\, g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{i}{6}\right) g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{i}{6}\, g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{i}{6}\right) g_3\, g_1\, G^2_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \frac{1}{6}\, g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{1}{6}\right) g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \frac{1}{6}\, g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{1}{6}\, g_3\, g_1\, G^3_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\left(-\frac{1}{6}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{1}{6}\right) g_3\, g_1\, G^4_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \frac{i}{6}\, g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{i}{6}\right) g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{i}{6}\, g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{i}{6}\right) g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \frac{i}{6}\, g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{i}{6}\right) g_3\, g_1\, G^5_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{1}{6}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{1}{6}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{1}{6}\right) g_3\, g_1\, G^6_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \frac{i}{6}\, g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{i}{6}\right) g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\frac{i}{6}\, g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{i}{6}\right) g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \frac{i}{6}\, g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{i}{6}\right) g_3\, g_1\, G^7_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$
$$+ \left(-\frac{\sqrt{3}}{18}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{\sqrt{3}}{18}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\frac{\sqrt{3}}{9}\, g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3}$$
$$+ \left(-\frac{\sqrt{3}}{18}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{3}}{18}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{3}}{9}\, g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$
$$+ \left(-\frac{\sqrt{3}}{18}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{\sqrt{3}}{18}\right) g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{3}}{9}\, g_3\, g_1\, G^8_\mu\, B_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^1_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3}$$
$$+ \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^1_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{1}{2}\, g_3\, g_2\, G^1_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\frac{1}{2}\, g_3\, g_2\, G^1_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{1}{2}\right) g_3\, g_2\, G^1_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{1}{2}\right) g_3\, g_2\, G^1_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^2_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^2_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{i}{2}\right) g_3\, g_2\, G^2_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{i}{2}\, g_3\, g_2\, G^2_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{i}{2}\, g_3\, g_2\, G^2_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{i}{2}\right) g_3\, g_2\, G^2_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^3_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^3_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{1}{2}\, g_3\, g_2\, G^3_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{r}\,\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{1}{2}\right)\, g_3\, g_2\, G^3_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{g}\,\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{2}\right)\, g_3\, g_2\, G^3_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{r}\,\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{1}{2}\, g_3\, g_2\, G^3_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{g}\,\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\,\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\,\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^4_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}i}{4}\right)\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\,\dagger}\, {\phi_5}_b^{+2/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\,\dagger}\, {\phi_5}_r^{+2/3}$$
$$+ \left(-\frac{\sqrt{2}i}{4}\right)\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^4_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-1/3} + \frac{1}{2}\, g_3\, g_2\, G^4_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{r}\,\dagger}\, {\phi_5}_b^{+2/3}$$

$$\frac{1}{2}\, g_3\, g_2\, G^4_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{b}\,\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{1}{2}\right)\, g_3\, g_2\, G^4_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{1}{2}\right)\, g_3\, g_2\, G^4_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}i}{4}\right)\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\,\dagger}\, {\phi_5}_b^{+2/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\,\dagger}\, {\phi_5}_r^{+2/3}$$
$$+ \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right)\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^5_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}}{4}\right)\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{b}\,\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\,\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{2}}{4}\right)\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\,\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^5_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{i}{2}\right) g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \frac{i}{2}\, g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3}$$

$$\frac{i}{2}\, g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{i}{2}\right) g_3\, g_2\, G^5_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^6_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^6_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{1}{2}\, g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \frac{1}{2}\, g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{2}\right) g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{1}{2}\right) g_3\, g_2\, G^6_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{\sqrt{2}i}{4}\right) g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}i}{4}\, g_3\, g_2\, G^7_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$
$$+ \left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{2}}{4}\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{2}}{4}\, g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$
$$+ \left(-\frac{\sqrt{2}}{4}\right) g_3\, g_2\, G^7_\mu\, W^2_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{i}{2}\right) g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\frac{i}{2}\, g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{i}{2}\, g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{i}{2}\right) g_3\, g_2\, G^7_\mu\, W^3_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{6}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{6}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{6}}{6}\right) g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{6}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{6}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{6}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\frac{\sqrt{6}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$
$$+ \left(-\frac{\sqrt{6}}{6}\right) g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{6}}{6}\right) g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{6}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{6}}{12}\, g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{6}}{6}\right) g_3\, g_2\, G^8_\mu\, W^1_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\left(-\frac{\sqrt{6}i}{12}\right) g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$
$$+ \left(-\frac{\sqrt{6}i}{12}\right) g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{6}i}{6}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{6}i}{12}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3}$$
$$+ \left(-\frac{\sqrt{6}i}{12}\right) g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \frac{\sqrt{6}i}{12}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3}$$

$$\left(-\frac{\sqrt{6}i}{12}\right) g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$
$$+ \left(-\frac{\sqrt{6}i}{6}\right) g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{6}i}{6}\, g_3\, g_2\, G^8_\mu\, W^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{6}i}{12}\,g_3\,g_2\,G^8_\mu\,W^2_\mu\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-1/3}+\frac{\sqrt{6}i}{12}\,g_3\,g_2\,G^8_\mu\,W^2_\mu\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_g^{-1/3}+\left(-\frac{\sqrt{6}i}{6}\right)\,g_3\,g_2\,G^8_\mu\,W^2_\mu\,{\phi_5}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{3}}{6}\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}+\frac{\sqrt{3}}{6}\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}+\left(-\frac{\sqrt{3}}{3}\right)\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_5}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right)\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\left(-\frac{\sqrt{3}}{6}\right)\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}+\frac{\sqrt{3}}{3}\,g_3\,g_2\,G^8_\mu\,W^3_\mu\,{\phi_5}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_5}^{-1/3,\bar{r}\dagger}\,{\phi_5}_r^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_5}^{-1/3,\bar{g}\dagger}\,{\phi_5}_g^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^1_\mu\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_g^{+2/3}+\frac{i}{4}\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_5}^{-1/3,\bar{r}\dagger}\,{\phi_5}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_5}^{-1/3,\bar{g}\dagger}\,{\phi_5}_g^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^2_\mu\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\left(-\frac{1}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_g^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_r^{+2/3}+\left(-\frac{1}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_5}^{-1/3,\bar{r}\dagger}\,{\phi_5}_g^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_5}^{-1/3,\bar{g}\dagger}\,{\phi_5}_r^{-1/3}+\left(-\frac{1}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_g^{-4/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^3_\mu\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_r^{-4/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^4_\mu\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_b^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^4_\mu\,{\phi_5}^{-1/3,\bar{g}\dagger}\,{\phi_5}_b^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^4_\mu\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_b^{-4/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^5_\mu\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_b^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^5_\mu\,{\phi_5}^{-1/3,\bar{g}\dagger}\,{\phi_5}_b^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^5_\mu\,{\phi_5}^{-4/3,\bar{g}\dagger}\,{\phi_5}_b^{-4/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^6_\mu\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_b^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^6_\mu\,{\phi_5}^{-1/3,\bar{r}\dagger}\,{\phi_5}_b^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^1_\mu\,G^6_\mu\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_b^{-4/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^7_\mu\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_b^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^7_\mu\,{\phi_5}^{-1/3,\bar{r}\dagger}\,{\phi_5}_b^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^1_\mu\,G^7_\mu\,{\phi_5}^{-4/3,\bar{r}\dagger}\,{\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^1_\mu\,G^8_\mu\,{\phi_5}^{+2/3,\bar{r}\dagger}\,{\phi_5}_g^{+2/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^1_\mu\,G^8_\mu\,{\phi_5}^{+2/3,\bar{g}\dagger}\,{\phi_5}_r^{+2/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^1_\mu\,G^8_\mu\,{\phi_5}^{-1/3,\bar{r}\dagger}\,{\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^1_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^1_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^1_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^2_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^2_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^2_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^2_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^2_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^2_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^2_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^2_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_3\, G^2_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^2_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^2_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^2_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^3_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^3_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^3_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^3_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^3_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^3_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \left(-\frac{\sqrt{3}}{12}\right) g_3\, g_3\, G^3_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^4_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^4_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^3_\mu\,\phi_5{}^{+2/3,\bar{b}\dagger}\,{\phi_5}_r^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^3_\mu\,\phi_5{}^{-1/3,\bar{b}\dagger}\,{\phi_5}_r^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^3_\mu\,\phi_5{}^{-4/3,\bar{b}\dagger}\,{\phi_5}_r^{-4/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^4_\mu\,\phi_5{}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^4_\mu\,\phi_5{}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^4_\mu\,\phi_5{}^{-1/3,\bar{r}\dagger}\,{\phi_5}_r^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^4_\mu\,\phi_5{}^{-1/3,\bar{b}\dagger}\,{\phi_5}_b^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^4_\mu\,\phi_5{}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^4_\mu\,\phi_5{}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^4_\mu\,G^5_\mu\,\phi_5{}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^4_\mu\,G^5_\mu\,\phi_5{}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}+\frac{i}{4}\,g_3\,g_3\,G^4_\mu\,G^5_\mu\,\phi_5{}^{-1/3,\bar{r}\dagger}\,{\phi_5}_r^{-1/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^4_\mu\,G^5_\mu\,\phi_5{}^{-1/3,\bar{b}\dagger}\,{\phi_5}_b^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^4_\mu\,G^5_\mu\,\phi_5{}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^4_\mu\,G^5_\mu\,\phi_5{}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^6_\mu\,\phi_5{}^{+2/3,\bar{r}\dagger}\,{\phi_5}_g^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^6_\mu\,\phi_5{}^{-1/3,\bar{r}\dagger}\,{\phi_5}_g^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^4_\mu\,G^6_\mu\,\phi_5{}^{-4/3,\bar{r}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^4_\mu\,G^7_\mu\,\phi_5{}^{+2/3,\bar{r}\dagger}\,{\phi_5}_g^{+2/3}+\frac{i}{4}\,g_3\,g_3\,G^4_\mu\,G^7_\mu\,\phi_5{}^{-1/3,\bar{r}\dagger}\,{\phi_5}_g^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^4_\mu\,G^7_\mu\,\phi_5{}^{-4/3,\bar{r}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right)\,g_3\,g_3\,G^4_\mu\,G^8_\mu\,\phi_5{}^{+2/3,\bar{r}\dagger}\,{\phi_5}_b^{+2/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^4_\mu\,G^8_\mu\,\phi_5{}^{+2/3,\bar{b}\dagger}\,{\phi_5}_r^{+2/3}$$
$$+\left(-\frac{\sqrt{3}}{6}\right)\,g_3\,g_3\,G^4_\mu\,G^8_\mu\,\phi_5{}^{-1/3,\bar{r}\dagger}\,{\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^4_\mu\,G^8_\mu\,\phi_5{}^{-1/3,\bar{b}\dagger}\,{\phi_5}_r^{-1/3}+\left(-\frac{\sqrt{3}}{6}\right)\,g_3\,g_3\,G^4_\mu\,G^8_\mu\,\phi_5{}^{-4/3,\bar{r}\dagger}\,{\phi_5}_b^{-4/3}+\frac{\sqrt{3}}{12}\,g_3\,g_3\,G^4_\mu\,G^8_\mu\,\phi_5{}^{-4/3,\bar{b}\dagger}\,{\phi_5}_r^{-4/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^1_\mu\,\phi_5{}^{+2/3,\bar{b}\dagger}\,{\phi_5}_g^{+2/3}+\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^1_\mu\,\phi_5{}^{-1/3,\bar{b}\dagger}\,{\phi_5}_g^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^1_\mu\,\phi_5{}^{-4/3,\bar{b}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^2_\mu\,\phi_5{}^{+2/3,\bar{b}\dagger}\,{\phi_5}_g^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^2_\mu\,\phi_5{}^{-1/3,\bar{b}\dagger}\,{\phi_5}_g^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^2_\mu\,\phi_5{}^{-4/3,\bar{b}\dagger}\,{\phi_5}_g^{-4/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^3_\mu\,\phi_5{}^{+2/3,\bar{b}\dagger}\,{\phi_5}_r^{+2/3}+\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^3_\mu\,\phi_5{}^{-1/3,\bar{b}\dagger}\,{\phi_5}_r^{-1/3}+\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^3_\mu\,\phi_5{}^{-4/3,\bar{b}\dagger}\,{\phi_5}_r^{-4/3}$$

$$\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^5_\mu\,G^4_\mu\,\phi_5{}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}+\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^4_\mu\,\phi_5{}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^5_\mu\,G^4_\mu\,\phi_5{}^{-1/3,\bar{r}\dagger}\,{\phi_5}_r^{-1/3}$$

$$\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^4_\mu\,\phi_5{}^{-1/3,\bar{b}\dagger}\,{\phi_5}_b^{-1/3}+\left(-\frac{i}{4}\right)\,g_3\,g_3\,G^5_\mu\,G^4_\mu\,\phi_5{}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\frac{i}{4}\,g_3\,g_3\,G^5_\mu\,G^4_\mu\,\phi_5{}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^5_\mu\,\phi_5{}^{+2/3,\bar{r}\dagger}\,{\phi_5}_r^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^5_\mu\,\phi_5{}^{+2/3,\bar{b}\dagger}\,{\phi_5}_b^{+2/3}+\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^5_\mu\,\phi_5{}^{-1/3,\bar{r}\dagger}\,{\phi_5}_r^{-1/3}$$

$$\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^5_\mu\,\phi_5{}^{-1/3,\bar{b}\dagger}\,{\phi_5}_b^{-1/3}+\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^5_\mu\,\phi_5{}^{-4/3,\bar{r}\dagger}\,{\phi_5}_r^{-4/3}+\frac{1}{4}\,g_3\,g_3\,G^5_\mu\,G^5_\mu\,\phi_5{}^{-4/3,\bar{b}\dagger}\,{\phi_5}_b^{-4/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^5_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^5_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^5_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^5_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^5_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{1}{4}\right) g_3\, g_3\, G^6_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^6_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{1}{4}\right) g_3\, g_3\, G^6_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^6_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\left(-\frac{i}{4}\right) g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \left(-\frac{i}{4}\right) g_3\, g_3\, G^6_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{\sqrt{3}}{6}\right) g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^6_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{1}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{1}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \left(-\frac{i}{4}\right)\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \frac{i}{4}\, g_3\, g_3\, G^7_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \frac{1}{4}\, g_3\, g_3\, G^7_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$

$$\frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{3}i}{6}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^7_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^1_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_g^{-1/3}$$

$$\frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_r^{-1/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_g^{-4/3} + \frac{\sqrt{3}i}{12}\, g_3\, g_3\, G^8_\mu\, G^2_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{\sqrt{3}}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3}$$
$$+ \left(-\frac{\sqrt{3}}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^3_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3}$$
$$+ \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^4_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_b^{+2/3}$$
$$+ \left(-\frac{\sqrt{3}i}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_r^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\left(-\frac{\sqrt{3}i}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_r^{-1/3}$$
$$+ \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{\sqrt{3}i}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^5_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_r^{-4/3}$$

$$\frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3} + \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3} + \frac{\sqrt{3}}{12}\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3}$$
$$+ \left(-\frac{\sqrt{3}}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^6_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_b^{+2/3}$$
$$+ \left(-\frac{\sqrt{3}i}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_g^{+2/3} + \left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\left(-\frac{\sqrt{3}i}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_g^{-1/3}$$

$$+\left(-\frac{\sqrt{3}i}{12}\right)\, g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_b^{-4/3} + \left(-\frac{\sqrt{3}i}{6}\right)\, g_3\, g_3\, G^8_\mu\, G^7_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_g^{-4/3}$$

$$\frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{r}\dagger}\, {\phi_5}_r^{+2/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{g}\dagger}\, {\phi_5}_g^{+2/3} + \frac{1}{3}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_5}^{+2/3,\bar{b}\dagger}\, {\phi_5}_b^{+2/3}$$

$$\frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{r}\dagger}\, {\phi_5}_r^{-1/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{g}\dagger}\, {\phi_5}_g^{-1/3} + \frac{1}{3}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_5}^{-1/3,\bar{b}\dagger}\, {\phi_5}_b^{-1/3}$$

$$\frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{r}\dagger}\, {\phi_5}_r^{-4/3} + \frac{1}{12}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{g}\dagger}\, {\phi_5}_g^{-4/3} + \frac{1}{3}\, g_3\, g_3\, G^8_\mu\, G^8_\mu\, {\phi_5}^{-4/3,\bar{b}\dagger}\, {\phi_5}_b^{-4/3}$$

$$-g_2\, (\partial_\mu W^1_\nu) W^{2\mu} W^{3\nu} + -g_2\, (\partial_\mu W^2_\nu) W^{3\mu} W^{1\nu} + -g_2\, (\partial_\mu W^3_\nu) W^{1\mu} W^{2\nu}$$

$$g_2\, (\partial_\mu W^2_\nu) W^{1\mu} W^{3\nu} + g_2\, (\partial_\mu W^3_\nu) W^{2\mu} W^{1\nu} + g_2\, (\partial_\mu W^1_\nu) W^{3\mu} W^{2\nu}$$

$$-\frac{g_2^2}{4}\, W^1_\mu W^2_\nu W^{1\mu} W^{2\nu} + \frac{g_2^2}{4}\, W^1_\mu W^2_\nu W^{2\mu} W^{1\nu} + -\frac{g_2^2}{4}\, W^2_\mu W^3_\nu W^{2\mu} W^{3\nu}$$

$$\frac{g_2^2}{4}\, W^2_\mu W^3_\nu W^{3\mu} W^{2\nu} + -\frac{g_2^2}{4}\, W^3_\mu W^1_\nu W^{3\mu} W^{1\nu} + \frac{g_2^2}{4}\, W^3_\mu W^1_\nu W^{1\mu} W^{3\nu}$$

$$\frac{g_2^2}{4}\, W^2_\mu W^1_\nu W^{1\mu} W^{2\nu} + -\frac{g_2^2}{4}\, W^2_\mu W^1_\nu W^{2\mu} W^{1\nu} + \frac{g_2^2}{4}\, W^3_\mu W^2_\nu W^{2\mu} W^{3\nu}$$

$$-\frac{g_2^2}{4}\, W^3_\mu W^2_\nu W^{3\mu} W^{2\nu} + \frac{g_2^2}{4}\, W^1_\mu W^3_\nu W^{3\mu} W^{1\nu} + -\frac{g_2^2}{4}\, W^1_\mu W^3_\nu W^{1\mu} W^{3\nu}$$

$$-g_3\, (\partial_\mu G^1_\nu) G^{2\mu} G^{3\nu} + g_3\, (\partial_\mu G^1_\nu) G^{3\mu} G^{2\nu} + -\frac{g_3}{2}\, (\partial_\mu G^1_\nu) G^{4\mu} G^{7\nu}$$

$$\frac{g_3}{2}\, (\partial_\mu G^1_\nu) G^{5\mu} G^{6\nu} + -\frac{g_3}{2}\, (\partial_\mu G^1_\nu) G^{6\mu} G^{5\nu} + \frac{g_3}{2}\, (\partial_\mu G^1_\nu) G^{7\mu} G^{4\nu}$$

$$g_3\, (\partial_\mu G^2_\nu) G^{1\mu} G^{3\nu} + -g_3\, (\partial_\mu G^2_\nu) G^{3\mu} G^{1\nu} + -\frac{g_3}{2}\, (\partial_\mu G^2_\nu) G^{4\mu} G^{6\nu}$$

$$-\frac{g_3}{2}\, (\partial_\mu G^2_\nu) G^{5\mu} G^{7\nu} + \frac{g_3}{2}\, (\partial_\mu G^2_\nu) G^{6\mu} G^{4\nu} + \frac{g_3}{2}\, (\partial_\mu G^2_\nu) G^{7\mu} G^{5\nu}$$

$$-g_3\, (\partial_\mu G^3_\nu) G^{1\mu} G^{2\nu} + g_3\, (\partial_\mu G^3_\nu) G^{2\mu} G^{1\nu} + -\frac{g_3}{2}\, (\partial_\mu G^3_\nu) G^{4\mu} G^{5\nu}$$

$$\frac{g_3}{2}\, (\partial_\mu G^3_\nu) G^{5\mu} G^{4\nu} + \frac{g_3}{2}\, (\partial_\mu G^3_\nu) G^{6\mu} G^{7\nu} + -\frac{g_3}{2}\, (\partial_\mu G^3_\nu) G^{7\mu} G^{6\nu}$$

$$\frac{g_3}{2}\, (\partial_\mu G^4_\nu) G^{1\mu} G^{7\nu} + \frac{g_3}{2}\, (\partial_\mu G^4_\nu) G^{2\mu} G^{6\nu} + \frac{g_3}{2}\, (\partial_\mu G^4_\nu) G^{3\mu} G^{5\nu}$$

$$-\frac{g_3}{2}\,(\partial_\mu G^4_\nu)G^{5\mu}G^{3\nu} + -\frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^4_\nu)G^{5\mu}G^{8\nu} + -\frac{g_3}{2}\,(\partial_\mu G^4_\nu)G^{6\mu}G^{2\nu}$$

$$-\frac{g_3}{2}\,(\partial_\mu G^4_\nu)G^{7\mu}G^{1\nu} + \frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^4_\nu)G^{8\mu}G^{5\nu} + -\frac{g_3}{2}\,(\partial_\mu G^5_\nu)G^{1\mu}G^{6\nu}$$

$$\frac{g_3}{2}\,(\partial_\mu G^5_\nu)G^{2\mu}G^{7\nu} + -\frac{g_3}{2}\,(\partial_\mu G^5_\nu)G^{3\mu}G^{4\nu} + \frac{g_3}{2}\,(\partial_\mu G^5_\nu)G^{4\mu}G^{3\nu}$$

$$\frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^5_\nu)G^{4\mu}G^{8\nu} + \frac{g_3}{2}\,(\partial_\mu G^5_\nu)G^{6\mu}G^{1\nu} + -\frac{g_3}{2}\,(\partial_\mu G^5_\nu)G^{7\mu}G^{2\nu}$$

$$-\frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^5_\nu)G^{8\mu}G^{4\nu} + \frac{g_3}{2}\,(\partial_\mu G^6_\nu)G^{1\mu}G^{5\nu} + -\frac{g_3}{2}\,(\partial_\mu G^6_\nu)G^{2\mu}G^{4\nu}$$

$$-\frac{g_3}{2}\,(\partial_\mu G^6_\nu)G^{3\mu}G^{7\nu} + \frac{g_3}{2}\,(\partial_\mu G^6_\nu)G^{4\mu}G^{2\nu} + -\frac{g_3}{2}\,(\partial_\mu G^6_\nu)G^{5\mu}G^{1\nu}$$

$$\frac{g_3}{2}\,(\partial_\mu G^6_\nu)G^{7\mu}G^{3\nu} + -\frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^6_\nu)G^{7\mu}G^{8\nu} + \frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^6_\nu)G^{8\mu}G^{7\nu}$$

$$-\frac{g_3}{2}\,(\partial_\mu G^7_\nu)G^{1\mu}G^{4\nu} + -\frac{g_3}{2}\,(\partial_\mu G^7_\nu)G^{2\mu}G^{5\nu} + \frac{g_3}{2}\,(\partial_\mu G^7_\nu)G^{3\mu}G^{6\nu}$$

$$\frac{g_3}{2}\,(\partial_\mu G^7_\nu)G^{4\mu}G^{1\nu} + \frac{g_3}{2}\,(\partial_\mu G^7_\nu)G^{5\mu}G^{2\nu} + -\frac{g_3}{2}\,(\partial_\mu G^7_\nu)G^{6\mu}G^{3\nu}$$

$$\frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^7_\nu)G^{6\mu}G^{8\nu} + -\frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^7_\nu)G^{8\mu}G^{6\nu} + -\frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^8_\nu)G^{4\mu}G^{5\nu}$$

$$\frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^8_\nu)G^{5\mu}G^{4\nu} + -\frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^8_\nu)G^{6\mu}G^{7\nu} + \frac{\sqrt{3}g_3}{2}\,(\partial_\mu G^8_\nu)G^{7\mu}G^{6\nu}$$

$$-\frac{g_3^2}{4}\,G^1_\mu G^2_\nu G^{1\mu}G^{2\nu} + \frac{g_3^2}{4}\,G^1_\mu G^2_\nu G^{2\mu}G^{1\nu} + -\frac{g_3^2}{8}\,G^1_\mu G^2_\nu G^{4\mu}G^{5\nu}$$

$$\frac{g_3^2}{8}\,G^1_\mu G^2_\nu G^{5\mu}G^{4\nu} + \frac{g_3^2}{8}\,G^1_\mu G^2_\nu G^{6\mu}G^{7\nu} + -\frac{g_3^2}{8}\,G^1_\mu G^2_\nu G^{7\mu}G^{6\nu}$$

$$-\frac{g_3^2}{4}\,G^1_\mu G^3_\nu G^{1\mu}G^{3\nu} + \frac{g_3^2}{4}\,G^1_\mu G^3_\nu G^{3\mu}G^{1\nu} + \frac{g_3^2}{8}\,G^1_\mu G^3_\nu G^{4\mu}G^{6\nu}$$

$$\frac{g_3^2}{8}\,G^1_\mu G^3_\nu G^{5\mu}G^{7\nu} + -\frac{g_3^2}{8}\,G^1_\mu G^3_\nu G^{6\mu}G^{4\nu} + -\frac{g_3^2}{8}\,G^1_\mu G^3_\nu G^{7\mu}G^{5\nu}$$

$$-\frac{g_3^2}{16}\,G^1_\mu G^4_\nu G^{1\mu}G^{4\nu} + -\frac{g_3^2}{16}\,G^1_\mu G^4_\nu G^{2\mu}G^{5\nu} + \frac{g_3^2}{16}\,G^1_\mu G^4_\nu G^{3\mu}G^{6\nu}$$

$$\frac{g_3^2}{16}\,G^1_\mu G^4_\nu G^{4\mu}G^{1\nu} + \frac{g_3^2}{16}\,G^1_\mu G^4_\nu G^{5\mu}G^{2\nu} + -\frac{g_3^2}{16}\,G^1_\mu G^4_\nu G^{6\mu}G^{3\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^1_\mu G^4_\nu G^{6\mu}G^{8\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^1_\mu G^4_\nu G^{8\mu}G^{6\nu} + -\frac{g_3^2}{16}\,G^1_\mu G^5_\nu G^{1\mu}G^{5\nu}$$

$$\frac{g_3^2}{16}\,G^1_\mu G^5_\nu G^{2\mu}G^{4\nu} + \frac{g_3^2}{16}\,G^1_\mu G^5_\nu G^{3\mu}G^{7\nu} + -\frac{g_3^2}{16}\,G^1_\mu G^5_\nu G^{4\mu}G^{2\nu}$$

$$\frac{g_3^2}{16}\,G^1_\mu G^5_\nu G^{5\mu}G^{1\nu} + -\frac{g_3^2}{16}\,G^1_\mu G^5_\nu G^{7\mu}G^{3\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^1_\mu G^5_\nu G^{7\mu}G^{8\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^1_\mu G^5_\nu G^{8\mu}G^{7\nu} + -\frac{g_3^2}{16}\,G^1_\mu G^6_\nu G^{1\mu}G^{6\nu} + \frac{g_3^2}{16}\,G^1_\mu G^6_\nu G^{2\mu}G^{7\nu}$$

$$-\frac{g_3^2}{16}\,G^1_\mu G^6_\nu G^{3\mu}G^{4\nu} + \frac{g_3^2}{16}\,G^1_\mu G^6_\nu G^{4\mu}G^{3\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^1_\mu G^6_\nu G^{4\mu}G^{8\nu}$$

$$\frac{g_3^2}{16}\,G^1_\mu G^6_\nu G^{6\mu}G^{1\nu} + -\frac{g_3^2}{16}\,G^1_\mu G^6_\nu G^{7\mu}G^{2\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^1_\mu G^6_\nu G^{8\mu}G^{4\nu}$$

$$-\frac{g_3^2}{16}\,G^1_\mu G^7_\nu G^{1\mu}G^{7\nu} + -\frac{g_3^2}{16}\,G^1_\mu G^7_\nu G^{2\mu}G^{6\nu} + -\frac{g_3^2}{16}\,G^1_\mu G^7_\nu G^{3\mu}G^{5\nu}$$

$$\frac{g_3^2}{16}\,G^1_\mu G^7_\nu G^{5\mu}G^{3\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^1_\mu G^7_\nu G^{5\mu}G^{8\nu} + \frac{g_3^2}{16}\,G^1_\mu G^7_\nu G^{6\mu}G^{2\nu}$$

$$\frac{g_3^2}{16}\,G^1_\mu G^7_\nu G^{7\mu}G^{1\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^1_\mu G^7_\nu G^{8\mu}G^{5\nu} + \frac{g_3^2}{4}\,G^2_\mu G^1_\nu G^{1\mu}G^{2\nu}$$

$$-\frac{g_3^2}{4}\,G^2_\mu G^1_\nu G^{2\mu}G^{1\nu} + \frac{g_3^2}{8}\,G^2_\mu G^1_\nu G^{4\mu}G^{5\nu} + -\frac{g_3^2}{8}\,G^2_\mu G^1_\nu G^{5\mu}G^{4\nu}$$

$$-\frac{g_3^2}{8}\,G^2_\mu G^1_\nu G^{6\mu}G^{7\nu} + \frac{g_3^2}{8}\,G^2_\mu G^1_\nu G^{7\mu}G^{6\nu} + -\frac{g_3^2}{4}\,G^2_\mu G^3_\nu G^{2\mu}G^{3\nu}$$

$$\frac{g_3^2}{4}\,G^2_\mu G^3_\nu G^{3\mu}G^{2\nu} + -\frac{g_3^2}{8}\,G^2_\mu G^3_\nu G^{4\mu}G^{7\nu} + \frac{g_3^2}{8}\,G^2_\mu G^3_\nu G^{5\mu}G^{6\nu}$$

$$-\frac{g_3^2}{8}\,G^2_\mu G^3_\nu G^{6\mu}G^{5\nu} + \frac{g_3^2}{8}\,G^2_\mu G^3_\nu G^{7\mu}G^{4\nu} + \frac{g_3^2}{16}\,G^2_\mu G^4_\nu G^{1\mu}G^{5\nu}$$

$$-\frac{g_3^2}{16}\,G^2_\mu G^4_\nu G^{2\mu}G^{4\nu} + -\frac{g_3^2}{16}\,G^2_\mu G^4_\nu G^{3\mu}G^{7\nu} + \frac{g_3^2}{16}\,G^2_\mu G^4_\nu G^{4\mu}G^{2\nu}$$

$$-\frac{g_3^2}{16}\,G^2_\mu G^4_\nu G^{5\mu}G^{1\nu} + \frac{g_3^2}{16}\,G^2_\mu G^4_\nu G^{7\mu}G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^2_\mu G^4_\nu G^{7\mu}G^{8\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^2_\mu G^4_\nu G^{8\mu}G^{7\nu} + -\frac{g_3^2}{16}\,G^2_\mu G^5_\nu G^{1\mu}G^{4\nu} + -\frac{g_3^2}{16}\,G^2_\mu G^5_\nu G^{2\mu}G^{5\nu}$$

$$\frac{g_3^2}{16}\,G^2_\mu G^5_\nu G^{3\mu}G^{6\nu} + \frac{g_3^2}{16}\,G^2_\mu G^5_\nu G^{4\mu}G^{1\nu} + \frac{g_3^2}{16}\,G^2_\mu G^5_\nu G^{5\mu}G^{2\nu}$$

$$-\frac{g_3^2}{16}\, G^2_\mu G^5_\nu G^{6\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16}\, G^2_\mu G^5_\nu G^{6\mu} G^{8\nu} + -\frac{\sqrt{3}g_3^2}{16}\, G^2_\mu G^5_\nu G^{8\mu} G^{6\nu}$$

$$-\frac{g_3^2}{16}\, G^2_\mu G^6_\nu G^{1\mu} G^{7\nu} + -\frac{g_3^2}{16}\, G^2_\mu G^6_\nu G^{2\mu} G^{6\nu} + -\frac{g_3^2}{16}\, G^2_\mu G^6_\nu G^{3\mu} G^{5\nu}$$

$$\frac{g_3^2}{16}\, G^2_\mu G^6_\nu G^{5\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16}\, G^2_\mu G^6_\nu G^{5\mu} G^{8\nu} + \frac{g_3^2}{16}\, G^2_\mu G^6_\nu G^{6\mu} G^{2\nu}$$

$$\frac{g_3^2}{16}\, G^2_\mu G^6_\nu G^{7\mu} G^{1\nu} + -\frac{\sqrt{3}g_3^2}{16}\, G^2_\mu G^6_\nu G^{8\mu} G^{5\nu} + \frac{g_3^2}{16}\, G^2_\mu G^7_\nu G^{1\mu} G^{6\nu}$$

$$-\frac{g_3^2}{16}\, G^2_\mu G^7_\nu G^{2\mu} G^{7\nu} + \frac{g_3^2}{16}\, G^2_\mu G^7_\nu G^{3\mu} G^{4\nu} + -\frac{g_3^2}{16}\, G^2_\mu G^7_\nu G^{4\mu} G^{3\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\, G^2_\mu G^7_\nu G^{4\mu} G^{8\nu} + -\frac{g_3^2}{16}\, G^2_\mu G^7_\nu G^{6\mu} G^{1\nu} + \frac{g_3^2}{16}\, G^2_\mu G^7_\nu G^{7\mu} G^{2\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\, G^2_\mu G^7_\nu G^{8\mu} G^{4\nu} + \frac{g_3^2}{4}\, G^3_\mu G^1_\nu G^{1\mu} G^{3\nu} + -\frac{g_3^2}{4}\, G^3_\mu G^1_\nu G^{3\mu} G^{1\nu}$$

$$-\frac{g_3^2}{8}\, G^3_\mu G^1_\nu G^{4\mu} G^{6\nu} + -\frac{g_3^2}{8}\, G^3_\mu G^1_\nu G^{5\mu} G^{7\nu} + \frac{g_3^2}{8}\, G^3_\mu G^1_\nu G^{6\mu} G^{4\nu}$$

$$\frac{g_3^2}{8}\, G^3_\mu G^1_\nu G^{7\mu} G^{5\nu} + \frac{g_3^2}{4}\, G^3_\mu G^2_\nu G^{2\mu} G^{3\nu} + -\frac{g_3^2}{4}\, G^3_\mu G^2_\nu G^{3\mu} G^{2\nu}$$

$$\frac{g_3^2}{8}\, G^3_\mu G^2_\nu G^{4\mu} G^{7\nu} + -\frac{g_3^2}{8}\, G^3_\mu G^2_\nu G^{5\mu} G^{6\nu} + \frac{g_3^2}{8}\, G^3_\mu G^2_\nu G^{6\mu} G^{5\nu}$$

$$-\frac{g_3^2}{8}\, G^3_\mu G^2_\nu G^{7\mu} G^{4\nu} + -\frac{g_3^2}{16}\, G^3_\mu G^4_\nu G^{1\mu} G^{6\nu} + \frac{g_3^2}{16}\, G^3_\mu G^4_\nu G^{2\mu} G^{7\nu}$$

$$-\frac{g_3^2}{16}\, G^3_\mu G^4_\nu G^{3\mu} G^{4\nu} + \frac{g_3^2}{16}\, G^3_\mu G^4_\nu G^{4\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16}\, G^3_\mu G^4_\nu G^{4\mu} G^{8\nu}$$

$$\frac{g_3^2}{16}\, G^3_\mu G^4_\nu G^{6\mu} G^{1\nu} + -\frac{g_3^2}{16}\, G^3_\mu G^4_\nu G^{7\mu} G^{2\nu} + -\frac{\sqrt{3}g_3^2}{16}\, G^3_\mu G^4_\nu G^{8\mu} G^{4\nu}$$

$$-\frac{g_3^2}{16}\, G^3_\mu G^5_\nu G^{1\mu} G^{7\nu} + -\frac{g_3^2}{16}\, G^3_\mu G^5_\nu G^{2\mu} G^{6\nu} + -\frac{g_3^2}{16}\, G^3_\mu G^5_\nu G^{3\mu} G^{5\nu}$$

$$\frac{g_3^2}{16}\, G^3_\mu G^5_\nu G^{5\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16}\, G^3_\mu G^5_\nu G^{5\mu} G^{8\nu} + \frac{g_3^2}{16}\, G^3_\mu G^5_\nu G^{6\mu} G^{2\nu}$$

$$\frac{g_3^2}{16}\, G^3_\mu G^5_\nu G^{7\mu} G^{1\nu} + -\frac{\sqrt{3}g_3^2}{16}\, G^3_\mu G^5_\nu G^{8\mu} G^{5\nu} + \frac{g_3^2}{16}\, G^3_\mu G^6_\nu G^{1\mu} G^{4\nu}$$

$$\frac{g_3^2}{16}\, G^3_\mu G^6_\nu G^{2\mu} G^{5\nu} + -\frac{g_3^2}{16}\, G^3_\mu G^6_\nu G^{3\mu} G^{6\nu} + -\frac{g_3^2}{16}\, G^3_\mu G^6_\nu G^{4\mu} G^{1\nu}$$

$$-\frac{g_3^2}{16}\,G^3_\mu G^6_\nu G^{5\mu}G^{2\nu}+\frac{g_3^2}{16}\,G^3_\mu G^6_\nu G^{6\mu}G^{3\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^3_\mu G^6_\nu G^{6\mu}G^{8\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^3_\mu G^6_\nu G^{8\mu}G^{6\nu}+\frac{g_3^2}{16}\,G^3_\mu G^7_\nu G^{1\mu}G^{5\nu}+-\frac{g_3^2}{16}\,G^3_\mu G^7_\nu G^{2\mu}G^{4\nu}$$

$$-\frac{g_3^2}{16}\,G^3_\mu G^7_\nu G^{3\mu}G^{7\nu}+\frac{g_3^2}{16}\,G^3_\mu G^7_\nu G^{4\mu}G^{2\nu}+-\frac{g_3^2}{16}\,G^3_\mu G^7_\nu G^{5\mu}G^{1\nu}$$

$$\frac{g_3^2}{16}\,G^3_\mu G^7_\nu G^{7\mu}G^{3\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^3_\mu G^7_\nu G^{7\mu}G^{8\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^3_\mu G^7_\nu G^{8\mu}G^{7\nu}$$

$$\frac{g_3^2}{16}\,G^4_\mu G^1_\nu G^{1\mu}G^{4\nu}+\frac{g_3^2}{16}\,G^4_\mu G^1_\nu G^{2\mu}G^{5\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^1_\nu G^{3\mu}G^{6\nu}$$

$$-\frac{g_3^2}{16}\,G^4_\mu G^1_\nu G^{4\mu}G^{1\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^1_\nu G^{5\mu}G^{2\nu}+\frac{g_3^2}{16}\,G^4_\mu G^1_\nu G^{6\mu}G^{3\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^1_\nu G^{6\mu}G^{8\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^1_\nu G^{8\mu}G^{6\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^2_\nu G^{1\mu}G^{5\nu}$$

$$\frac{g_3^2}{16}\,G^4_\mu G^2_\nu G^{2\mu}G^{4\nu}+\frac{g_3^2}{16}\,G^4_\mu G^2_\nu G^{3\mu}G^{7\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^2_\nu G^{4\mu}G^{2\nu}$$

$$\frac{g_3^2}{16}\,G^4_\mu G^2_\nu G^{5\mu}G^{1\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^2_\nu G^{7\mu}G^{3\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^2_\nu G^{7\mu}G^{8\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^2_\nu G^{8\mu}G^{7\nu}+\frac{g_3^2}{16}\,G^4_\mu G^3_\nu G^{1\mu}G^{6\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^3_\nu G^{2\mu}G^{7\nu}$$

$$\frac{g_3^2}{16}\,G^4_\mu G^3_\nu G^{3\mu}G^{4\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^3_\nu G^{4\mu}G^{3\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^3_\nu G^{4\mu}G^{8\nu}$$

$$-\frac{g_3^2}{16}\,G^4_\mu G^3_\nu G^{6\mu}G^{1\nu}+\frac{g_3^2}{16}\,G^4_\mu G^3_\nu G^{7\mu}G^{2\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^3_\nu G^{8\mu}G^{4\nu}$$

$$-\frac{g_3^2}{8}\,G^4_\mu G^5_\nu G^{1\mu}G^{2\nu}+\frac{g_3^2}{8}\,G^4_\mu G^5_\nu G^{2\mu}G^{1\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^5_\nu G^{4\mu}G^{5\nu}$$

$$\frac{g_3^2}{16}\,G^4_\mu G^5_\nu G^{5\mu}G^{4\nu}+\frac{g_3^2}{16}\,G^4_\mu G^5_\nu G^{6\mu}G^{7\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^5_\nu G^{7\mu}G^{6\nu}$$

$$-\frac{3g_3^2}{16}\,G^4_\mu G^5_\nu G^{4\mu}G^{5\nu}+\frac{3g_3^2}{16}\,G^4_\mu G^5_\nu G^{5\mu}G^{4\nu}+-\frac{3g_3^2}{16}\,G^4_\mu G^5_\nu G^{6\mu}G^{7\nu}$$

$$\frac{3g_3^2}{16}\,G^4_\mu G^5_\nu G^{7\mu}G^{6\nu}+\frac{g_3^2}{8}\,G^4_\mu G^6_\nu G^{1\mu}G^{3\nu}+-\frac{g_3^2}{8}\,G^4_\mu G^6_\nu G^{3\mu}G^{1\nu}$$

$$-\frac{g_3^2}{16}\,G^4_\mu G^6_\nu G^{4\mu}G^{6\nu}+-\frac{g_3^2}{16}\,G^4_\mu G^6_\nu G^{5\mu}G^{7\nu}+\frac{g_3^2}{16}\,G^4_\mu G^6_\nu G^{6\mu}G^{4\nu}$$

$$\frac{g_3^2}{16}\,G^4_\mu G^6_\nu G^{7\mu}G^{5\nu} + -\frac{g_3^2}{8}\,G^4_\mu G^7_\nu G^{2\mu}G^{3\nu} + \frac{g_3^2}{8}\,G^4_\mu G^7_\nu G^{3\mu}G^{2\nu}$$

$$-\frac{g_3^2}{16}\,G^4_\mu G^7_\nu G^{4\mu}G^{7\nu} + \frac{g_3^2}{16}\,G^4_\mu G^7_\nu G^{5\mu}G^{6\nu} + -\frac{g_3^2}{16}\,G^4_\mu G^7_\nu G^{6\mu}G^{5\nu}$$

$$\frac{g_3^2}{16}\,G^4_\mu G^7_\nu G^{7\mu}G^{4\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^8_\nu G^{1\mu}G^{6\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^8_\nu G^{2\mu}G^{7\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^8_\nu G^{3\mu}G^{4\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^8_\nu G^{4\mu}G^{3\nu} + -\frac{3g_3^2}{16}\,G^4_\mu G^8_\nu G^{4\mu}G^{8\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^8_\nu G^{6\mu}G^{1\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^4_\mu G^8_\nu G^{7\mu}G^{2\nu} + \frac{3g_3^2}{16}\,G^4_\mu G^8_\nu G^{8\mu}G^{4\nu}$$

$$\frac{g_3^2}{16}\,G^5_\mu G^1_\nu G^{1\mu}G^{5\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^1_\nu G^{2\mu}G^{4\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^1_\nu G^{3\mu}G^{7\nu}$$

$$\frac{g_3^2}{16}\,G^5_\mu G^1_\nu G^{4\mu}G^{2\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^1_\nu G^{5\mu}G^{1\nu} + \frac{g_3^2}{16}\,G^5_\mu G^1_\nu G^{7\mu}G^{3\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^1_\nu G^{7\mu}G^{8\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^1_\nu G^{8\mu}G^{7\nu} + \frac{g_3^2}{16}\,G^5_\mu G^2_\nu G^{1\mu}G^{4\nu}$$

$$\frac{g_3^2}{16}\,G^5_\mu G^2_\nu G^{2\mu}G^{5\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^2_\nu G^{3\mu}G^{6\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^2_\nu G^{4\mu}G^{1\nu}$$

$$-\frac{g_3^2}{16}\,G^5_\mu G^2_\nu G^{5\mu}G^{2\nu} + \frac{g_3^2}{16}\,G^5_\mu G^2_\nu G^{6\mu}G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^2_\nu G^{6\mu}G^{8\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^2_\nu G^{8\mu}G^{6\nu} + \frac{g_3^2}{16}\,G^5_\mu G^3_\nu G^{1\mu}G^{7\nu} + \frac{g_3^2}{16}\,G^5_\mu G^3_\nu G^{2\mu}G^{6\nu}$$

$$\frac{g_3^2}{16}\,G^5_\mu G^3_\nu G^{3\mu}G^{5\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^3_\nu G^{5\mu}G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^3_\nu G^{5\mu}G^{8\nu}$$

$$-\frac{g_3^2}{16}\,G^5_\mu G^3_\nu G^{6\mu}G^{2\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^3_\nu G^{7\mu}G^{1\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^3_\nu G^{8\mu}G^{5\nu}$$

$$\frac{g_3^2}{8}\,G^5_\mu G^4_\nu G^{1\mu}G^{2\nu} + -\frac{g_3^2}{8}\,G^5_\mu G^4_\nu G^{2\mu}G^{1\nu} + \frac{g_3^2}{16}\,G^5_\mu G^4_\nu G^{4\mu}G^{5\nu}$$

$$-\frac{g_3^2}{16}\,G^5_\mu G^4_\nu G^{5\mu}G^{4\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^4_\nu G^{6\mu}G^{7\nu} + \frac{g_3^2}{16}\,G^5_\mu G^4_\nu G^{7\mu}G^{6\nu}$$

$$\frac{3g_3^2}{16}\,G^5_\mu G^4_\nu G^{4\mu}G^{5\nu} + -\frac{3g_3^2}{16}\,G^5_\mu G^4_\nu G^{5\mu}G^{4\nu} + \frac{3g_3^2}{16}\,G^5_\mu G^4_\nu G^{6\mu}G^{7\nu}$$

$$-\frac{3g_3^2}{16}\,G^5_\mu G^4_\nu G^{7\mu}G^{6\nu} + \frac{g_3^2}{8}\,G^5_\mu G^6_\nu G^{2\mu}G^{3\nu} + -\frac{g_3^2}{8}\,G^5_\mu G^6_\nu G^{3\mu}G^{2\nu}$$

$$\frac{g_3^2}{16}\,G^5_\mu G^6_\nu G^{4\mu} G^{7\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^6_\nu G^{5\mu} G^{6\nu} + \frac{g_3^2}{16}\,G^5_\mu G^6_\nu G^{6\mu} G^{5\nu}$$

$$-\frac{g_3^2}{16}\,G^5_\mu G^6_\nu G^{7\mu} G^{4\nu} + \frac{g_3^2}{8}\,G^5_\mu G^7_\nu G^{1\mu} G^{3\nu} + -\frac{g_3^2}{8}\,G^5_\mu G^7_\nu G^{3\mu} G^{1\nu}$$

$$-\frac{g_3^2}{16}\,G^5_\mu G^7_\nu G^{4\mu} G^{6\nu} + -\frac{g_3^2}{16}\,G^5_\mu G^7_\nu G^{5\mu} G^{7\nu} + \frac{g_3^2}{16}\,G^5_\mu G^7_\nu G^{6\mu} G^{4\nu}$$

$$\frac{g_3^2}{16}\,G^5_\mu G^7_\nu G^{7\mu} G^{5\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^8_\nu G^{1\mu} G^{7\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^8_\nu G^{2\mu} G^{6\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^8_\nu G^{3\mu} G^{5\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^8_\nu G^{5\mu} G^{3\nu} + -\frac{3g_3^2}{16}\,G^5_\mu G^8_\nu G^{5\mu} G^{8\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^8_\nu G^{6\mu} G^{2\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^5_\mu G^8_\nu G^{7\mu} G^{1\nu} + \frac{3g_3^2}{16}\,G^5_\mu G^8_\nu G^{8\mu} G^{5\nu}$$

$$\frac{g_3^2}{16}\,G^6_\mu G^1_\nu G^{1\mu} G^{6\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^1_\nu G^{2\mu} G^{7\nu} + \frac{g_3^2}{16}\,G^6_\mu G^1_\nu G^{3\mu} G^{4\nu}$$

$$-\frac{g_3^2}{16}\,G^6_\mu G^1_\nu G^{4\mu} G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^1_\nu G^{4\mu} G^{8\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^1_\nu G^{6\mu} G^{1\nu}$$

$$\frac{g_3^2}{16}\,G^6_\mu G^1_\nu G^{7\mu} G^{2\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^1_\nu G^{8\mu} G^{4\nu} + \frac{g_3^2}{16}\,G^6_\mu G^2_\nu G^{1\mu} G^{7\nu}$$

$$\frac{g_3^2}{16}\,G^6_\mu G^2_\nu G^{2\mu} G^{6\nu} + \frac{g_3^2}{16}\,G^6_\mu G^2_\nu G^{3\mu} G^{5\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^2_\nu G^{5\mu} G^{3\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^2_\nu G^{5\mu} G^{8\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^2_\nu G^{6\mu} G^{2\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^2_\nu G^{7\mu} G^{1\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^2_\nu G^{8\mu} G^{5\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^3_\nu G^{1\mu} G^{4\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^3_\nu G^{2\mu} G^{5\nu}$$

$$\frac{g_3^2}{16}\,G^6_\mu G^3_\nu G^{3\mu} G^{6\nu} + \frac{g_3^2}{16}\,G^6_\mu G^3_\nu G^{4\mu} G^{1\nu} + \frac{g_3^2}{16}\,G^6_\mu G^3_\nu G^{5\mu} G^{2\nu}$$

$$-\frac{g_3^2}{16}\,G^6_\mu G^3_\nu G^{6\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^3_\nu G^{6\mu} G^{8\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^3_\nu G^{8\mu} G^{6\nu}$$

$$-\frac{g_3^2}{8}\,G^6_\mu G^4_\nu G^{1\mu} G^{3\nu} + \frac{g_3^2}{8}\,G^6_\mu G^4_\nu G^{3\mu} G^{1\nu} + \frac{g_3^2}{16}\,G^6_\mu G^4_\nu G^{4\mu} G^{6\nu}$$

$$\frac{g_3^2}{16}\,G^6_\mu G^4_\nu G^{5\mu} G^{7\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^4_\nu G^{6\mu} G^{4\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^4_\nu G^{7\mu} G^{5\nu}$$

$$-\frac{g_3^2}{8}\,G^6_\mu G^5_\nu G^{2\mu} G^{3\nu} + \frac{g_3^2}{8}\,G^6_\mu G^5_\nu G^{3\mu} G^{2\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^5_\nu G^{4\mu} G^{7\nu}$$

$$\frac{g_3^2}{16}\,G^6_\mu G^5_\nu G^{5\mu}G^{6\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^5_\nu G^{6\mu}G^{5\nu} + \frac{g_3^2}{16}\,G^6_\mu G^5_\nu G^{7\mu}G^{4\nu}$$

$$\frac{g_3^2}{8}\,G^6_\mu G^7_\nu G^{1\mu}G^{2\nu} + -\frac{g_3^2}{8}\,G^6_\mu G^7_\nu G^{2\mu}G^{1\nu} + \frac{g_3^2}{16}\,G^6_\mu G^7_\nu G^{4\mu}G^{5\nu}$$

$$-\frac{g_3^2}{16}\,G^6_\mu G^7_\nu G^{5\mu}G^{4\nu} + -\frac{g_3^2}{16}\,G^6_\mu G^7_\nu G^{6\mu}G^{7\nu} + \frac{g_3^2}{16}\,G^6_\mu G^7_\nu G^{7\mu}G^{6\nu}$$

$$-\frac{3g_3^2}{16}\,G^6_\mu G^7_\nu G^{4\mu}G^{5\nu} + \frac{3g_3^2}{16}\,G^6_\mu G^7_\nu G^{5\mu}G^{4\nu} + -\frac{3g_3^2}{16}\,G^6_\mu G^7_\nu G^{6\mu}G^{7\nu}$$

$$\frac{3g_3^2}{16}\,G^6_\mu G^7_\nu G^{7\mu}G^{6\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^8_\nu G^{1\mu}G^{4\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^8_\nu G^{2\mu}G^{5\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^8_\nu G^{3\mu}G^{6\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^8_\nu G^{4\mu}G^{1\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^8_\nu G^{5\mu}G^{2\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^6_\mu G^8_\nu G^{6\mu}G^{3\nu} + -\frac{3g_3^2}{16}\,G^6_\mu G^8_\nu G^{6\mu}G^{8\nu} + \frac{3g_3^2}{16}\,G^6_\mu G^8_\nu G^{8\mu}G^{6\nu}$$

$$\frac{g_3^2}{16}\,G^7_\mu G^1_\nu G^{1\mu}G^{7\nu} + \frac{g_3^2}{16}\,G^7_\mu G^1_\nu G^{2\mu}G^{6\nu} + \frac{g_3^2}{16}\,G^7_\mu G^1_\nu G^{3\mu}G^{5\nu}$$

$$-\frac{g_3^2}{16}\,G^7_\mu G^1_\nu G^{5\mu}G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^1_\nu G^{5\mu}G^{8\nu} + -\frac{g_3^2}{16}\,G^7_\mu G^1_\nu G^{6\mu}G^{2\nu}$$

$$-\frac{g_3^2}{16}\,G^7_\mu G^1_\nu G^{7\mu}G^{1\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^1_\nu G^{8\mu}G^{5\nu} + -\frac{g_3^2}{16}\,G^7_\mu G^2_\nu G^{1\mu}G^{6\nu}$$

$$\frac{g_3^2}{16}\,G^7_\mu G^2_\nu G^{2\mu}G^{7\nu} + -\frac{g_3^2}{16}\,G^7_\mu G^2_\nu G^{3\mu}G^{4\nu} + \frac{g_3^2}{16}\,G^7_\mu G^2_\nu G^{4\mu}G^{3\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^2_\nu G^{4\mu}G^{8\nu} + \frac{g_3^2}{16}\,G^7_\mu G^2_\nu G^{6\mu}G^{1\nu} + -\frac{g_3^2}{16}\,G^7_\mu G^2_\nu G^{7\mu}G^{2\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^2_\nu G^{8\mu}G^{4\nu} + -\frac{g_3^2}{16}\,G^7_\mu G^3_\nu G^{1\mu}G^{5\nu} + \frac{g_3^2}{16}\,G^7_\mu G^3_\nu G^{2\mu}G^{4\nu}$$

$$\frac{g_3^2}{16}\,G^7_\mu G^3_\nu G^{3\mu}G^{7\nu} + -\frac{g_3^2}{16}\,G^7_\mu G^3_\nu G^{4\mu}G^{2\nu} + \frac{g_3^2}{16}\,G^7_\mu G^3_\nu G^{5\mu}G^{1\nu}$$

$$-\frac{g_3^2}{16}\,G^7_\mu G^3_\nu G^{7\mu}G^{3\nu} + \frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^3_\nu G^{7\mu}G^{8\nu} + -\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^3_\nu G^{8\mu}G^{7\nu}$$

$$\frac{g_3^2}{8}\,G^7_\mu G^4_\nu G^{2\mu}G^{3\nu} + -\frac{g_3^2}{8}\,G^7_\mu G^4_\nu G^{3\mu}G^{2\nu} + \frac{g_3^2}{16}\,G^7_\mu G^4_\nu G^{4\mu}G^{7\nu}$$

$$-\frac{g_3^2}{16}\,G^7_\mu G^4_\nu G^{5\mu}G^{6\nu} + \frac{g_3^2}{16}\,G^7_\mu G^4_\nu G^{6\mu}G^{5\nu} + -\frac{g_3^2}{16}\,G^7_\mu G^4_\nu G^{7\mu}G^{4\nu}$$

$$-\frac{g_3^2}{8}\,G^7_\mu G^5_\nu G^{1\mu}G^{3\nu}+\frac{g_3^2}{8}\,G^7_\mu G^5_\nu G^{3\mu}G^{1\nu}+\frac{g_3^2}{16}\,G^7_\mu G^5_\nu G^{4\mu}G^{6\nu}$$

$$\frac{g_3^2}{16}\,G^7_\mu G^5_\nu G^{5\mu}G^{7\nu}+-\frac{g_3^2}{16}\,G^7_\mu G^5_\nu G^{6\mu}G^{4\nu}+-\frac{g_3^2}{16}\,G^7_\mu G^5_\nu G^{7\mu}G^{5\nu}$$

$$-\frac{g_3^2}{8}\,G^7_\mu G^6_\nu G^{1\mu}G^{2\nu}+\frac{g_3^2}{8}\,G^7_\mu G^6_\nu G^{2\mu}G^{1\nu}+-\frac{g_3^2}{16}\,G^7_\mu G^6_\nu G^{4\mu}G^{5\nu}$$

$$\frac{g_3^2}{16}\,G^7_\mu G^6_\nu G^{5\mu}G^{4\nu}+\frac{g_3^2}{16}\,G^7_\mu G^6_\nu G^{6\mu}G^{7\nu}+-\frac{g_3^2}{16}\,G^7_\mu G^6_\nu G^{7\mu}G^{6\nu}$$

$$\frac{3g_3^2}{16}\,G^7_\mu G^6_\nu G^{4\mu}G^{5\nu}+-\frac{3g_3^2}{16}\,G^7_\mu G^6_\nu G^{5\mu}G^{4\nu}+\frac{3g_3^2}{16}\,G^7_\mu G^6_\nu G^{6\mu}G^{7\nu}$$

$$-\frac{3g_3^2}{16}\,G^7_\mu G^6_\nu G^{7\mu}G^{6\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^8_\nu G^{1\mu}G^{5\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^8_\nu G^{2\mu}G^{4\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^8_\nu G^{3\mu}G^{7\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^8_\nu G^{4\mu}G^{2\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^8_\nu G^{5\mu}G^{1\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^7_\mu G^8_\nu G^{7\mu}G^{3\nu}+-\frac{3g_3^2}{16}\,G^7_\mu G^8_\nu G^{7\mu}G^{8\nu}+\frac{3g_3^2}{16}\,G^7_\mu G^8_\nu G^{8\mu}G^{7\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^4_\nu G^{1\mu}G^{6\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^4_\nu G^{2\mu}G^{7\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^4_\nu G^{3\mu}G^{4\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^4_\nu G^{4\mu}G^{3\nu}+\frac{3g_3^2}{16}\,G^8_\mu G^4_\nu G^{4\mu}G^{8\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^4_\nu G^{6\mu}G^{1\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^4_\nu G^{7\mu}G^{2\nu}+-\frac{3g_3^2}{16}\,G^8_\mu G^4_\nu G^{8\mu}G^{4\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^5_\nu G^{1\mu}G^{7\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^5_\nu G^{2\mu}G^{6\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^5_\nu G^{3\mu}G^{5\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^5_\nu G^{5\mu}G^{3\nu}$$

$$\frac{3g_3^2}{16}\,G^8_\mu G^5_\nu G^{5\mu}G^{8\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^5_\nu G^{6\mu}G^{2\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^5_\nu G^{7\mu}G^{1\nu}$$

$$-\frac{3g_3^2}{16}\,G^8_\mu G^5_\nu G^{8\mu}G^{5\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^6_\nu G^{1\mu}G^{4\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^6_\nu G^{2\mu}G^{5\nu}$$

$$\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^6_\nu G^{3\mu}G^{6\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^6_\nu G^{4\mu}G^{1\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^6_\nu G^{5\mu}G^{2\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^6_\nu G^{6\mu}G^{3\nu}+\frac{3g_3^2}{16}\,G^8_\mu G^6_\nu G^{6\mu}G^{8\nu}+-\frac{3g_3^2}{16}\,G^8_\mu G^6_\nu G^{8\mu}G^{6\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^7_\nu G^{1\mu}G^{5\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^7_\nu G^{2\mu}G^{4\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^7_\nu G^{3\mu}G^{7\nu}$$

$$-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^7_\nu G^{4\mu}G^{2\nu}+\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^7_\nu G^{5\mu}G^{1\nu}+-\frac{\sqrt{3}g_3^2}{16}\,G^8_\mu G^7_\nu G^{7\mu}G^{3\nu}$$

$$\frac{3g_3^2}{16}\,G^8_\mu G^7_\nu G^{7\mu}G^{8\nu}+-\frac{3g_3^2}{16}\,G^8_\mu G^7_\nu G^{8\mu}G^{7\nu}$$

4 EWSB Mass Matrices

4.1 VEV Substitutions

$$H^0 \to H^0 + \frac{\sqrt{2}v}{2}$$

$$H^{0\dagger} \to H^{0\dagger} + \frac{\sqrt{2}v}{2}$$

4.2 Stationary Condition

$$\frac{\partial V}{\partial v} = v\left(\lambda_{11}v^2 + m_1^2\right)$$
$$= 0$$

4.3 Neutral Scalar Mass Matrix

Neutral basis

$$b_1 = \mathrm{Re}(H^0)$$

$$b_2 = \mathrm{Im}(H^0)$$

Neutral scalar mass-matrix entries

$$M^2_{\text{neutral},11} = 3\lambda_{11}v^2 + m_1^2$$

$$M^2_{\text{neutral},12} = 0$$

$$M^2_{\text{neutral},21} = 0$$

$$M^2_{\text{neutral},22} = \lambda_{11}v^2 + m_1^2$$

4.4 Scalar Hermitian Blocks

$$M_1^2[(H^+)^\dagger; H^+] =$$
$$\left[\lambda_{11}v^2 + m_1^2\right]$$

$$M_2^2[\left({\phi_1}_b^{-1/3}\right)^\dagger, \left({\phi_4}_b^{-1/3}\right)^\dagger, \left({\phi_5}_b^{-1/3}\right)^\dagger; {\phi_1}_b^{-1/3}, {\phi_4}_b^{-1/3}, {\phi_5}_b^{-1/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{14}v^2}{2} + m_2^2 & \frac{\sqrt{2}v\overline{\mu_7}}{2} & -\frac{v^2\overline{\lambda_{15}}}{2} \\ \frac{\sqrt{2}\mu_7 v}{2} & \frac{\lambda_{19,c1}v^2}{2} + m_5^2 & -\frac{\sqrt{2}\mu_8 v}{2} \\ -\frac{\lambda_{15}v^2}{2} & -\frac{\sqrt{2}v\overline{\mu_8}}{2} & \frac{\lambda_{21,c2}v^2}{2} + m_6^2 \end{bmatrix}$$

$$M_3^2[\left({\phi_1}_g^{-1/3}\right)^\dagger, \left({\phi_4}_g^{-1/3}\right)^\dagger, \left({\phi_5}_g^{-1/3}\right)^\dagger; {\phi_1}_g^{-1/3}, {\phi_4}_g^{-1/3}, {\phi_5}_g^{-1/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{14}v^2}{2} + m_2^2 & \frac{\sqrt{2}v\overline{\mu_7}}{2} & -\frac{v^2\overline{\lambda_{15}}}{2} \\ \frac{\sqrt{2}\mu_7 v}{2} & \frac{\lambda_{19,c1}v^2}{2} + m_5^2 & -\frac{\sqrt{2}\mu_8 v}{2} \\ -\frac{\lambda_{15}v^2}{2} & -\frac{\sqrt{2}v\overline{\mu_8}}{2} & \frac{\lambda_{21,c2}v^2}{2} + m_6^2 \end{bmatrix}$$

$$M_4^2[\left(\phi_{1r}^{-1/3}\right)^\dagger, \left(\phi_{4r}^{-1/3}\right)^\dagger, \left(\phi_{5r}^{-1/3}\right)^\dagger; \phi_{1r}^{-1/3}, \phi_{4r}^{-1/3}, \phi_{5r}^{-1/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{14}v^2}{2} + m_2^2 & \frac{\sqrt{2}v\overline{\mu_7}}{2} & -\frac{v^2\overline{\lambda_{15}}}{2} \\ \frac{\sqrt{2}\mu_7 v}{2} & \frac{\lambda_{19,c1}v^2}{2} + m_5^2 & -\frac{\sqrt{2}\mu_8 v}{2} \\ -\frac{\lambda_{15}v^2}{2} & -\frac{\sqrt{2}v\overline{\mu_8}}{2} & \frac{\lambda_{21,c2}v^2}{2} + m_6^2 \end{bmatrix}$$

$$M_5^2[\left(\phi_{2b}^{-4/3}\right)^\dagger, \left(\phi_{5b}^{-4/3}\right)^\dagger; \phi_{2b}^{-4/3}, \phi_{5b}^{-4/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{16}v^2}{2} + m_3^2 & \frac{v^2\overline{\lambda_{12}}}{2} \\ \frac{\lambda_{12}v^2}{2} & \frac{\lambda_{21,c1}v^2}{2} + m_6^2 \end{bmatrix}$$

$$M_6^2[\left(\phi_{2g}^{-4/3}\right)^\dagger, \left(\phi_{5g}^{-4/3}\right)^\dagger; \phi_{2g}^{-4/3}, \phi_{5g}^{-4/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{16}v^2}{2} + m_3^2 & \frac{v^2\overline{\lambda_{12}}}{2} \\ \frac{\lambda_{12}v^2}{2} & \frac{\lambda_{21,c1}v^2}{2} + m_6^2 \end{bmatrix}$$

$$M_7^2[\left(\phi_{2r}^{-4/3}\right)^\dagger, \left(\phi_{5r}^{-4/3}\right)^\dagger; \phi_{2r}^{-4/3}, \phi_{5r}^{-4/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{16}v^2}{2} + m_3^2 & \frac{v^2\overline{\lambda_{12}}}{2} \\ \frac{\lambda_{12}v^2}{2} & \frac{\lambda_{21,c1}v^2}{2} + m_6^2 \end{bmatrix}$$

$$M_8^2[\left(\phi_{3b}^{+2/3}\right)^\dagger, \left(\phi_{4b}^{+2/3}\right)^\dagger, \left(\phi_{5b}^{+2/3}\right)^\dagger; \phi_{3b}^{+2/3}, \phi_{4b}^{+2/3}, \phi_{5b}^{+2/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{17,c1}v^2}{2} + m_4^2 & -\frac{\lambda_{13}v^2}{2} & 0 \\ -\frac{v^2\overline{\lambda_{13}}}{2} & \frac{\lambda_{19,c2}v^2}{2} + m_5^2 & -\mu_8 v \\ 0 & -v\overline{\mu_8} & -\frac{\lambda_{21,c1}v^2}{2} + \lambda_{21,c2}v^2 + m_6^2 \end{bmatrix}$$

$$M_9^2[\left(\phi_{3g}^{+2/3}\right)^\dagger, \left(\phi_{4g}^{+2/3}\right)^\dagger, \left(\phi_{5g}^{+2/3}\right)^\dagger; \phi_{3g}^{+2/3}, \phi_{4g}^{+2/3}, \phi_{5g}^{+2/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{17,c1}v^2}{2} + m_4^2 & -\frac{\lambda_{13}v^2}{2} & 0 \\ -\frac{v^2\overline{\lambda_{13}}}{2} & \frac{\lambda_{19,c2}v^2}{2} + m_5^2 & -\mu_8 v \\ 0 & -v\overline{\mu_8} & -\frac{\lambda_{21,c1}v^2}{2} + \lambda_{21,c2}v^2 + m_6^2 \end{bmatrix}$$

$$M_{10}^2[\left(\phi_{3r}^{+2/3}\right)^\dagger, \left(\phi_{4r}^{+2/3}\right)^\dagger, \left(\phi_{5r}^{+2/3}\right)^\dagger; \phi_{3r}^{+2/3}, \phi_{4r}^{+2/3}, \phi_{5r}^{+2/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{17,c1}v^2}{2} + m_4^2 & -\frac{\lambda_{13}v^2}{2} & 0 \\ -\frac{v^2\overline{\lambda_{13}}}{2} & \frac{\lambda_{19,c2}v^2}{2} + m_5^2 & -\mu_8 v \\ 0 & -v\overline{\mu_8} & -\frac{\lambda_{21,c1}v^2}{2} + \lambda_{21,c2}v^2 + m_6^2 \end{bmatrix}$$

$$M_{11}^2[\left(\phi_{3b}^{+5/3}\right)^\dagger; \phi_{3b}^{+5/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{17,c2}v^2}{2} + m_4^2 \end{bmatrix}$$

$$M_{12}^2[\left(\phi_{3g}^{+5/3}\right)^\dagger; \phi_{3g}^{+5/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{17,c2}v^2}{2} + m_4^2 \end{bmatrix}$$

$$M_{13}^2[\left(\phi_{3r}^{+5/3}\right)^\dagger; \phi_{3r}^{+5/3}] =$$
$$\begin{bmatrix} \frac{\lambda_{17,c2}v^2}{2} + m_4^2 \end{bmatrix}$$

4.5 Scalar Holomorphic Blocks

No blocks generated.

4.6 Fermion Blocks

$$M_1[d^{C+1/3,\bar{b}}; q_b^{-1/3}] = \left[\frac{\sqrt{2}Y_5 v}{2}\right]$$

$$M_2[d^{C+1/3,\bar{g}}; q_g^{-1/3}] = \left[\frac{\sqrt{2}Y_5 v}{2}\right]$$

$$M_3[d^{C+1/3,\bar{r}}; q_r^{-1/3}] = \left[\frac{\sqrt{2}Y_5 v}{2}\right]$$

$$M_4[e^{C+}; l^-] = \left[\frac{\sqrt{2}Y_{10} v}{2}\right]$$

$$M_5[u^{C-2/3,\bar{b}}; q_b^{+2/3}] = \left[\frac{\sqrt{2}Y_6 v}{2}\right]$$

$$M_6[u^{C-2/3,\bar{g}}; q_g^{+2/3}] = \left[\frac{\sqrt{2}Y_6 v}{2}\right]$$

$$M_7[u^{C-2/3,\bar{r}}; q_r^{+2/3}] = \left[\frac{\sqrt{2}Y_6 v}{2}\right]$$